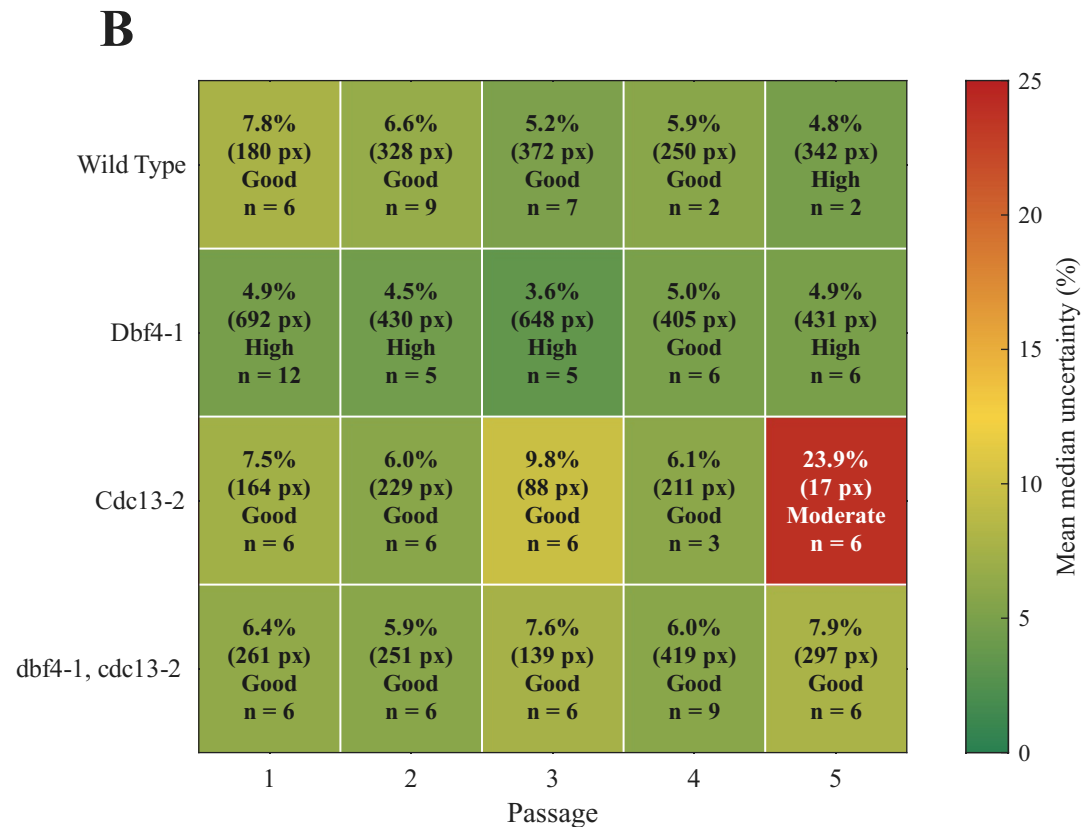
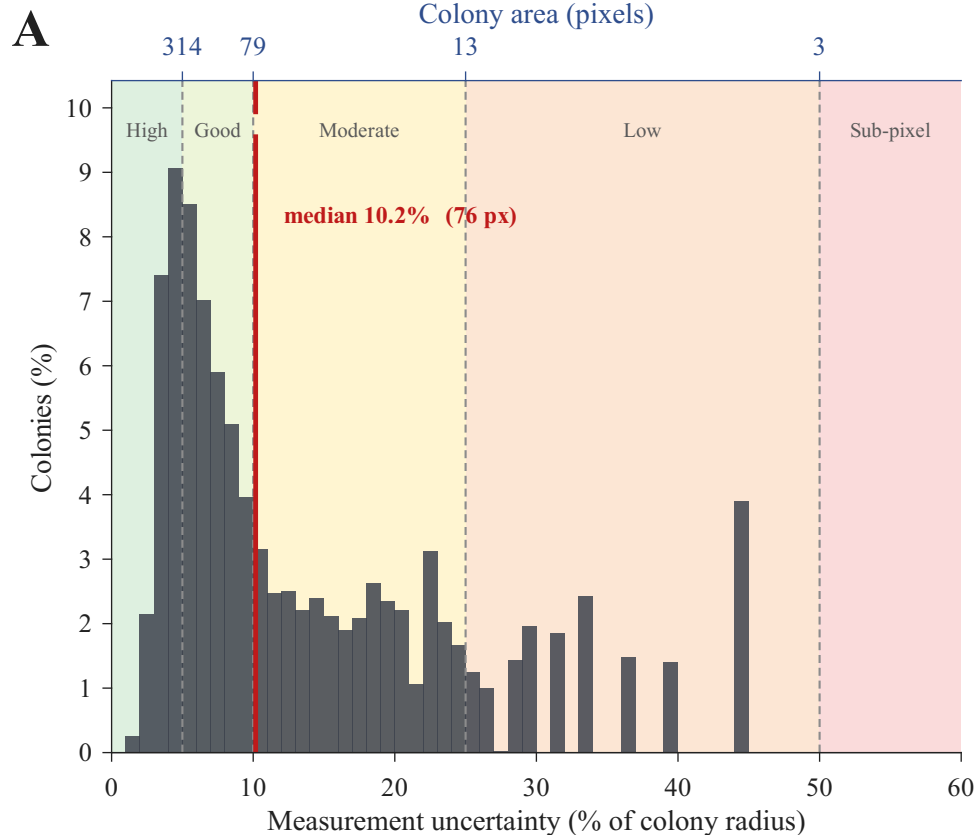


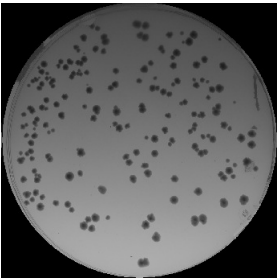
Measurement Fidelity: Rescue (25300 colonies, 121 Colony Sets)



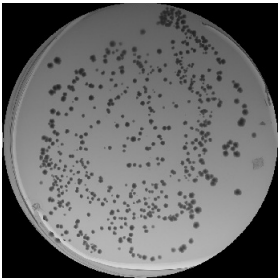
wild_{type}, Passage 1

Original Source Images

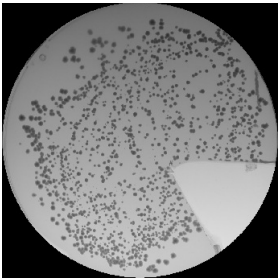
Std 6d



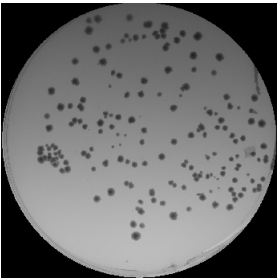
Colonies 132
Capture Ratio 59.1%
Rep A
HCE 2304
Plate 90 mm



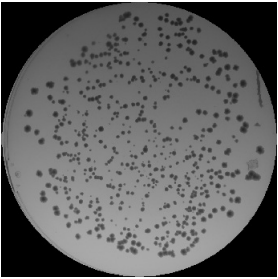
261
43.5%
A
11520
90 mm



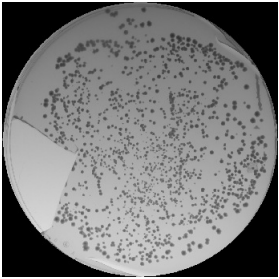
523
37.0%
A
57600
90 mm



114
70.8%
B
2304
90 mm



Colonies 327
Capture Ratio 54.8%
Rep B
HCE 11520
Plate 90 mm

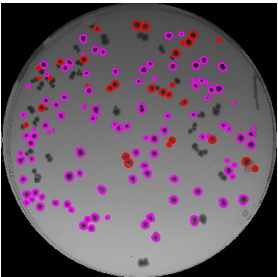


593
41.1%
B
57600
90 mm

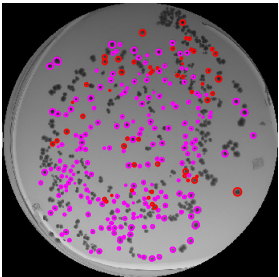
wild_type, Passage 1

Source Images with Colony Overlays

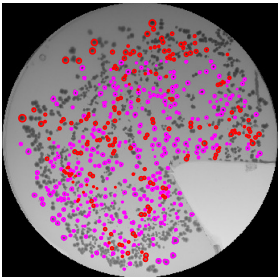
Std 6d



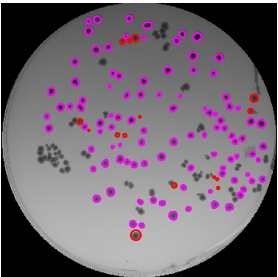
Colonies 132
Capture Ratio 59.1%
Rep A
HCE 2304
Plate 90 mm



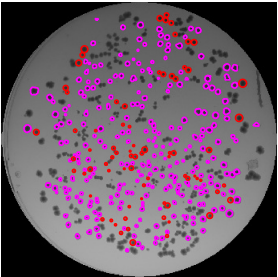
261
43.5%
A
11520
90 mm



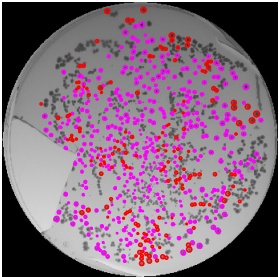
523
37.0%
A
57600
90 mm



114
70.8%
B
2304
90 mm



Colonies 327
Capture Ratio 54.8%
Rep B
HCE 11520
Plate 90 mm

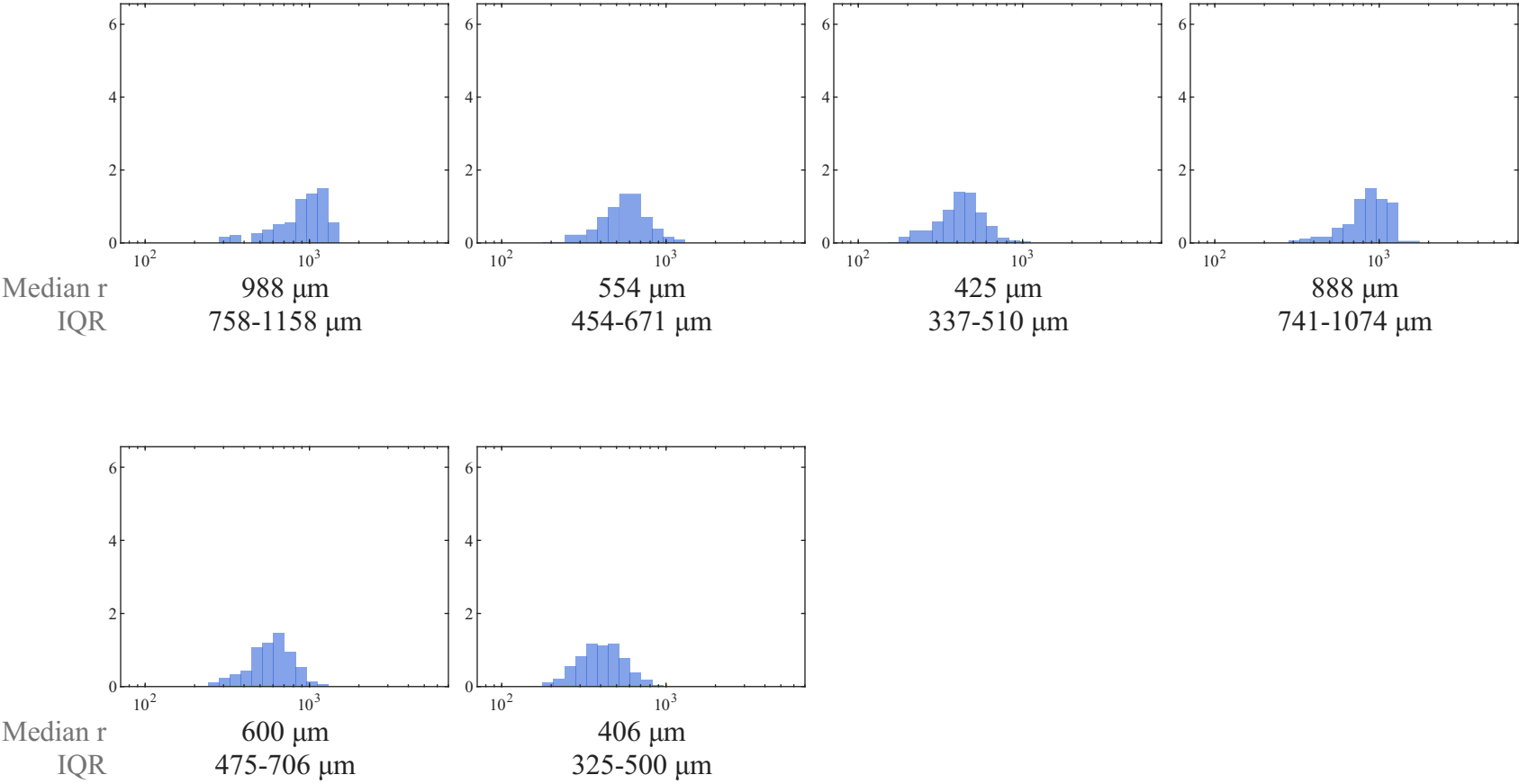


593
41.1%
B
57600
90 mm

Density Histograms

X axis: Colony Radius μm , Y axis: Density

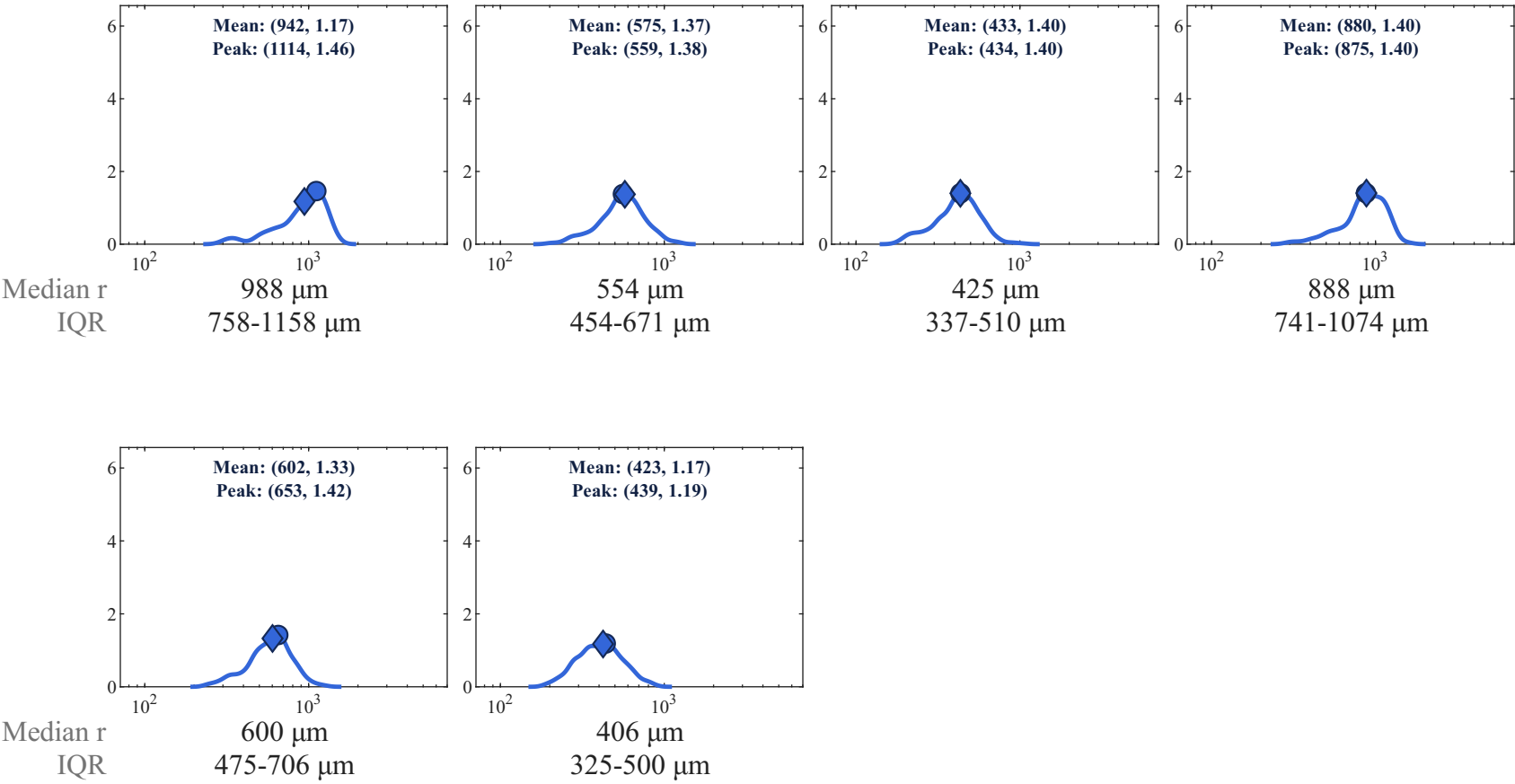
Std 6d



Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

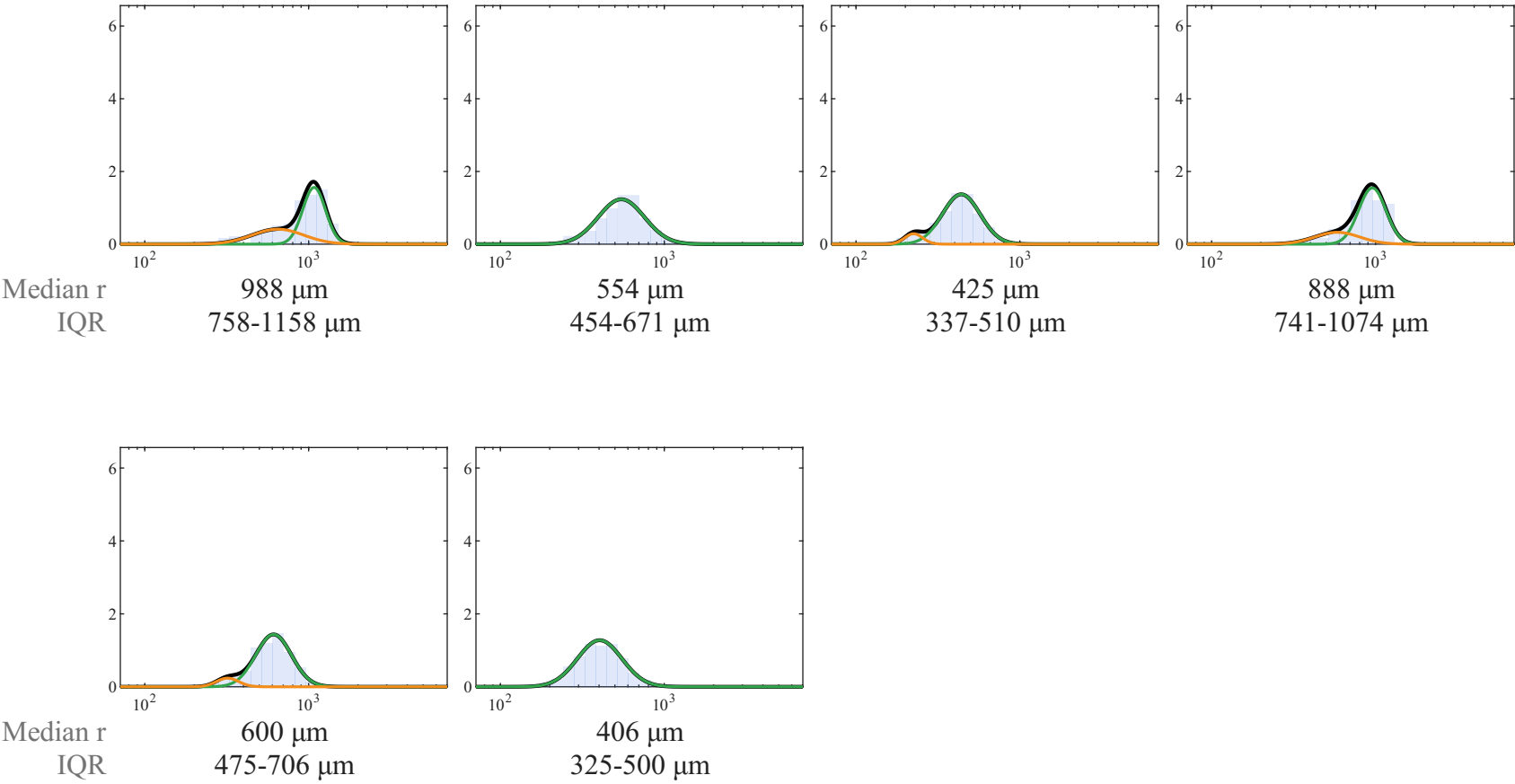
Std 6d



LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

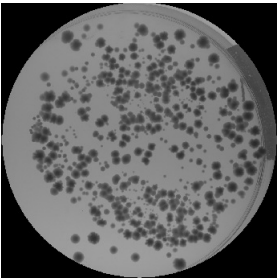
Std 6d



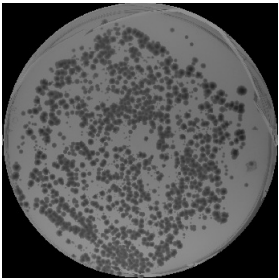
wild_type, Passage 2

Original Source Images

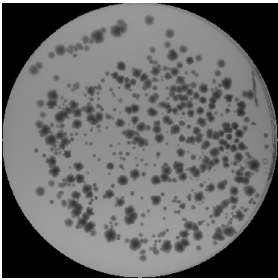
Std 6d



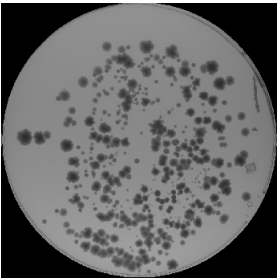
Colonies 215
Capture Ratio 28.6%
Rep A
HCE 144
Plate 90 mm



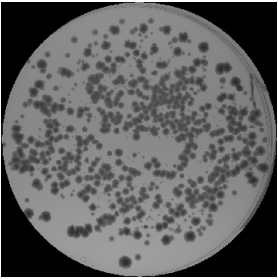
108
5.1%
A
752
90 mm



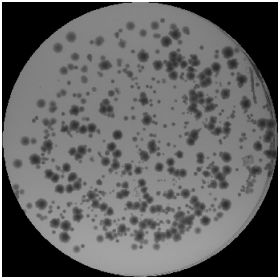
164
19.4%
A
3776
90 mm



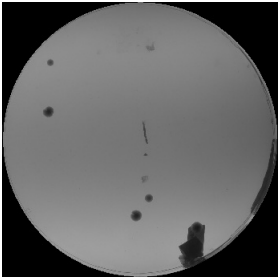
208
44.8%
B
144
90 mm



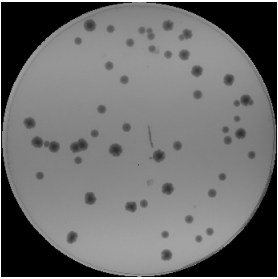
Colonies 181
Capture Ratio 19.2%
Rep B
HCE 752
Plate 90 mm



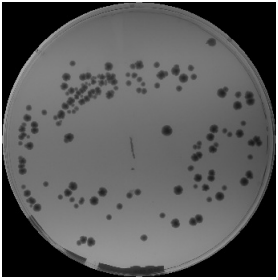
248
37.4%
B
3776
90 mm



4
15.1%
—
6.4
140 mm



48
79.6%
—
32
140 mm

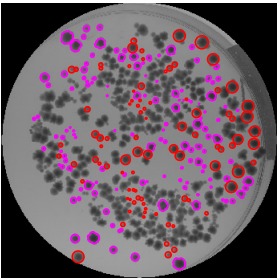


Colonies 65
Capture Ratio 40.0%
Rep —
HCE 160
Plate 140 mm

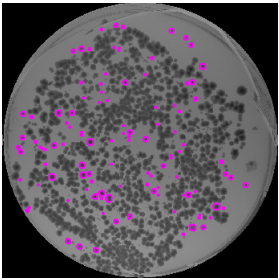
wild_type, Passage 2

Source Images with Colony Overlays

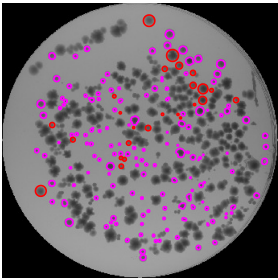
Std 6d



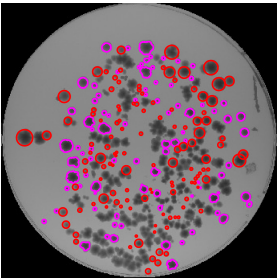
Colonies 215
Capture Ratio 28.6%
Rep A
HCE 144
Plate 90 mm



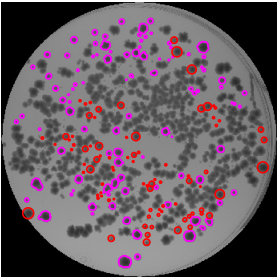
108
5.1%
A
752
90 mm



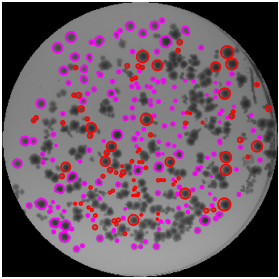
164
19.4%
A
3776
90 mm



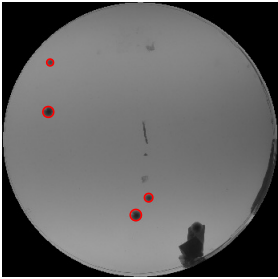
208
44.8%
B
144
90 mm



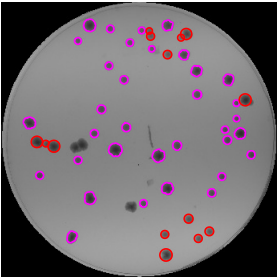
Colonies 181
Capture Ratio 19.2%
Rep B
HCE 752
Plate 90 mm



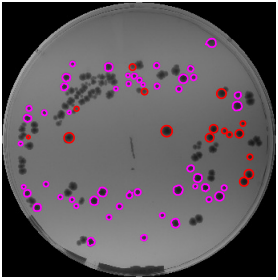
248
37.4%
B
3776
90 mm



4
15.1%
—
6.4
140 mm



48
79.6%
—
32
140 mm

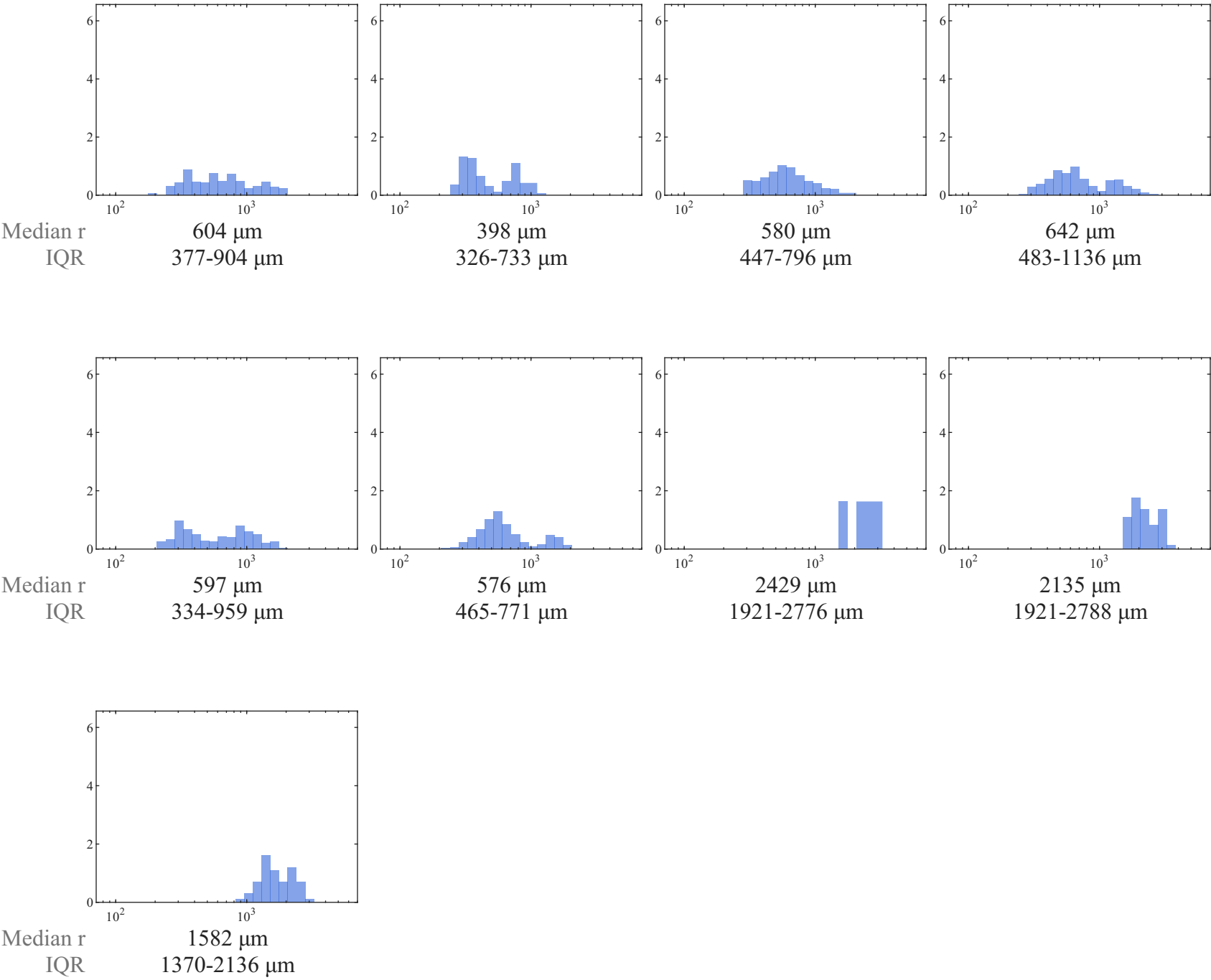


Colonies 65
Capture Ratio 40.0%
Rep —
HCE 160
Plate 140 mm

Density Histograms

X axis: Colony Radius μm , Y axis: Density

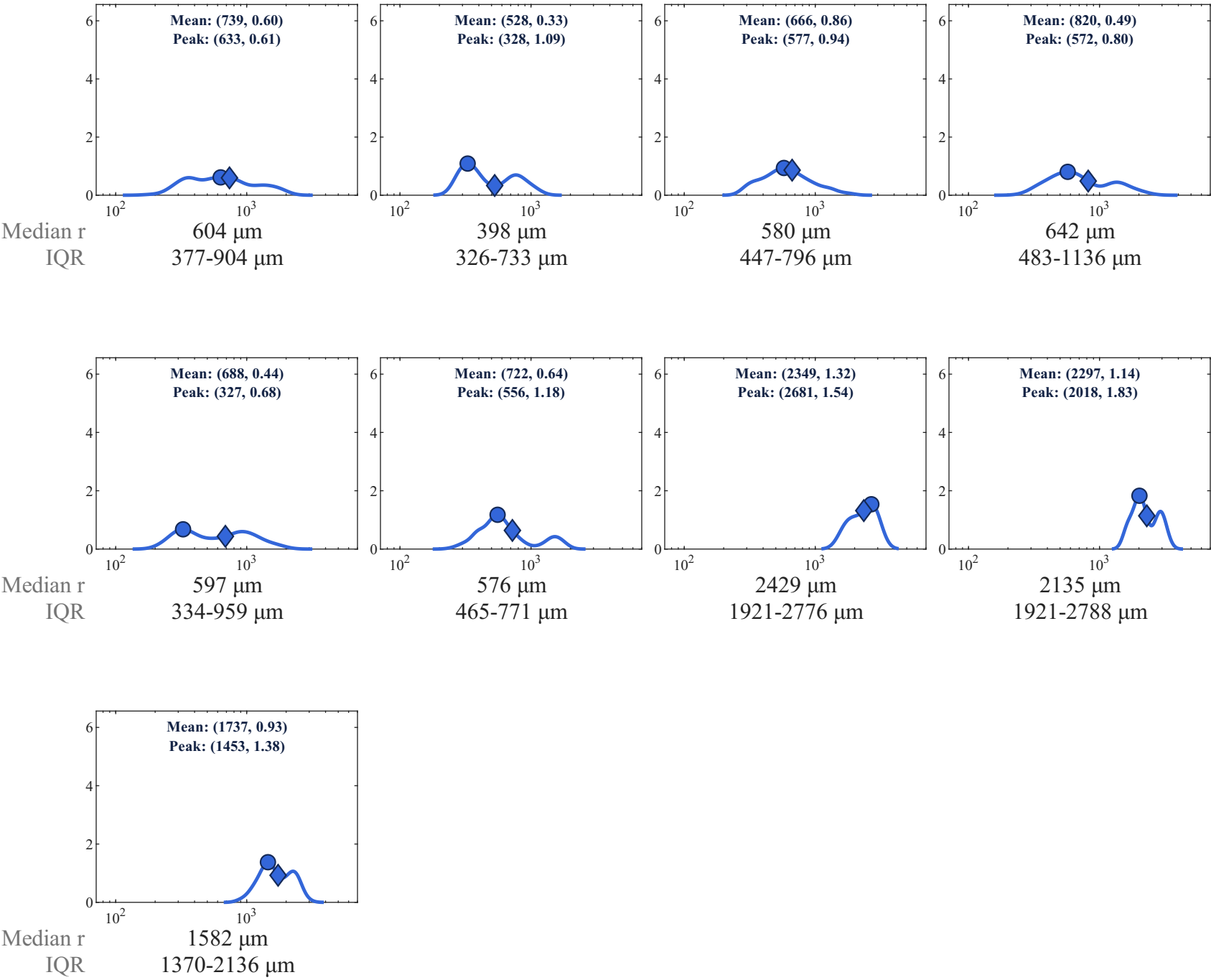
Std 6d



Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d



LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

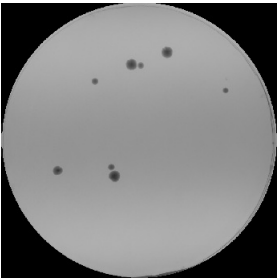
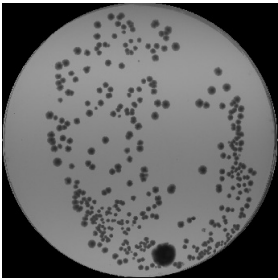
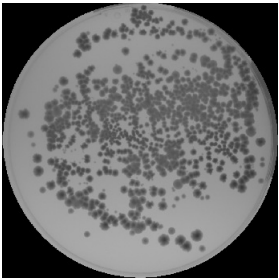
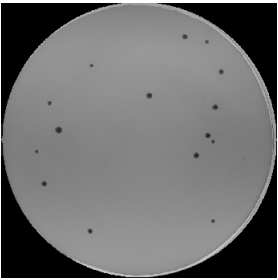
Std 6d



wild_type, Passage 3

Original Source Images

Std 6d

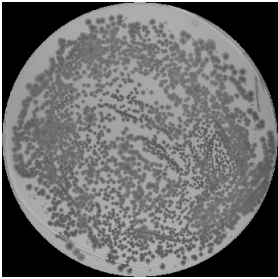
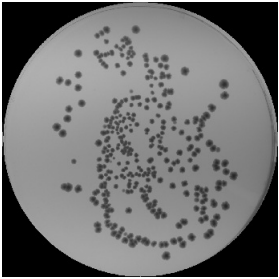
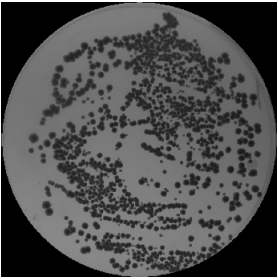


Colonies 15
Capture Ratio 65.7%
Rep A
HCE 160
Plate 140 mm

83
13.3%
A
768
90 mm

130
39.0%
A
800
140 mm

8
101.9%
B
160
140 mm



Colonies 171
Capture Ratio 17.2%
Rep B
HCE 768
Plate 90 mm

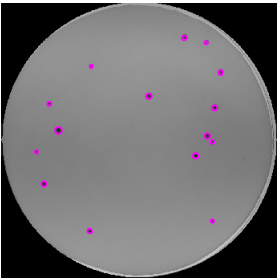
88
33.2%
B
800
140 mm

113
4.9%
B
3856
90 mm

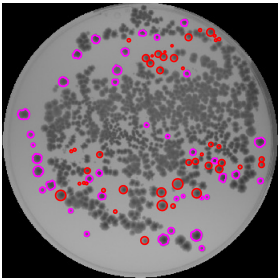
wild_type, Passage 3

Source Images with Colony Overlays

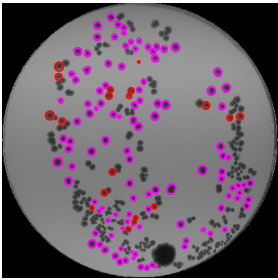
Std 6d



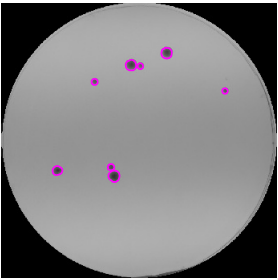
Colonies 15
Capture Ratio 65.7%
Rep A
HCE 160
Plate 140 mm



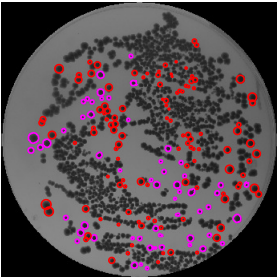
83
13.3%
A
768
90 mm



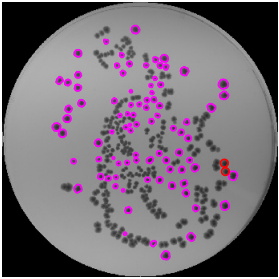
130
39.0%
A
800
140 mm



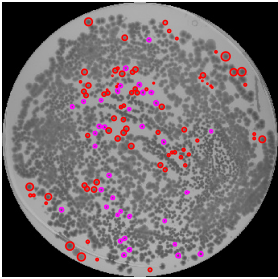
8
101.9%
B
160
140 mm



Colonies 171
Capture Ratio 17.2%
Rep B
HCE 768
Plate 90 mm



88
33.2%
B
800
140 mm

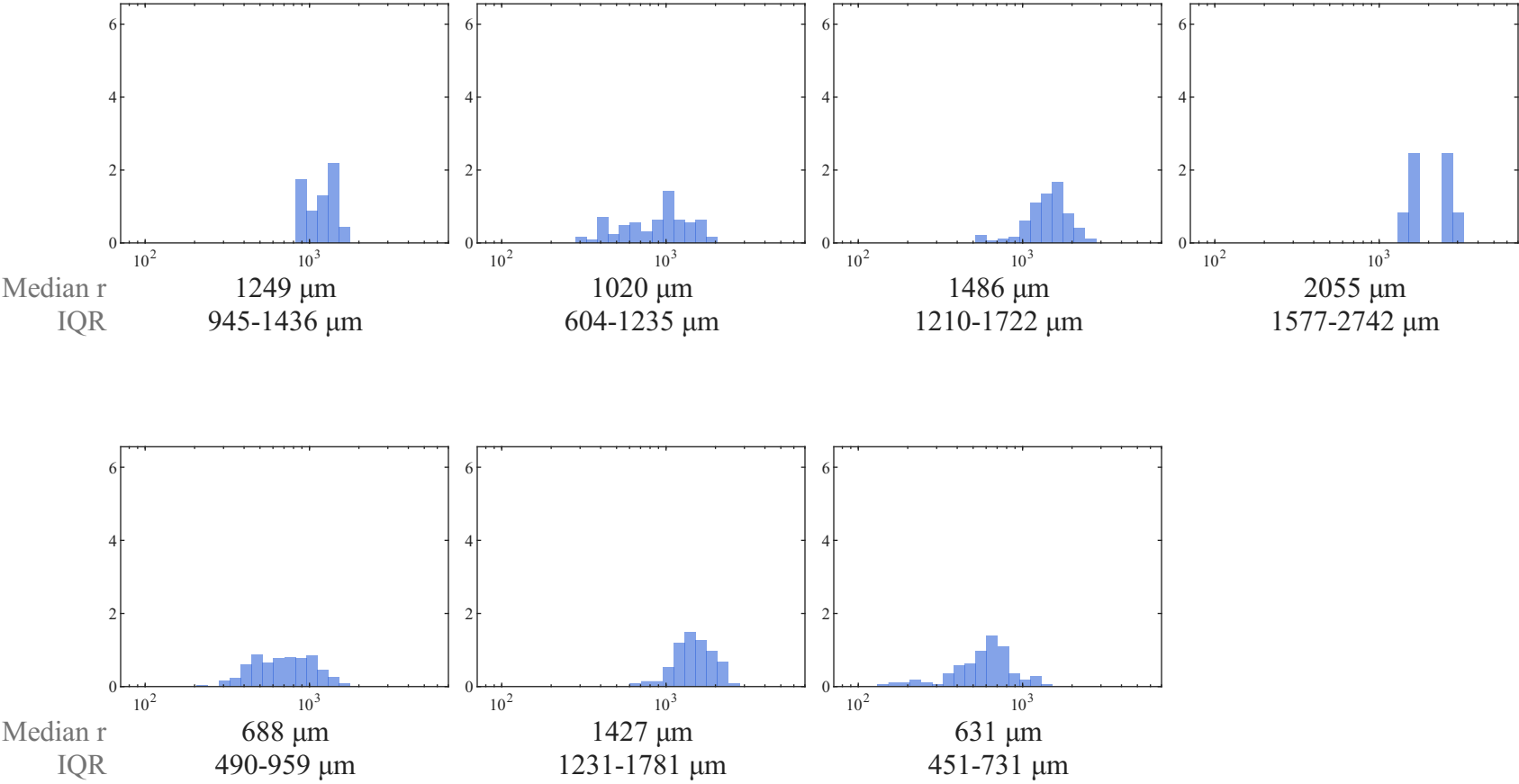


113
4.9%
B
3856
90 mm

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

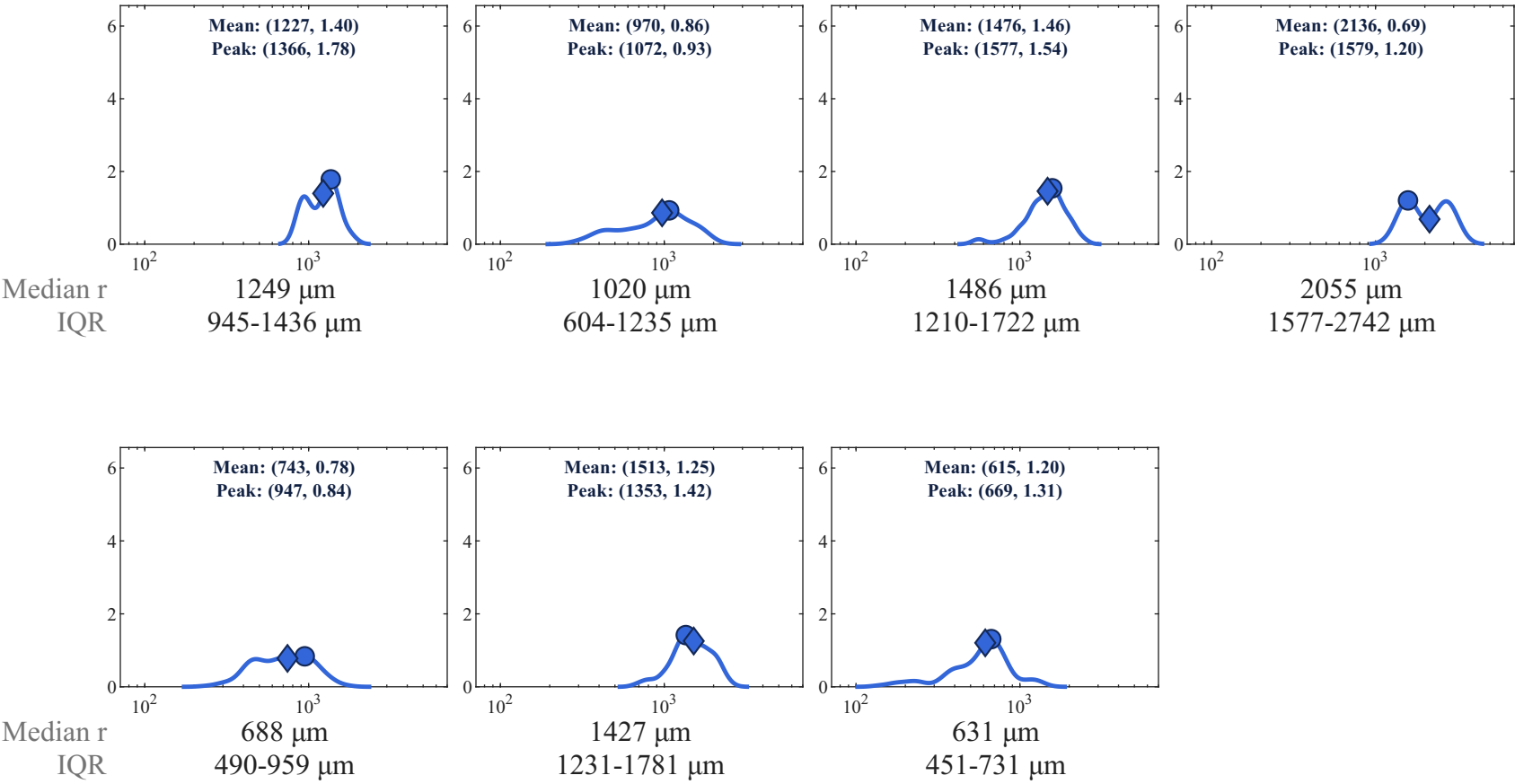


wild_type, Passage 3

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

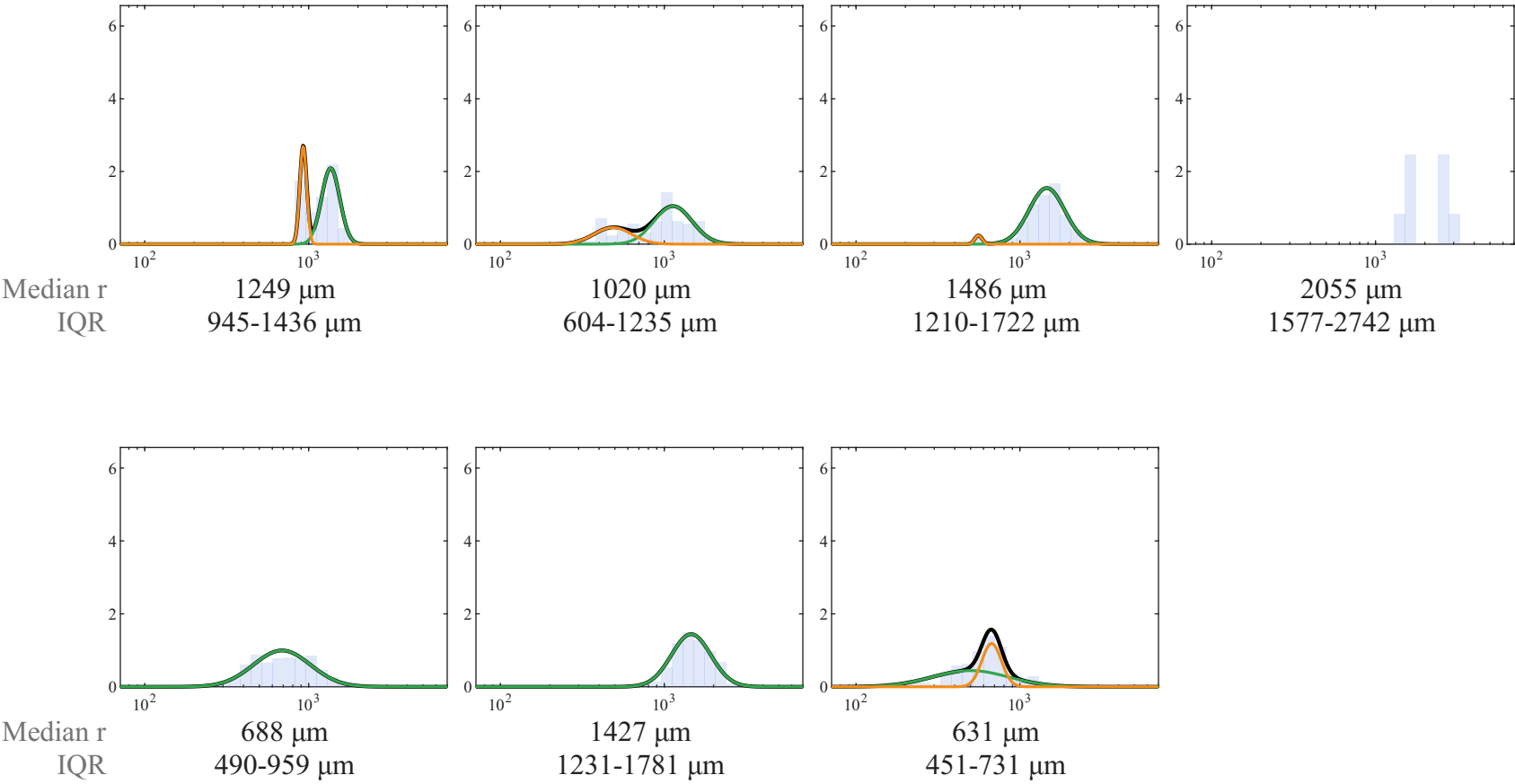
Std 6d



LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

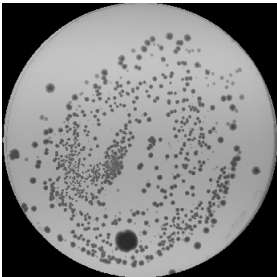
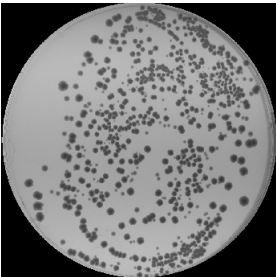
Std 6d



wild_type, Passage 4

Original Source Images

Std 6d



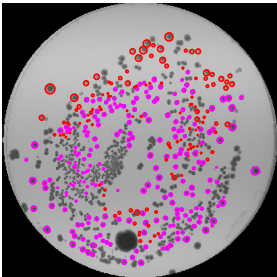
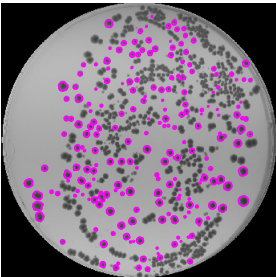
Colonies 203
Capture Ratio 29.1%
Rep A
HCE 800
Plate 140 mm

341
35.9%
B
800
140 mm

wild_type, Passage 4

Source Images with Colony Overlays

Std 6d



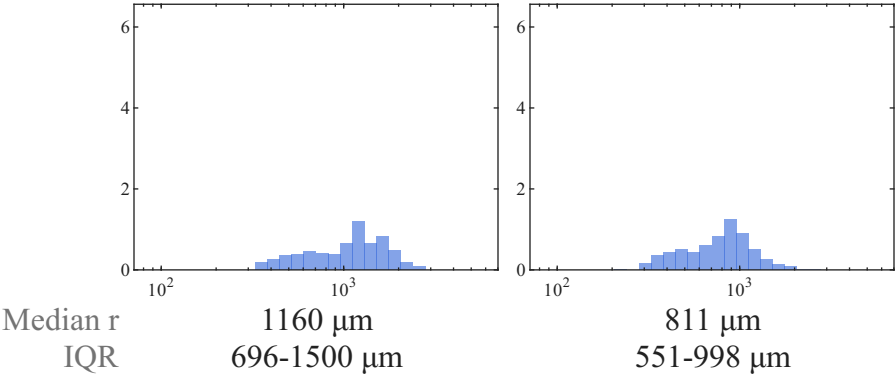
Colonies 203
Capture Ratio 29.1%
Rep A
HCE 800
Plate 140 mm

341
35.9%
B
800
140 mm

Density Histograms

X axis: Colony Radius μm , Y axis: Density

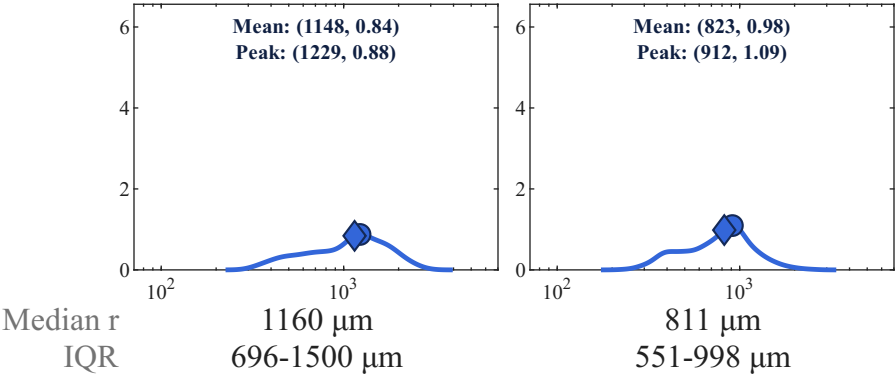
Std 6d



Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

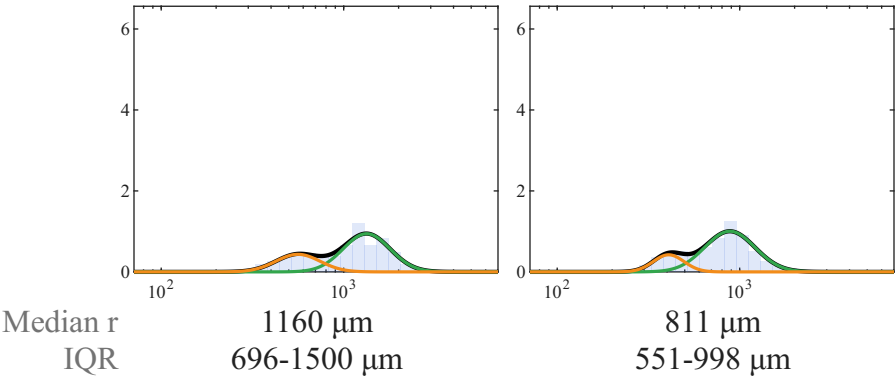
Std 6d



LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

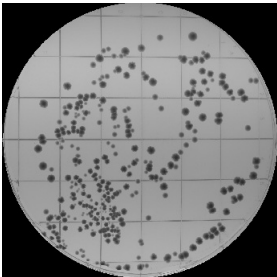
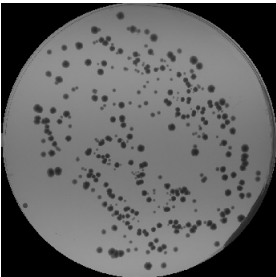
Std 6d



wild_type, Passage 5

Original Source Images

Std 6d

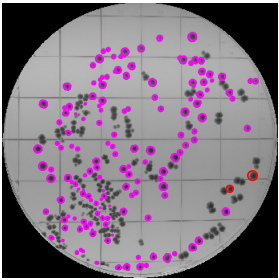
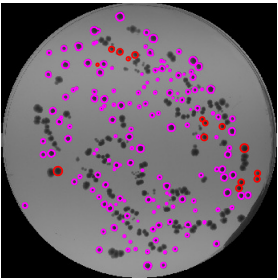


Colonies 169
Capture Ratio 42.4%
Rep A
HCE 800
Plate 140 mm

149
42.2%
B
800
140 mm

Source Images with Colony Overlays

Std 6d



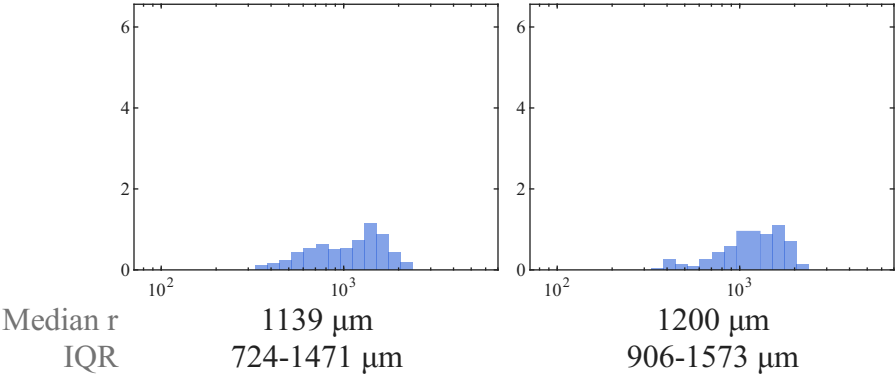
Colonies 169
Capture Ratio 42.4%
Rep A
HCE 800
Plate 140 mm

149
42.2%
B
800
140 mm

Density Histograms

X axis: Colony Radius μm , Y axis: Density

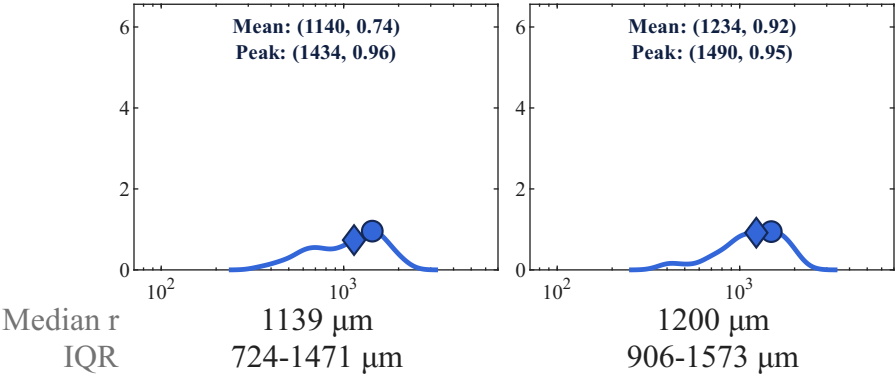
Std 6d



Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

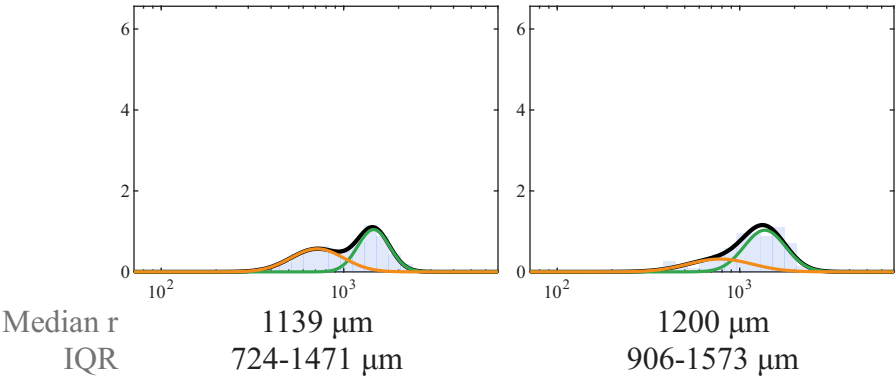
Std 6d



LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

Std 6d



wild_type: Colony Radius by Plate, Across Passages

X axis: Passage, Y axis: Colony Radius μm

n	132	261	523	114	327	593	215	108	164	208	181	248	4	48	65	15	83	130	8	171	88	113	203	341	169	149	
Median μm	988	554	425	888	600	406	604	398	580	642	597	576	242	921	351	582	1249	1020	1486	2055	688	1427	631	1160	811	1139	1200

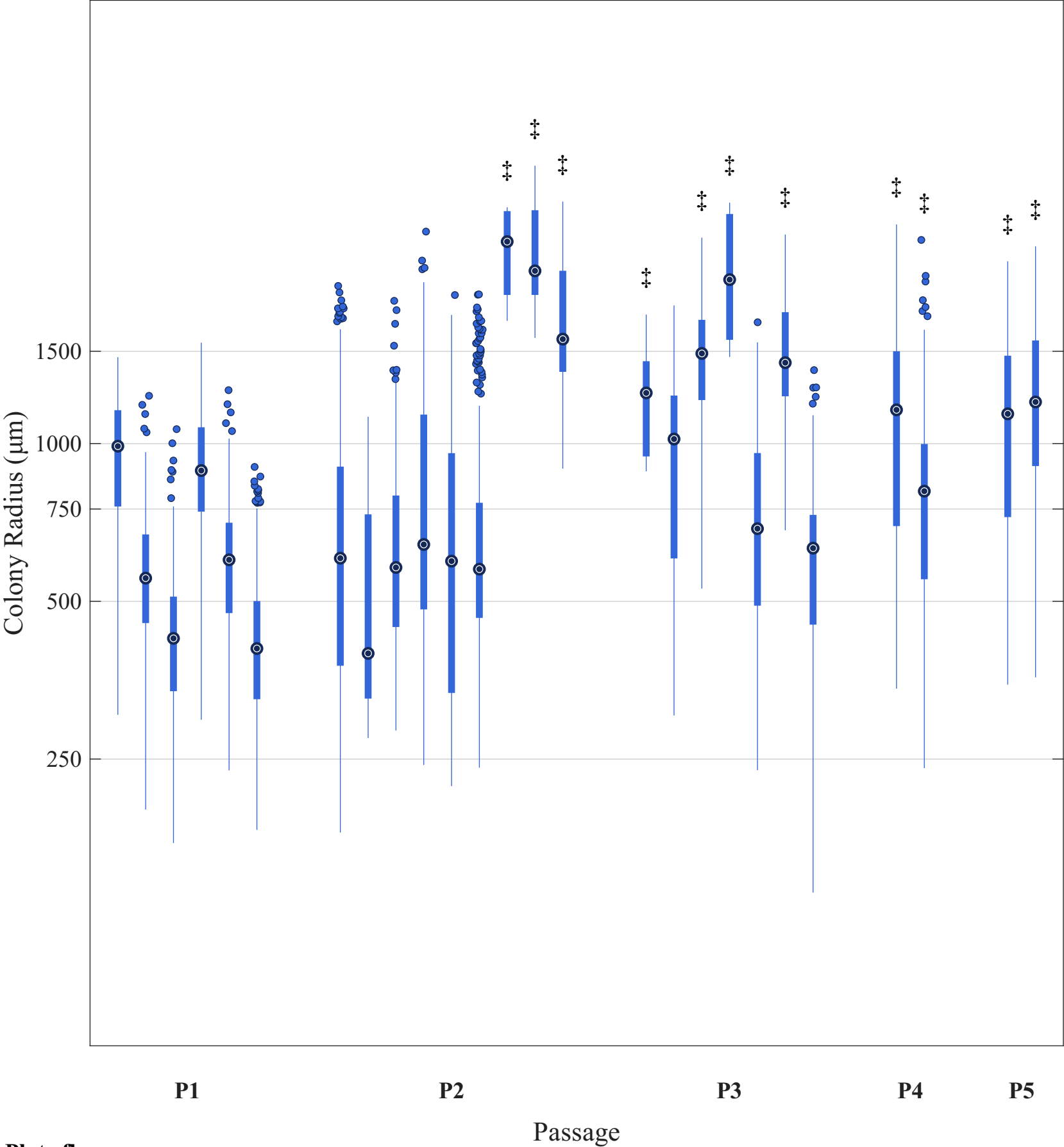
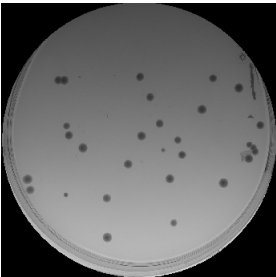


Plate flags:
‡ 140 mm plate

dbf4-1, Passage 1

Original Source Images

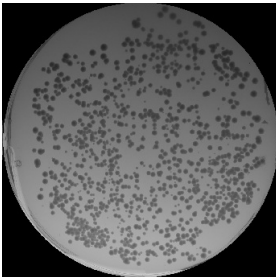
Std 6d



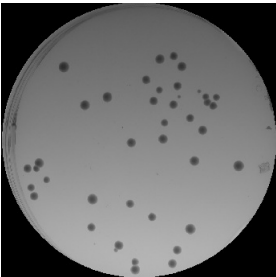
Colonies 25
Capture Ratio 54.3%
Rep A
HCE 3264
Plate 90 mm



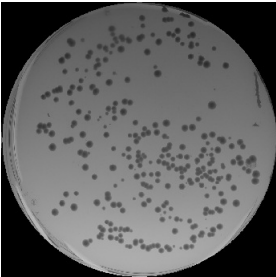
160
36.7%
A
16320
90 mm



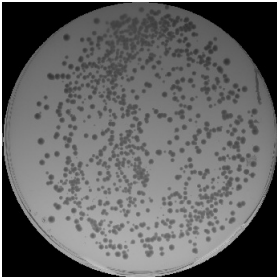
301
26.2%
A
81600
90 mm



39
82.8%
B
3264
90 mm

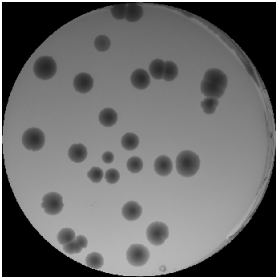


Colonies 200
Capture Ratio 66.6%
Rep B
HCE 16320
Plate 90 mm

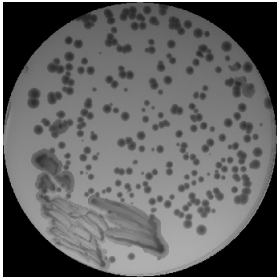


161
16.6%
B
81600
90 mm

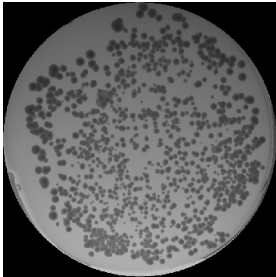
Std 21d



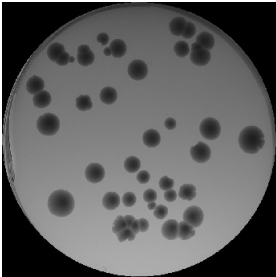
Colonies 35
Capture Ratio 115.4%
Rep A
HCE 3264
Plate 90 mm



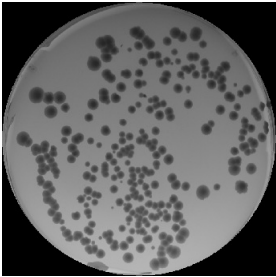
112
28.4%
A
16320
90 mm



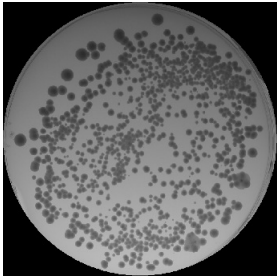
158
13.6%
A
81600
90 mm



29
64.3%
B
3264
90 mm



Colonies 93
Capture Ratio 38.0%
Rep B
HCE 16320
Plate 90 mm

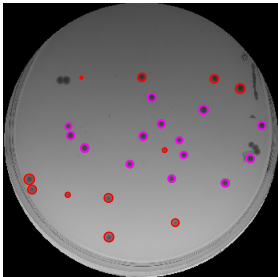


174
23.2%
B
81600
90 mm

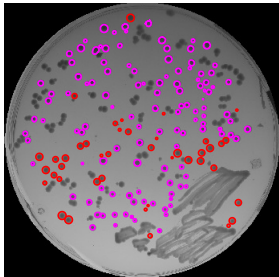
dbf4-1, Passage 1

Source Images with Colony Overlays

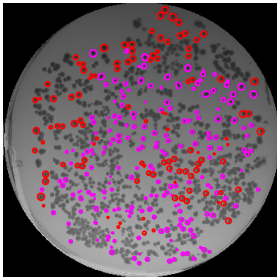
Std 6d



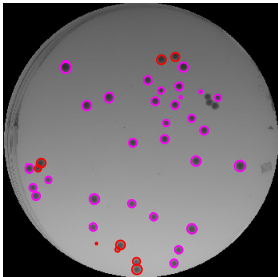
Colonies 25
Capture Ratio 54.3%
Rep A
HCE 3264
Plate 90 mm



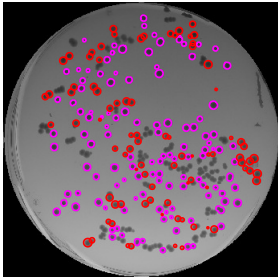
160
36.7%
A
16320
90 mm



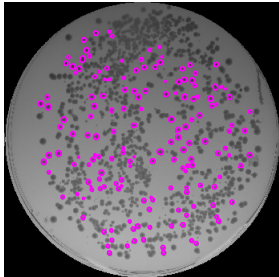
301
26.2%
A
81600
90 mm



39
82.8%
B
3264
90 mm

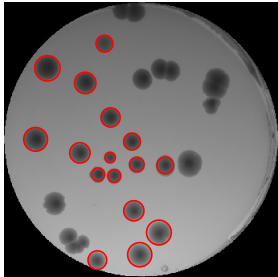


Colonies 200
Capture Ratio 66.6%
Rep B
HCE 16320
Plate 90 mm

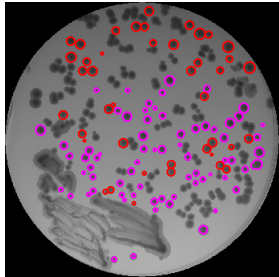


161
16.6%
B
81600
90 mm

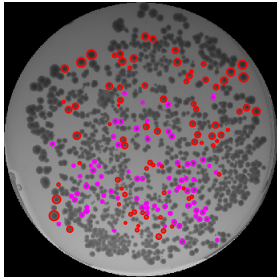
Std 21d



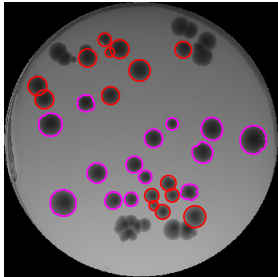
Colonies 35
Capture Ratio 115.4%
Rep A
HCE 3264
Plate 90 mm



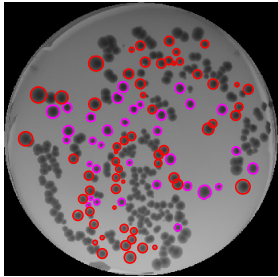
112
28.4%
A
16320
90 mm



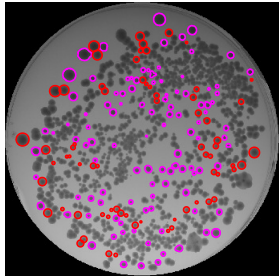
158
13.6%
A
81600
90 mm



29
64.3%
B
3264
90 mm



Colonies 93
Capture Ratio 38.0%
Rep B
HCE 16320
Plate 90 mm



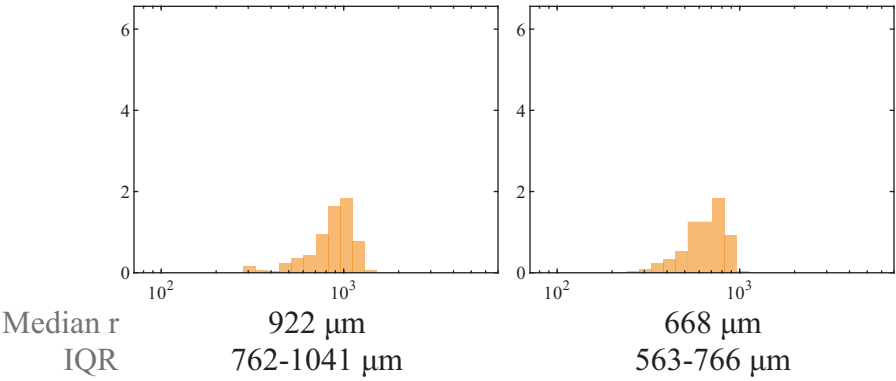
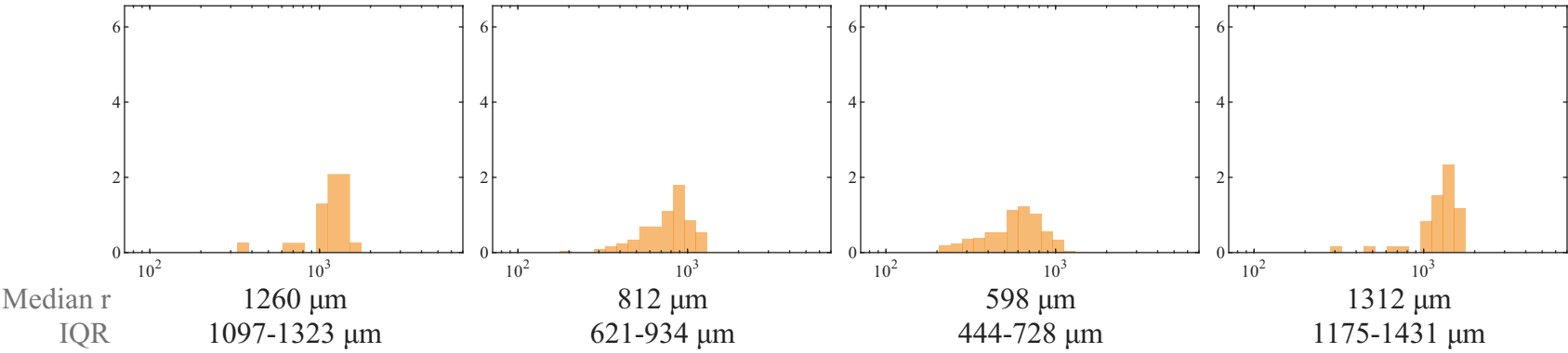
174
23.2%
B
81600
90 mm

dbf4-1, Passage 1

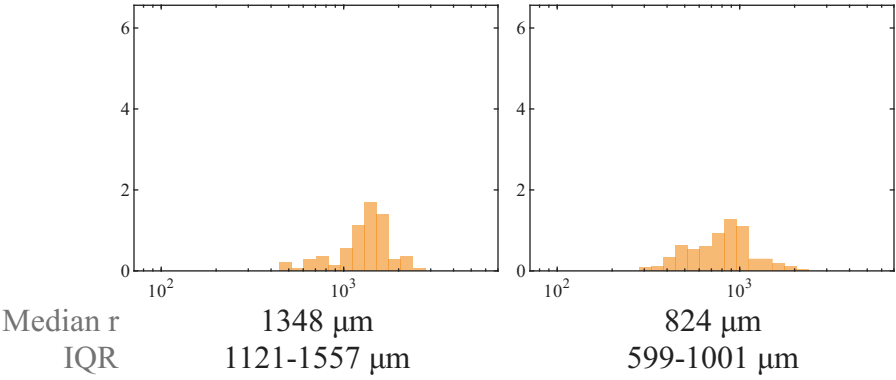
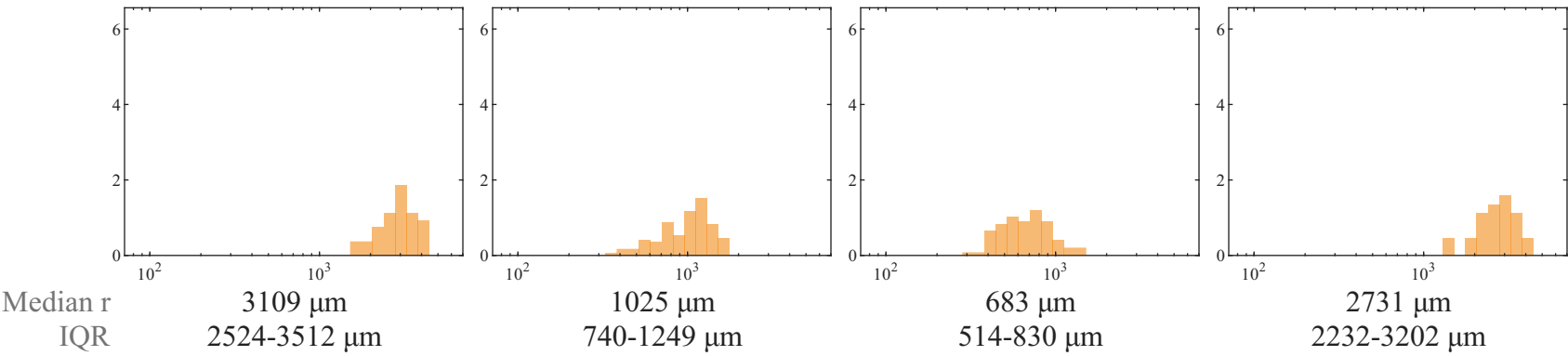
Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d



Std 21d

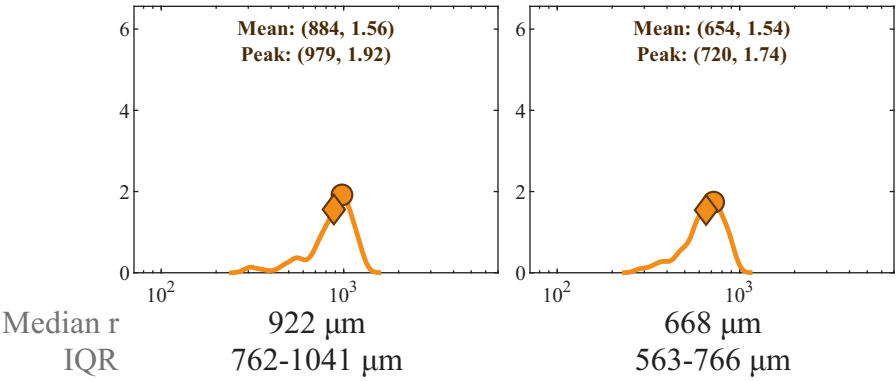
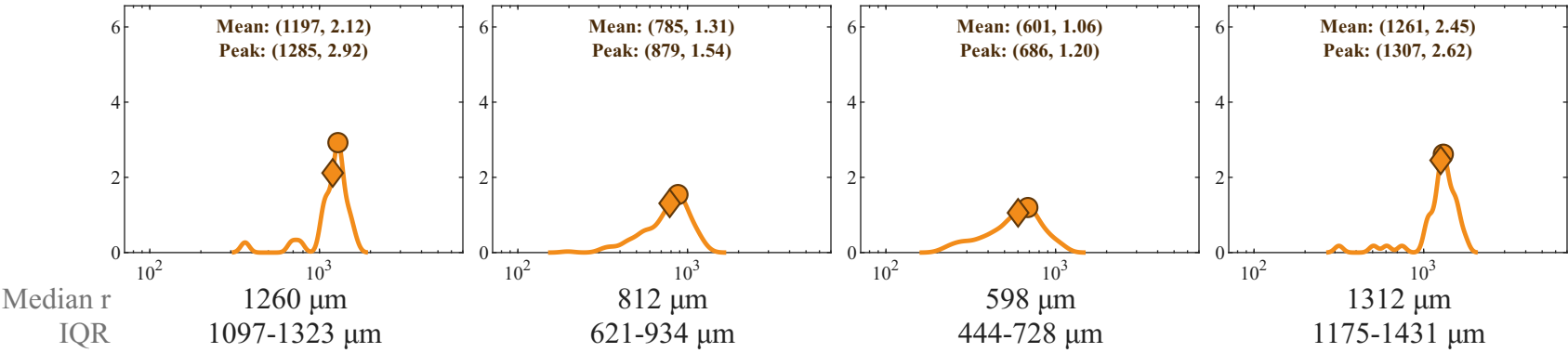


dbf4-1, Passage 1

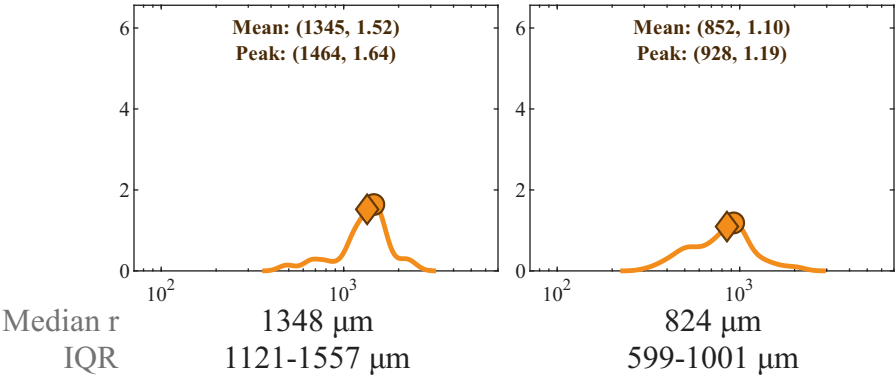
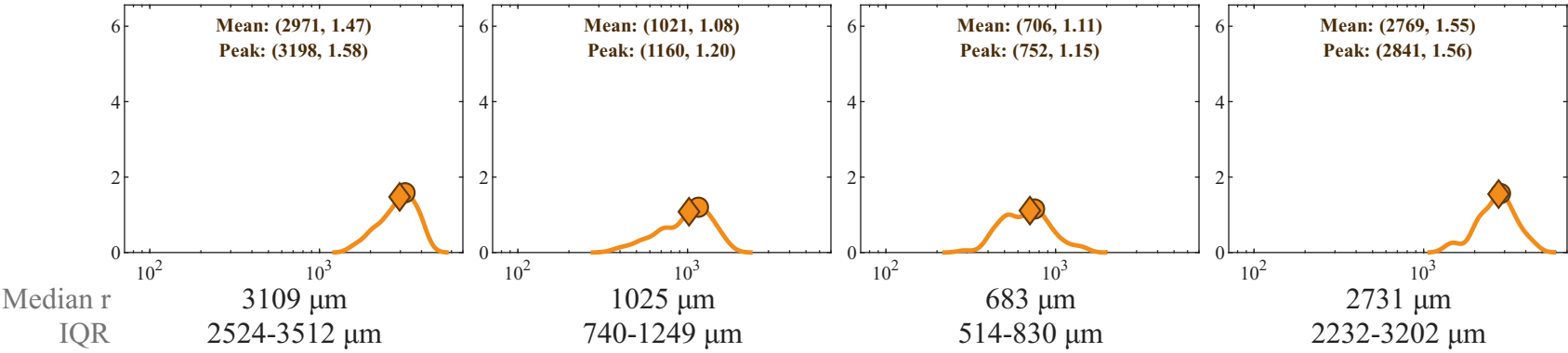
Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d



Std 21d

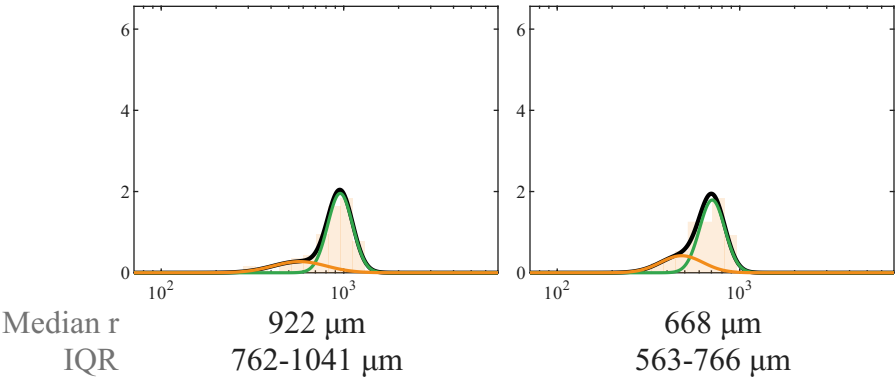
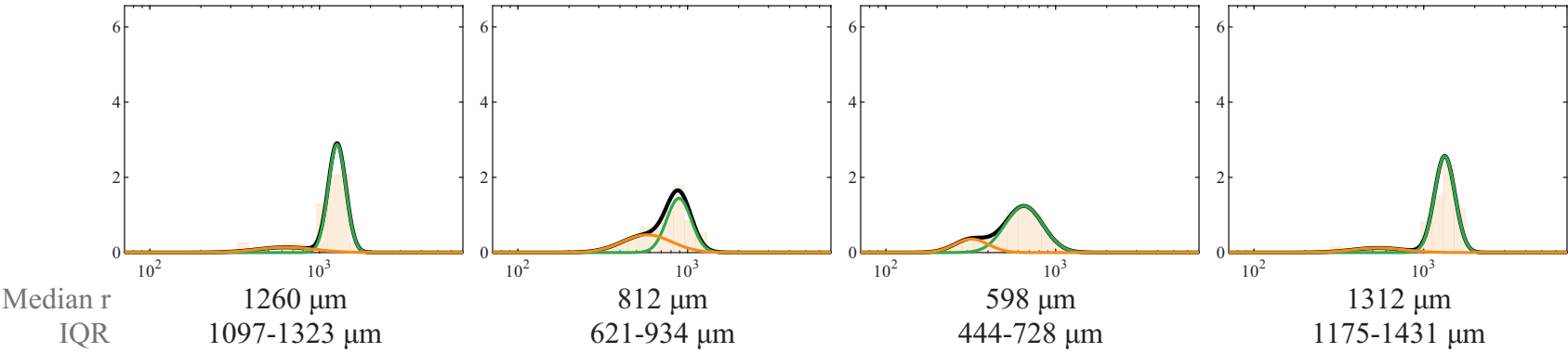


dbf4-1, Passage 1

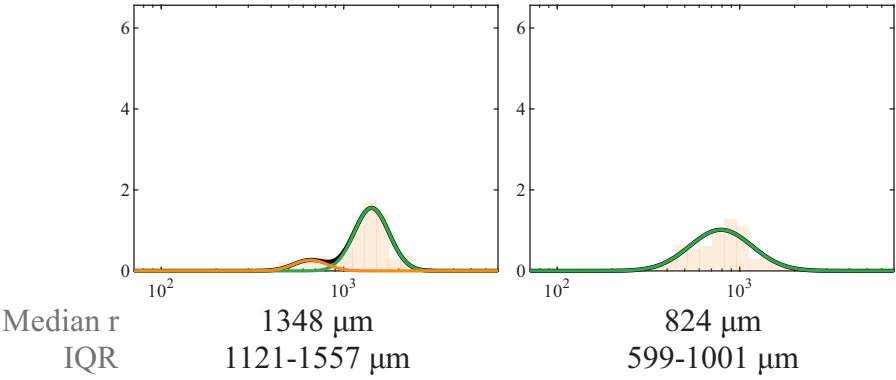
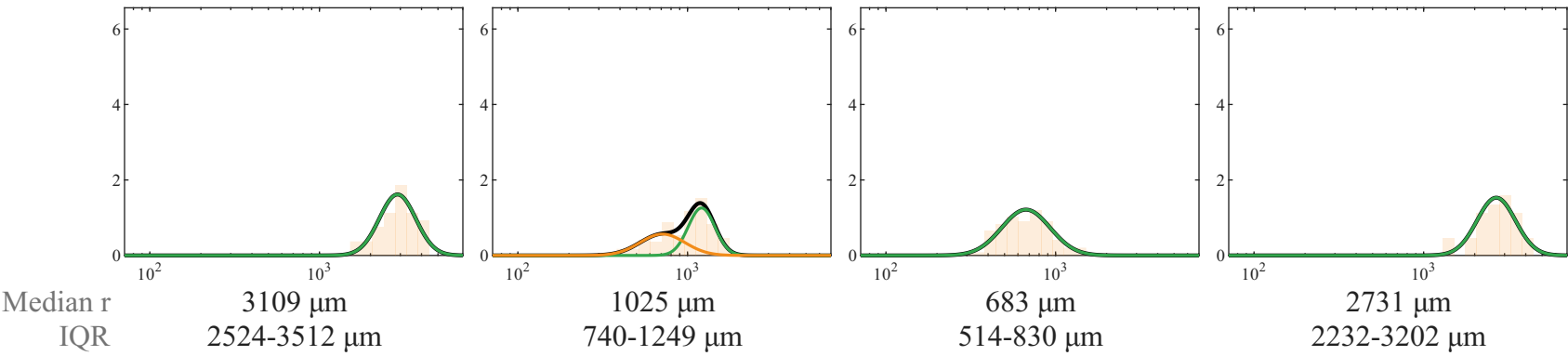
LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

Std 6d



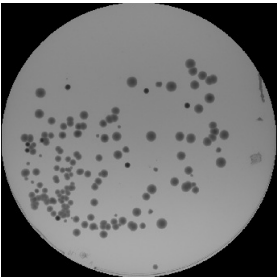
Std 21d



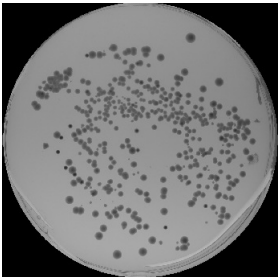
dbf4-1, Passage 2

Original Source Images

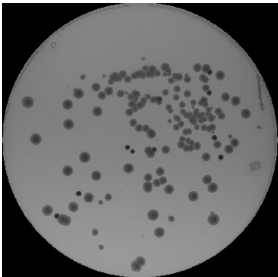
Std 6d



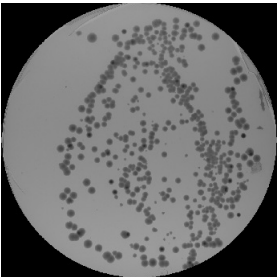
Colonies 61
Capture Ratio 47.7%
Rep A
HCE 144
Plate 90 mm



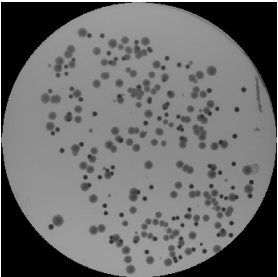
Colonies 138
Capture Ratio 34.6%
Rep A
HCE 768
Plate 90 mm



Colonies 51
Capture Ratio 44.2%
Rep B
HCE 144
Plate 90 mm



Colonies 148
Capture Ratio 34.4%
Rep B
HCE 768
Plate 90 mm

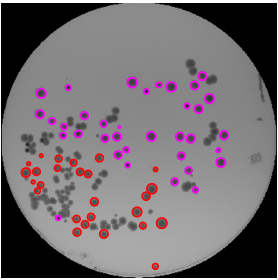


Colonies 182
Capture Ratio 77.1%
Rep B
HCE 3840
Plate 90 mm

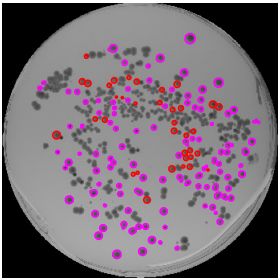
dbf4-1, Passage 2

Source Images with Colony Overlays

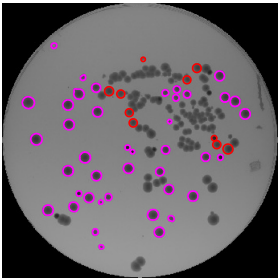
Std 6d



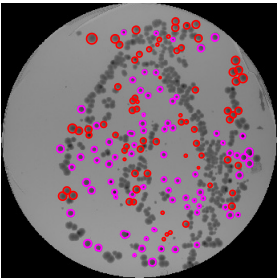
Colonies 61
Capture Ratio 47.7%
Rep A
HCE 144
Plate 90 mm



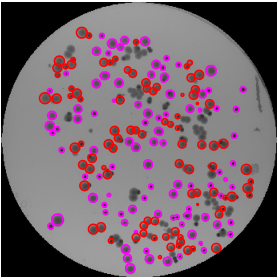
Colonies 138
Capture Ratio 34.6%
Rep A
HCE 768
Plate 90 mm



Colonies 51
Capture Ratio 44.2%
Rep B
HCE 144
Plate 90 mm



Colonies 148
Capture Ratio 34.4%
Rep B
HCE 768
Plate 90 mm

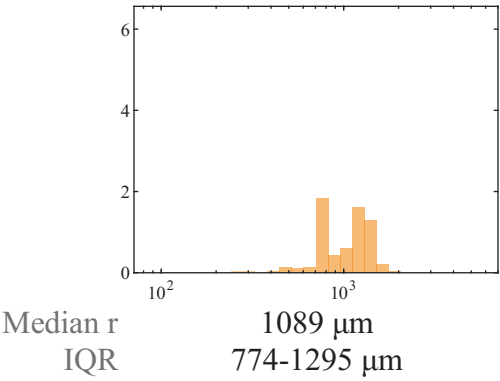
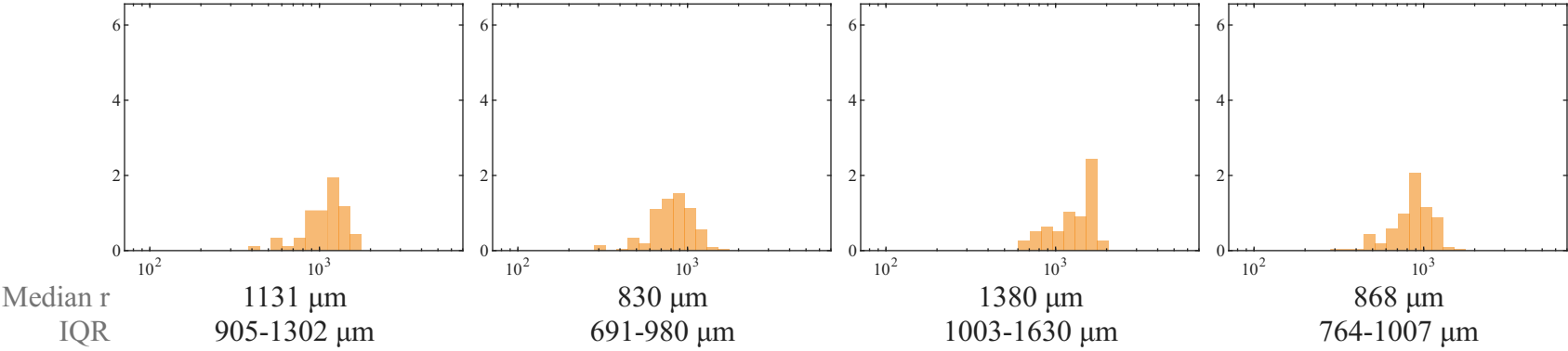


Colonies 182
Capture Ratio 77.1%
Rep B
HCE 3840
Plate 90 mm

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

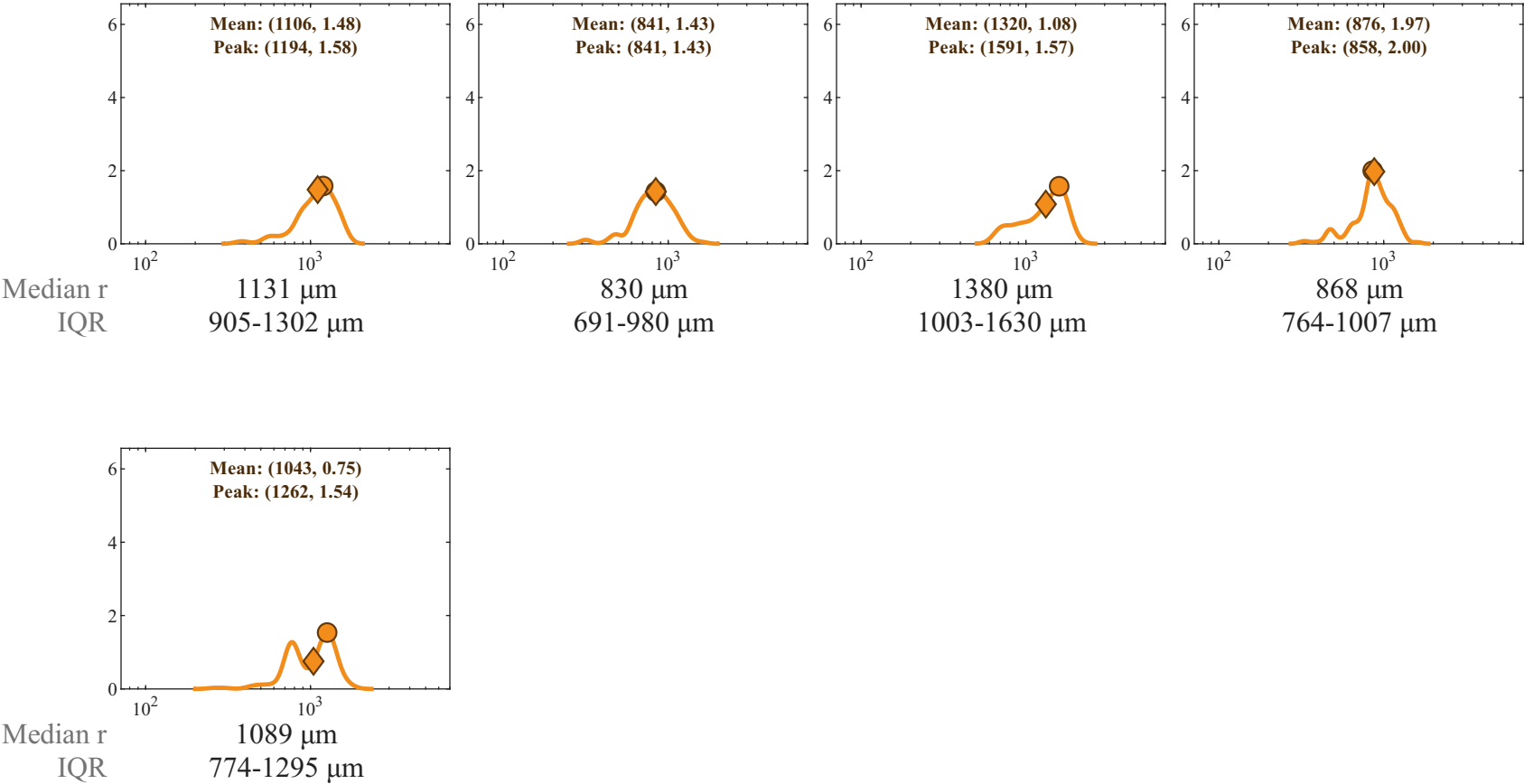


dbf4-1, Passage 2

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

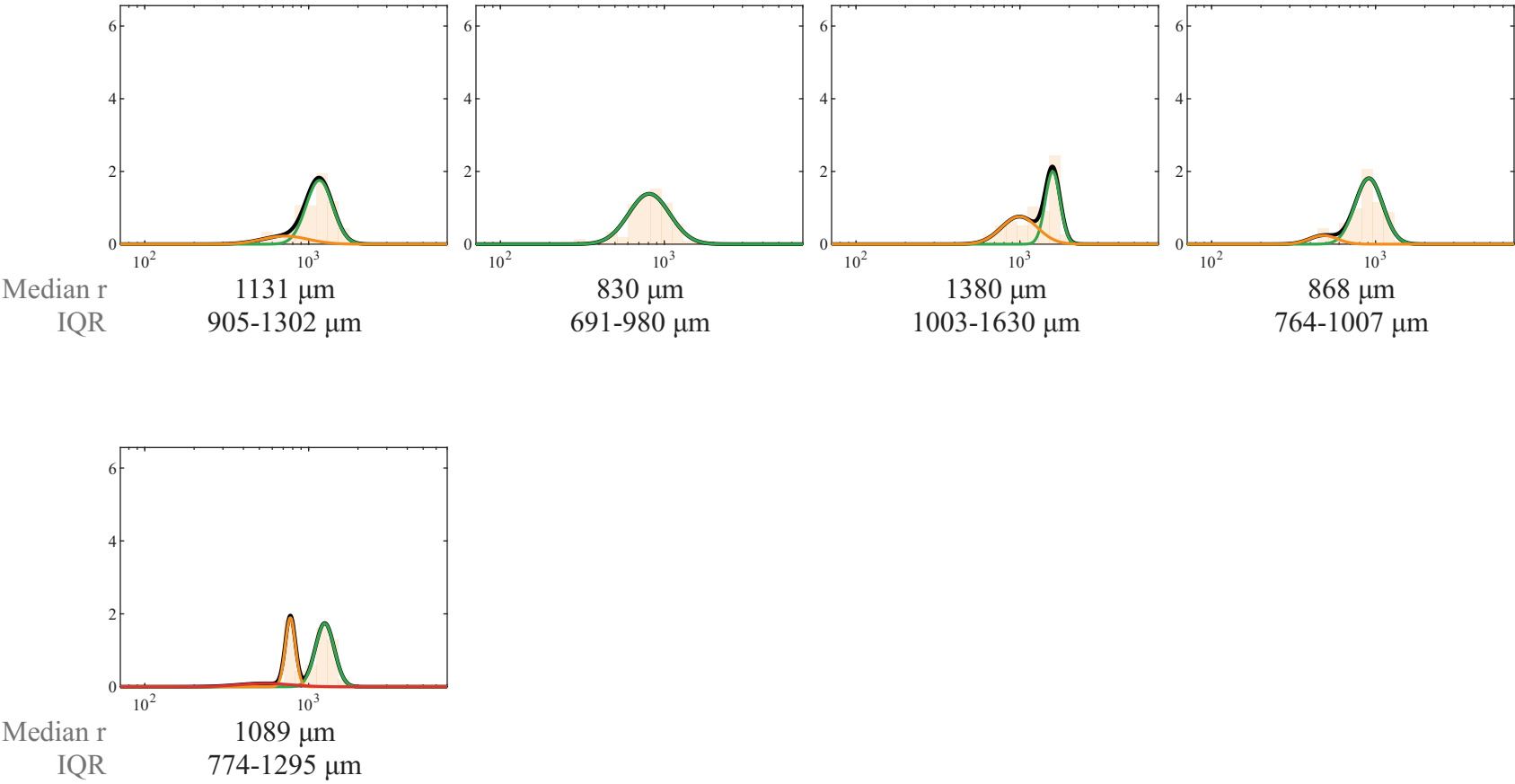
Std 6d



LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

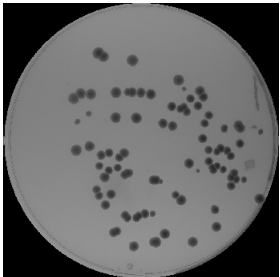
Std 6d



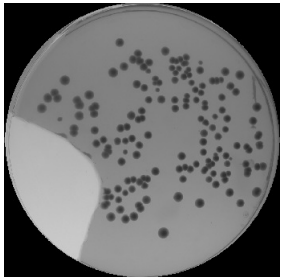
dbf4-1, Passage 3

Original Source Images

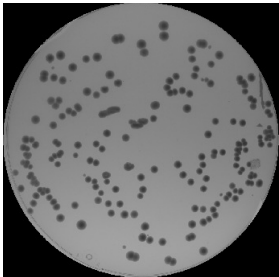
Std 6d



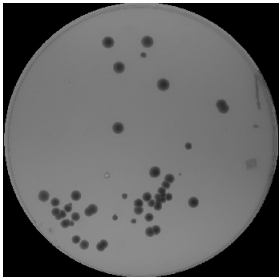
Colonies 40
Capture Ratio 47.9%
Rep A
HCE 160
Plate 90 mm



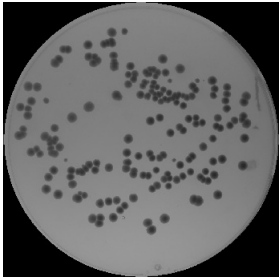
79
48.9%
A
816
90 mm



103
52.5%
A
4096
90 mm



20
49.5%
B
160
90 mm

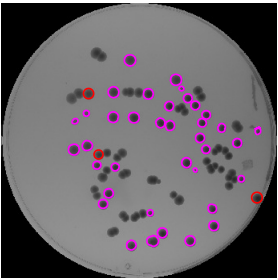


Colonies 88
Capture Ratio 60.2%
Rep B
HCE 816
Plate 90 mm

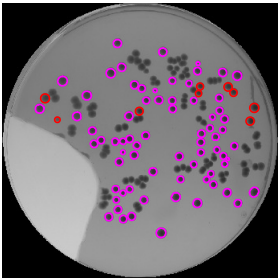
dbf4-1, Passage 3

Source Images with Colony Overlays

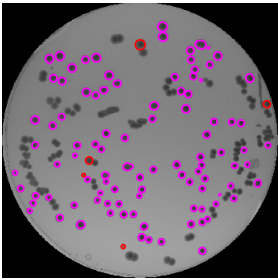
Std 6d



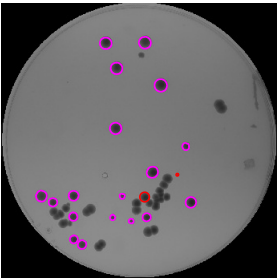
Colonies 40
Capture Ratio 47.9%
Rep A
HCE 160
Plate 90 mm



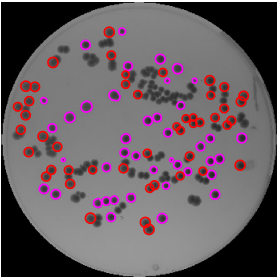
79
48.9%
A
816
90 mm



103
52.5%
A
4096
90 mm



20
49.5%
B
160
90 mm



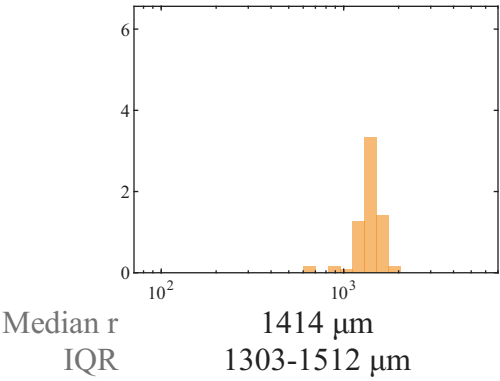
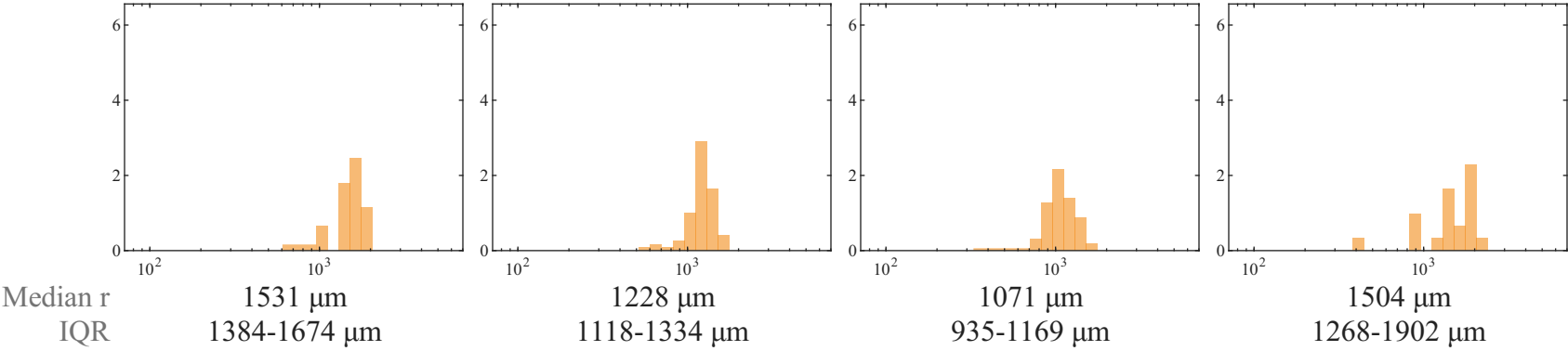
Colonies 88
Capture Ratio 60.2%
Rep B
HCE 816
Plate 90 mm

dbf4-1, Passage 3

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

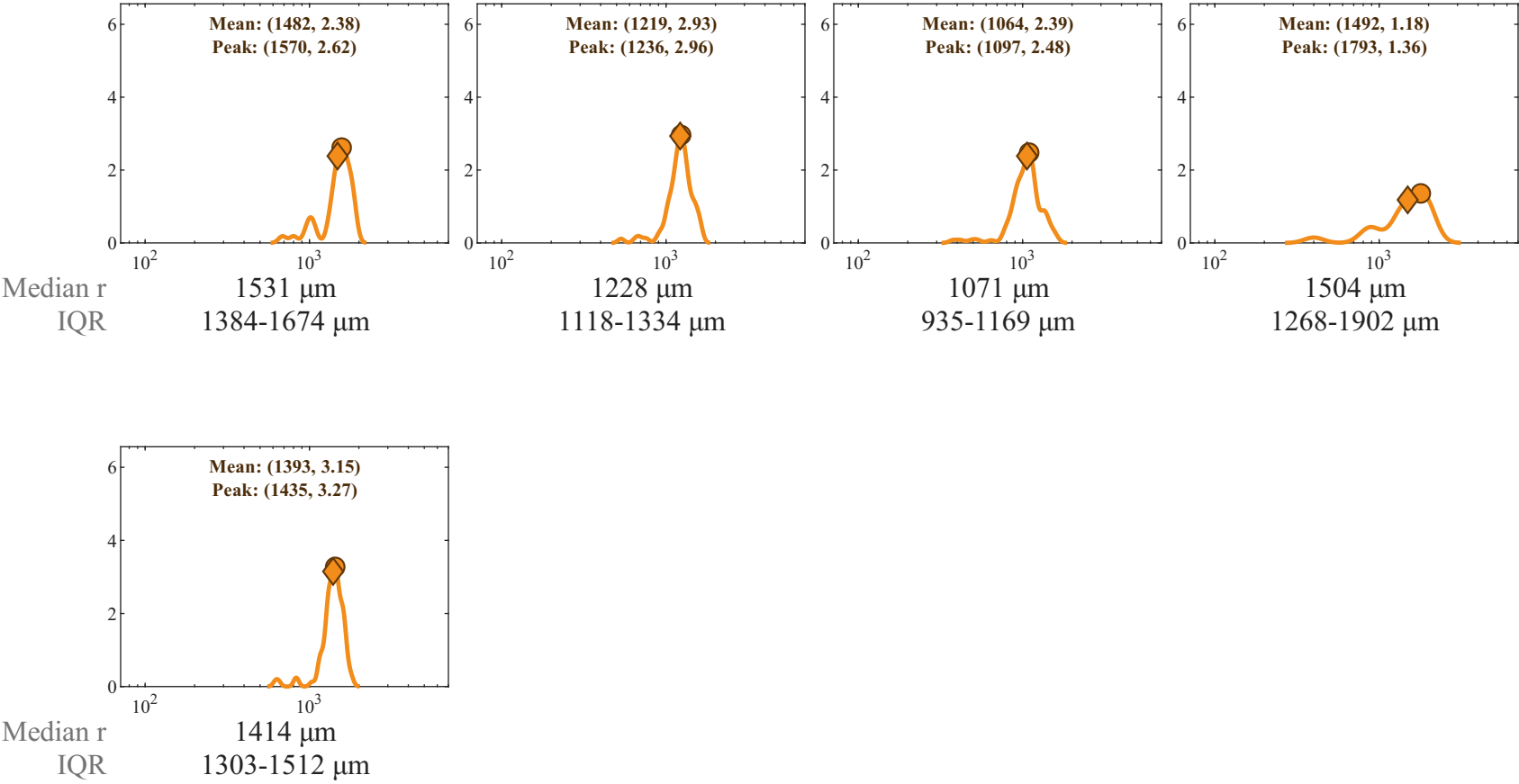


dbf4-1, Passage 3

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

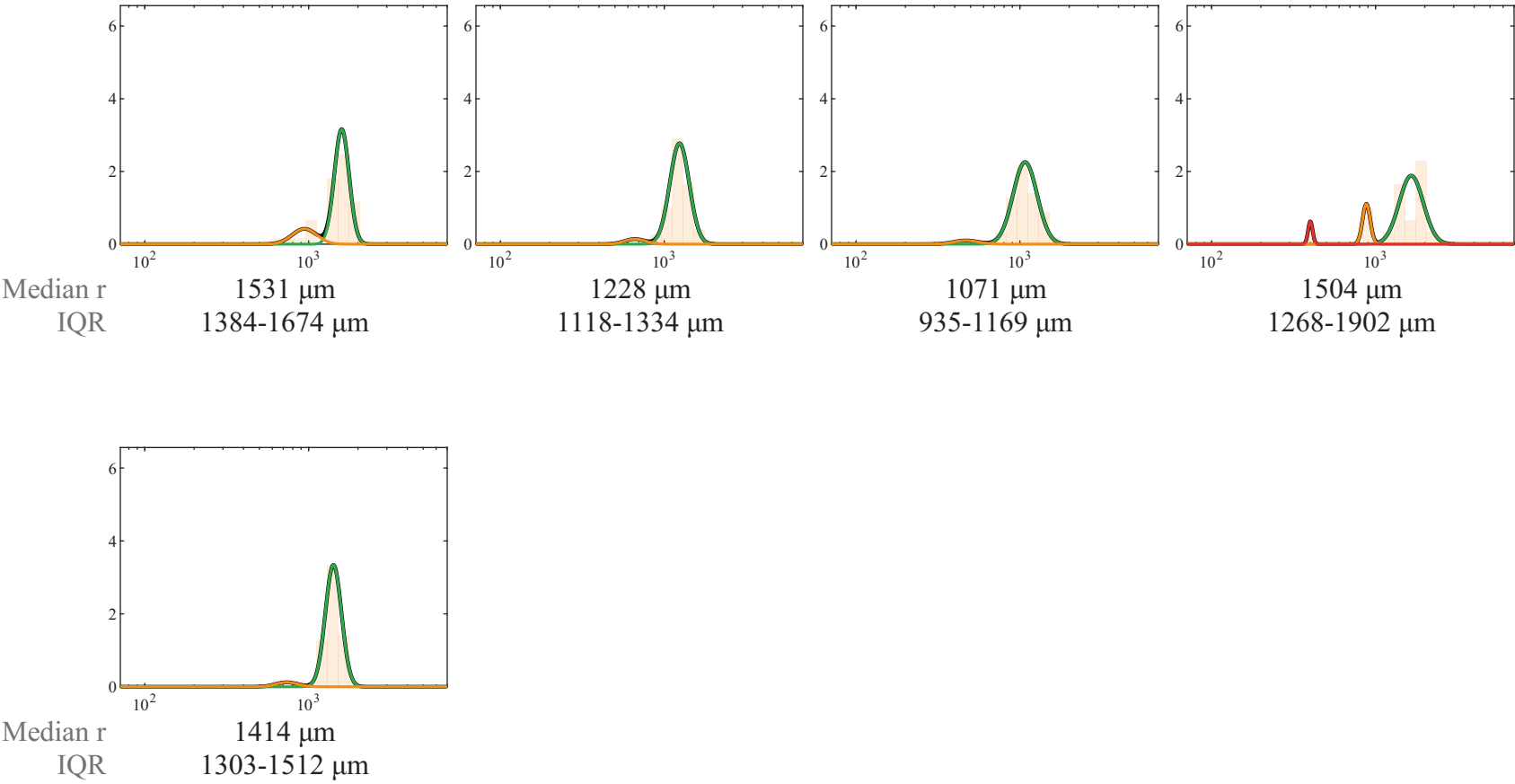


dbf4-1, Passage 3

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

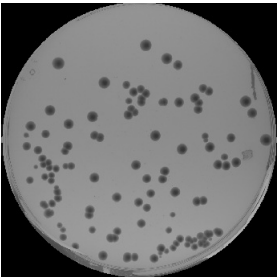
Std 6d



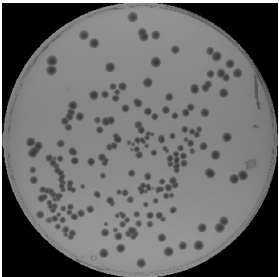
dbf4-1, Passage 4

Original Source Images

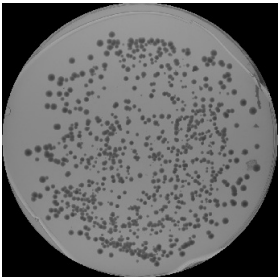
Std 6d



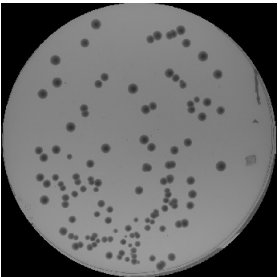
Colonies 55
Capture Ratio 45.1%
Rep A
HCE 144
Plate 90 mm



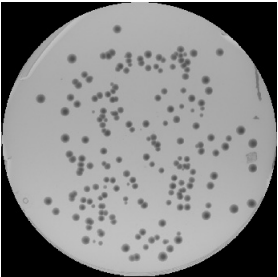
Colonies 111
Capture Ratio 49.3%
Rep A
HCE 688
Plate 90 mm



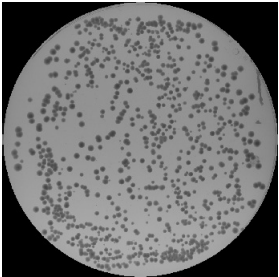
Colonies 366
Capture Ratio 46.3%
Rep A
HCE 3456
Plate 90 mm



Colonies 63
Capture Ratio 52.1%
Rep B
HCE 144
Plate 90 mm



Colonies 116
Capture Ratio 68.3%
Rep B
HCE 688
Plate 90 mm

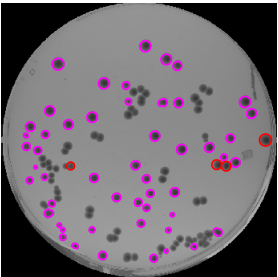


Colonies 366
Capture Ratio 42.3%
Rep B
HCE 3456
Plate 90 mm

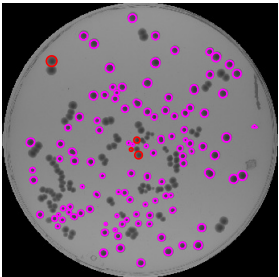
dbf4-1, Passage 4

Source Images with Colony Overlays

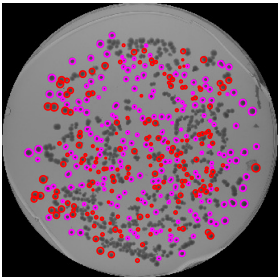
Std 6d



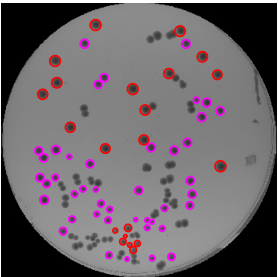
Colonies 55
Capture Ratio 45.1%
Rep A
HCE 144
Plate 90 mm



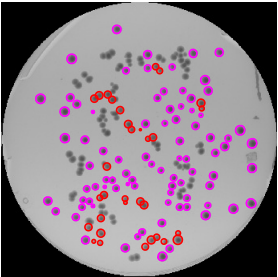
Colonies 111
Capture Ratio 49.3%
Rep A
HCE 688
Plate 90 mm



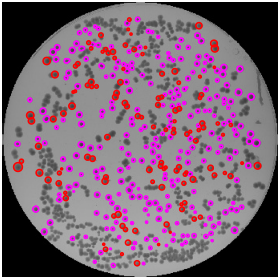
Colonies 366
Capture Ratio 46.3%
Rep A
HCE 3456
Plate 90 mm



Colonies 63
Capture Ratio 52.1%
Rep B
HCE 144
Plate 90 mm



Colonies 116
Capture Ratio 68.3%
Rep B
HCE 688
Plate 90 mm

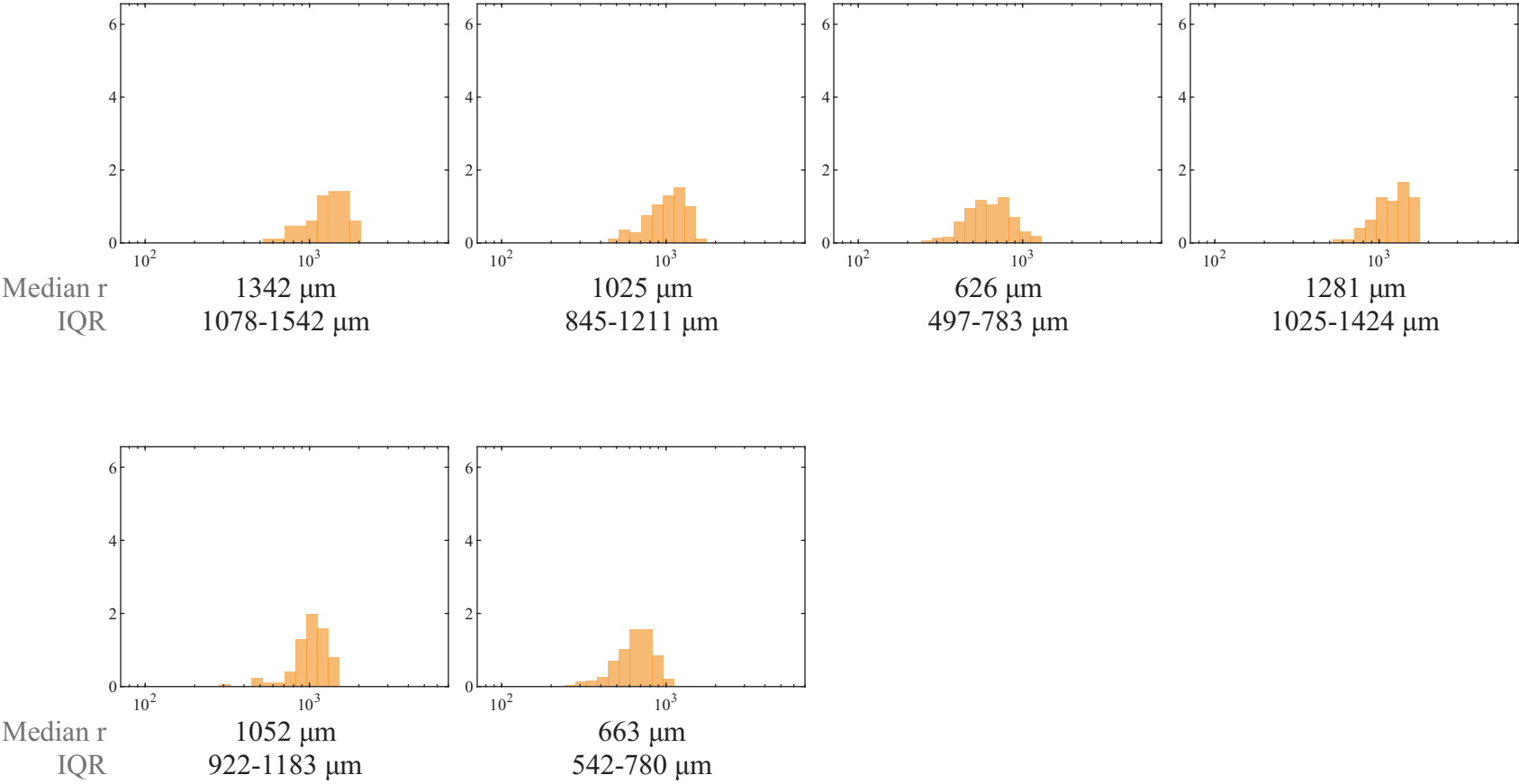


Colonies 366
Capture Ratio 42.3%
Rep B
HCE 3456
Plate 90 mm

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

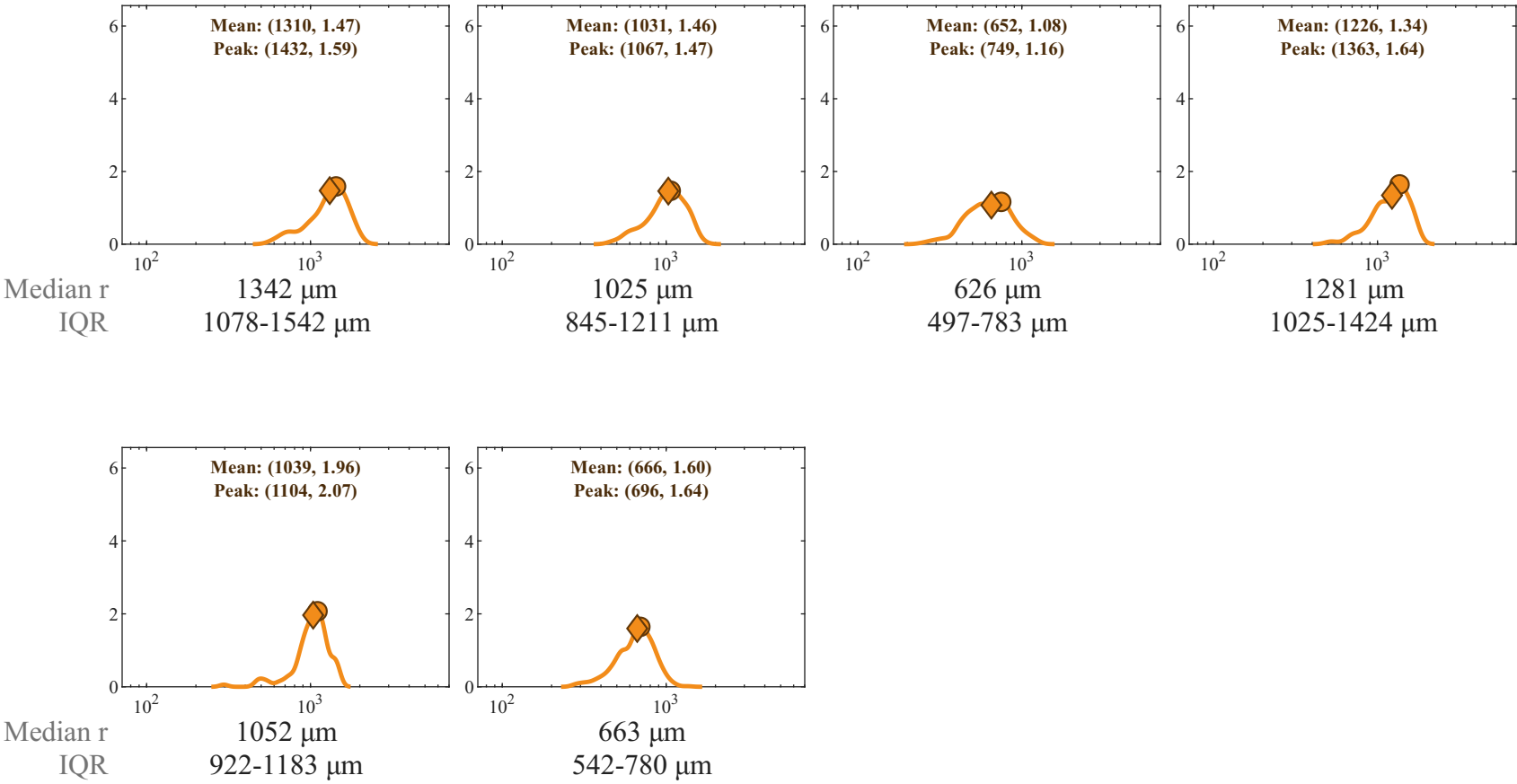


dbf4-1, Passage 4

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

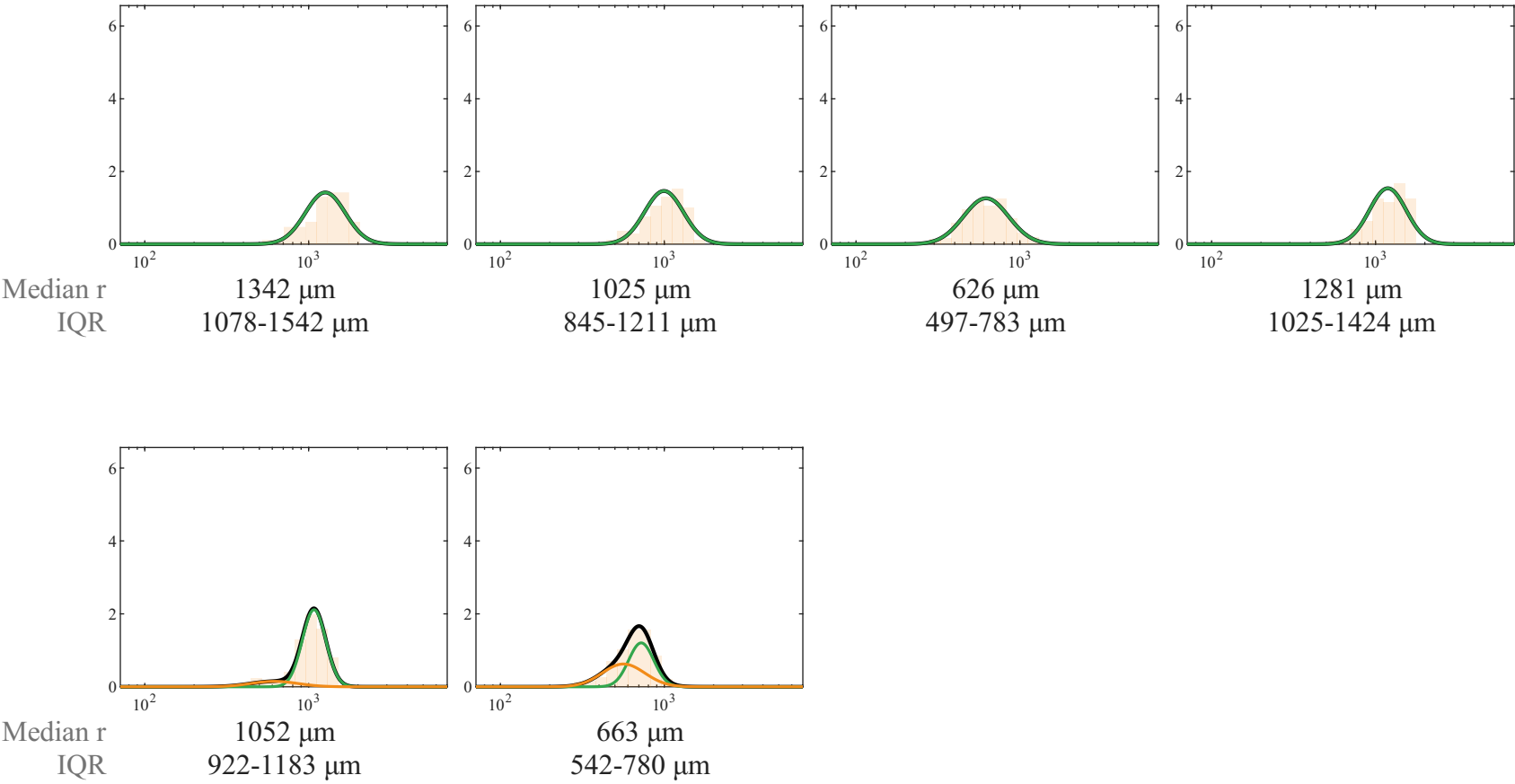
Std 6d



LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

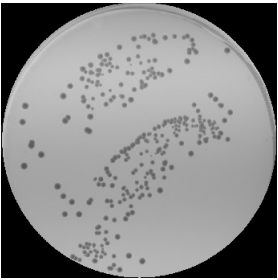
Std 6d



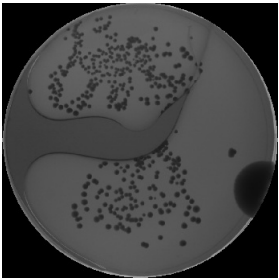
dbf4-1, Passage 5

Original Source Images

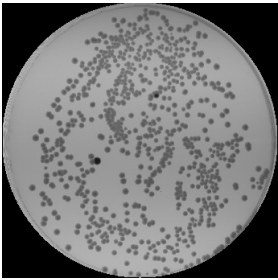
Std 6d



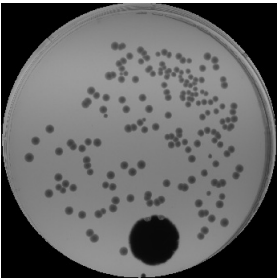
Colonies 113
Capture Ratio 42.3%
Rep A
HCE 160
Plate 140 mm



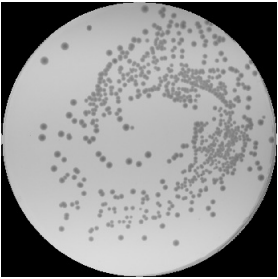
111
13.2%
A
800
90 mm



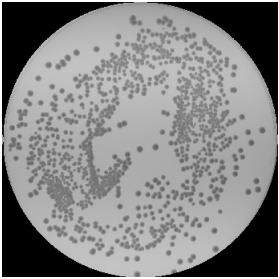
254
39.6%
A
4000
140 mm



89
36.3%
B
160
140 mm



Colonies 142
Capture Ratio 33.6%
Rep B
HCE 800
Plate 140 mm

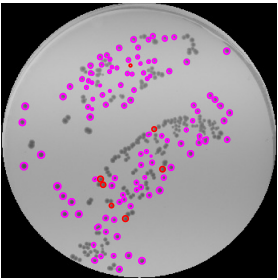


255
31.8%
B
4000
140 mm

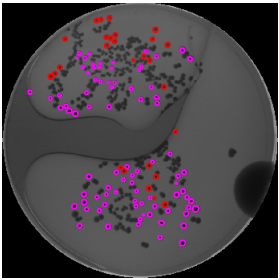
dbf4-1, Passage 5

Source Images with Colony Overlays

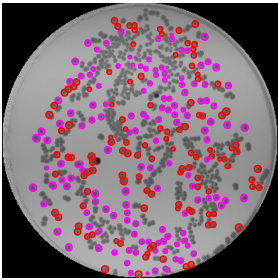
Std 6d



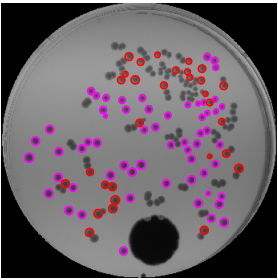
Colonies 113
Capture Ratio 42.3%
Rep A
HCE 160
Plate 140 mm



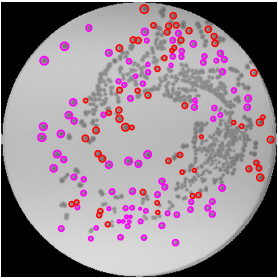
Colonies 111
Capture Ratio 13.2%
Rep A
HCE 800
Plate 90 mm



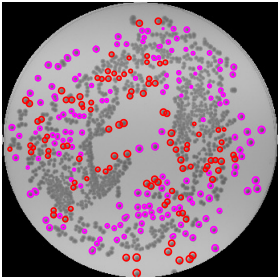
Colonies 254
Capture Ratio 39.6%
Rep A
HCE 4000
Plate 140 mm



Colonies 89
Capture Ratio 36.3%
Rep B
HCE 160
Plate 140 mm



Colonies 142
Capture Ratio 33.6%
Rep B
HCE 800
Plate 140 mm

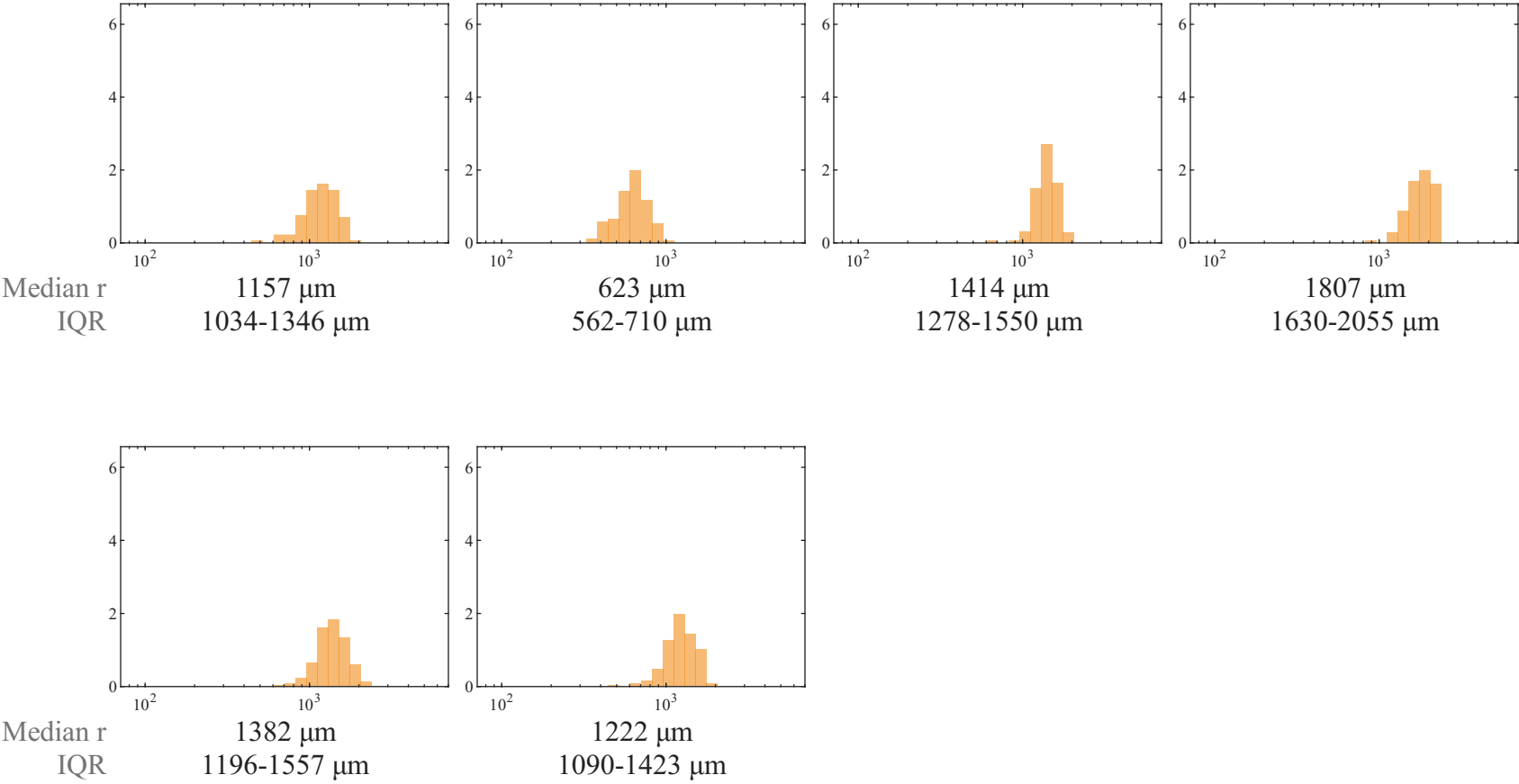


Colonies 255
Capture Ratio 31.8%
Rep B
HCE 4000
Plate 140 mm

Density Histograms

X axis: Colony Radius μm , Y axis: Density

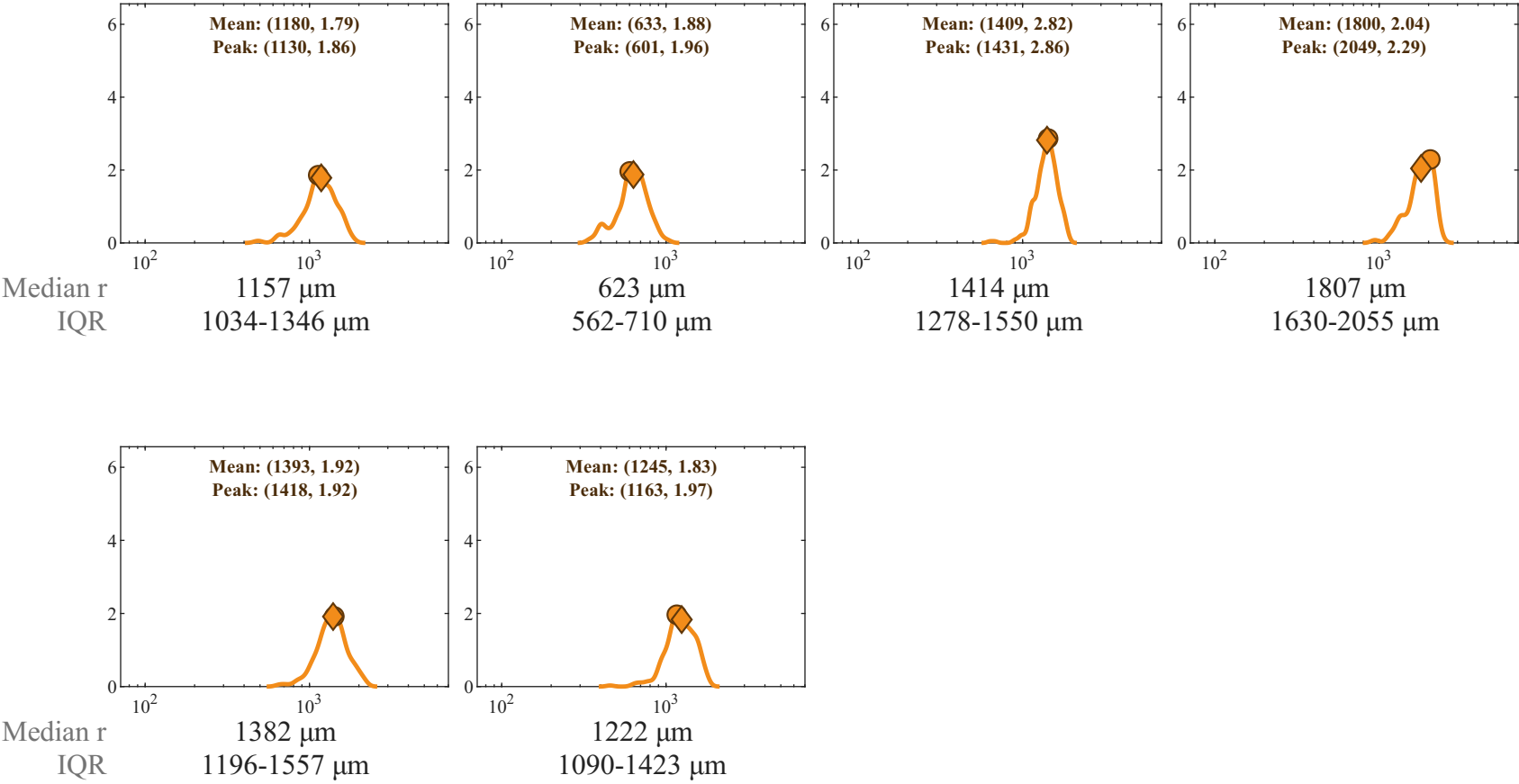
Std 6d



Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

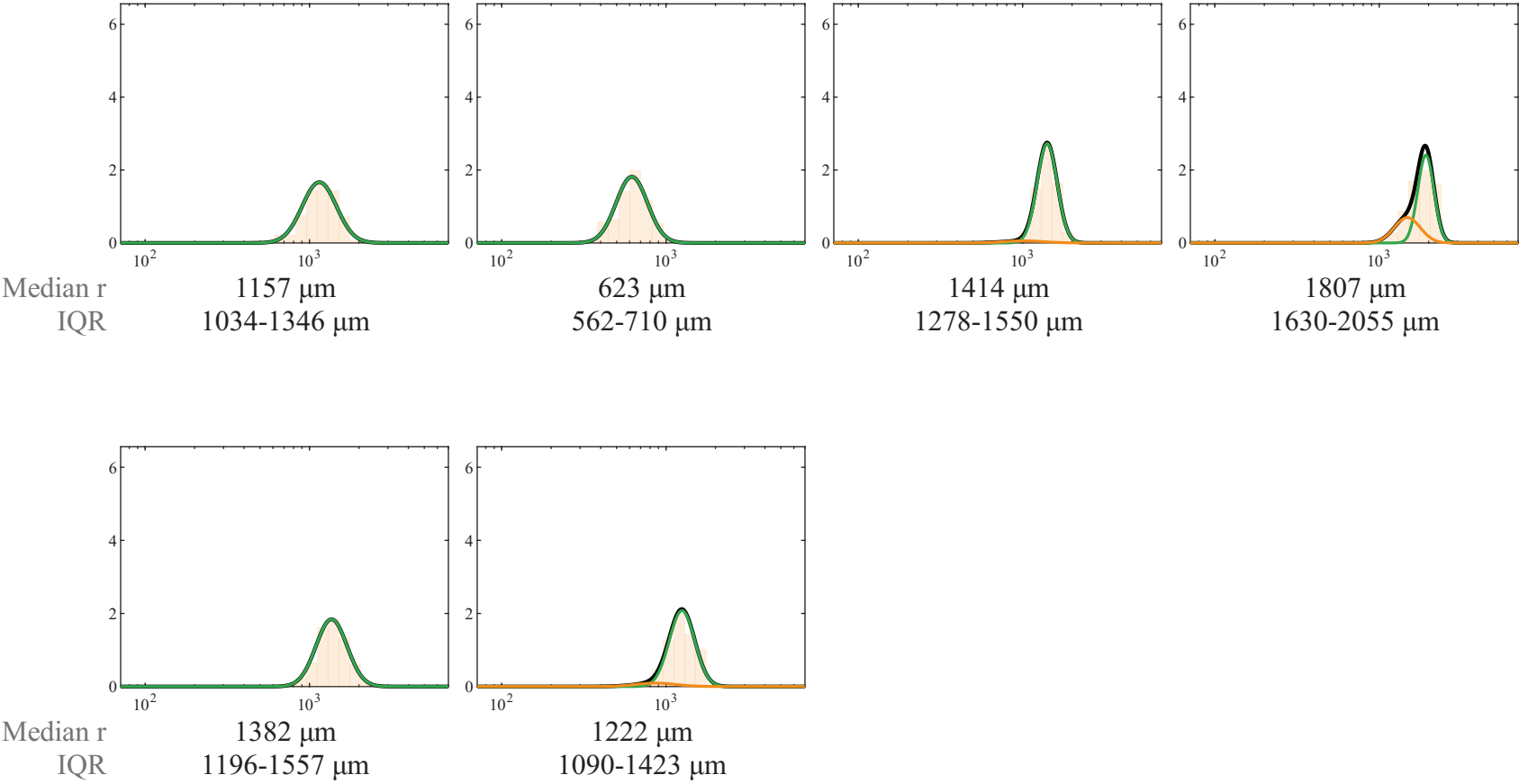
Std 6d



LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

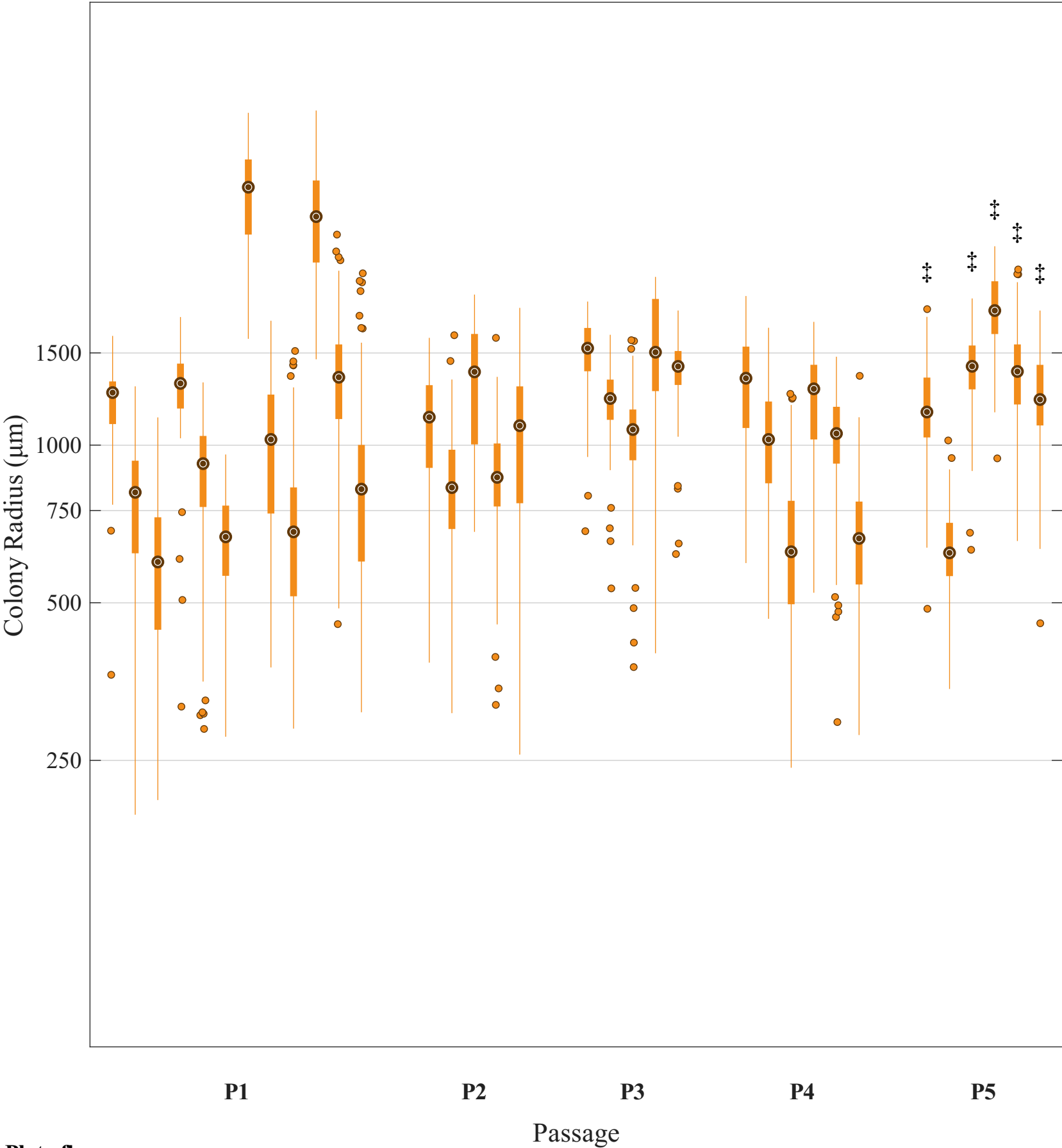
Std 6d



dbf4-1: Colony Radius by Plate, Across Passages

X axis: Passage, Y axis: Colony Radius μm

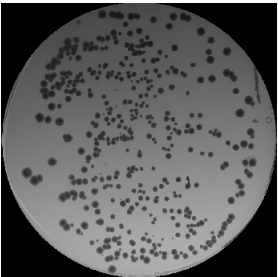
n	25	160	301	39	200	161	35	112	158	29	93	174	61	138	51	148	182	40	79	103	20	88	55	111	366	63	116	366	113	111	125	4	89	142	255
Median μm	1266	812	598	831	922	668	1090	258	373	1348	824	1138	301	380	681	089	1531	228	071	504	414	1342	202	362	628	281	056	63	1157	623	414	807	381	222	



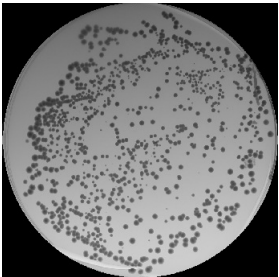
cdc13-2, Passage 1

Original Source Images

Std 6d



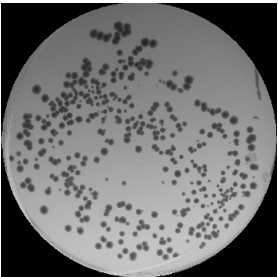
Colonies 193
Capture Ratio 44.1%
Rep A
HCE 3024
Plate 90 mm



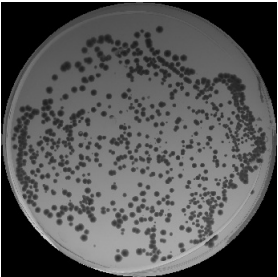
406
40.9%
A
15104
90 mm



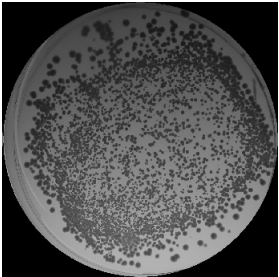
102
6.6%
A
75520
90 mm



147
35.1%
B
3024
90 mm



Colonies 230
Capture Ratio 27.7%
Rep B
HCE 15104
Plate 90 mm

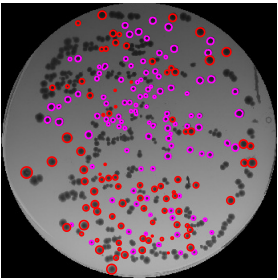


389
12.6%
B
75520
90 mm

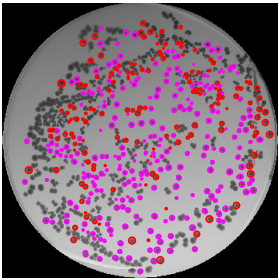
cdc13-2, Passage 1

Source Images with Colony Overlays

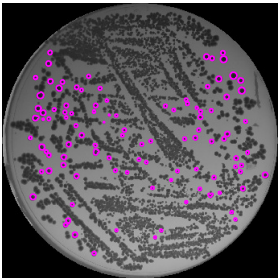
Std 6d



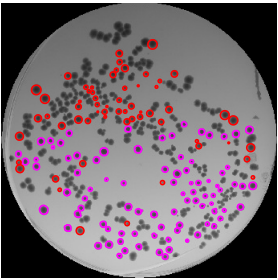
Colonies 193
Capture Ratio 44.1%
Rep A
HCE 3024
Plate 90 mm



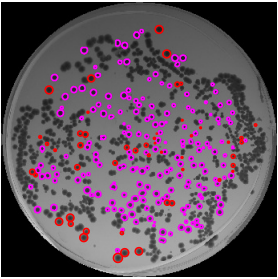
406
40.9%
A
15104
90 mm



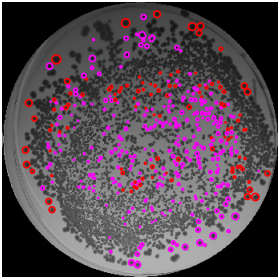
102
6.6%
A
75520
90 mm



147
35.1%
B
3024
90 mm



Colonies 230
Capture Ratio 27.7%
Rep B
HCE 15104
Plate 90 mm



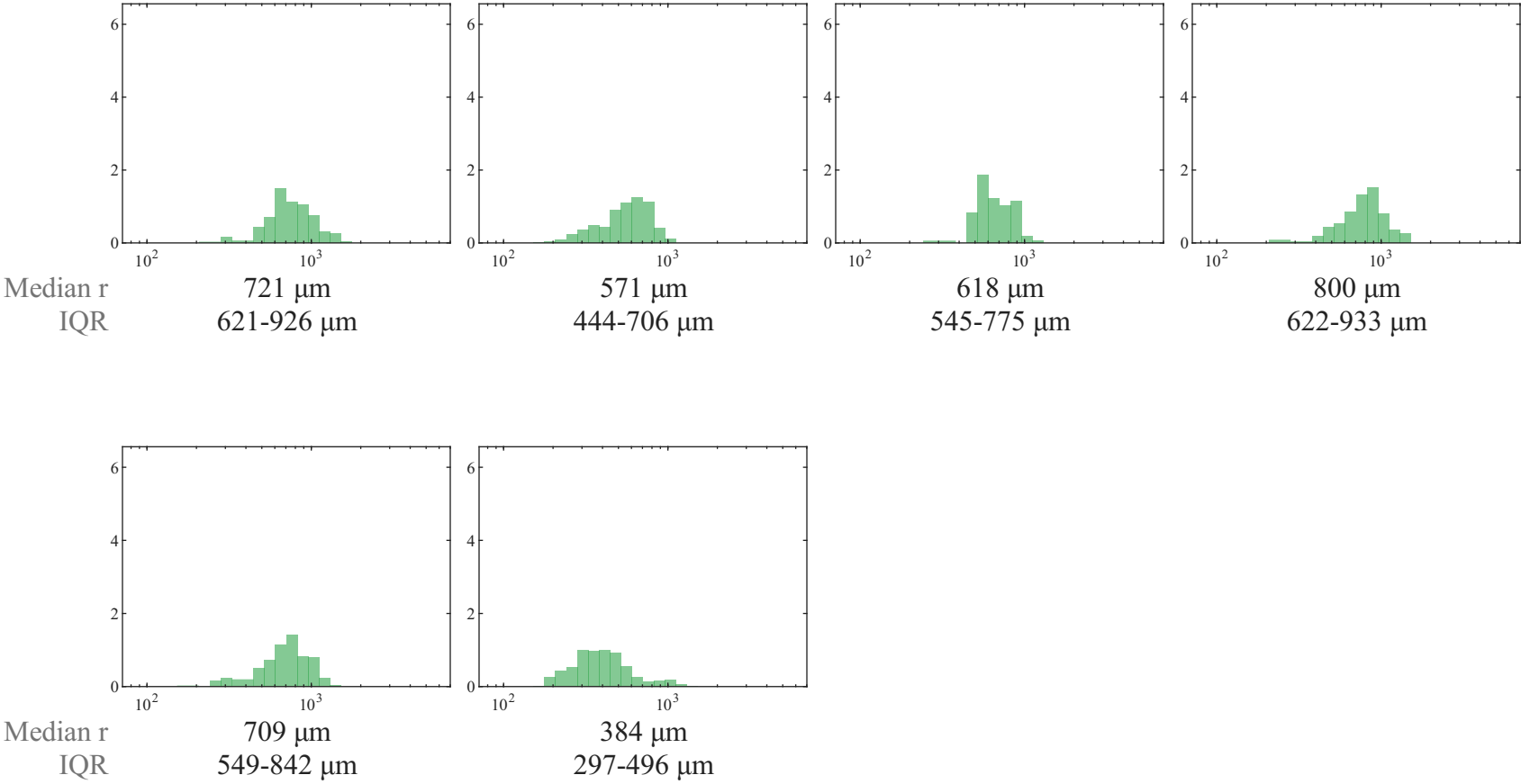
389
12.6%
B
75520
90 mm

cdc13-2, Passage 1

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

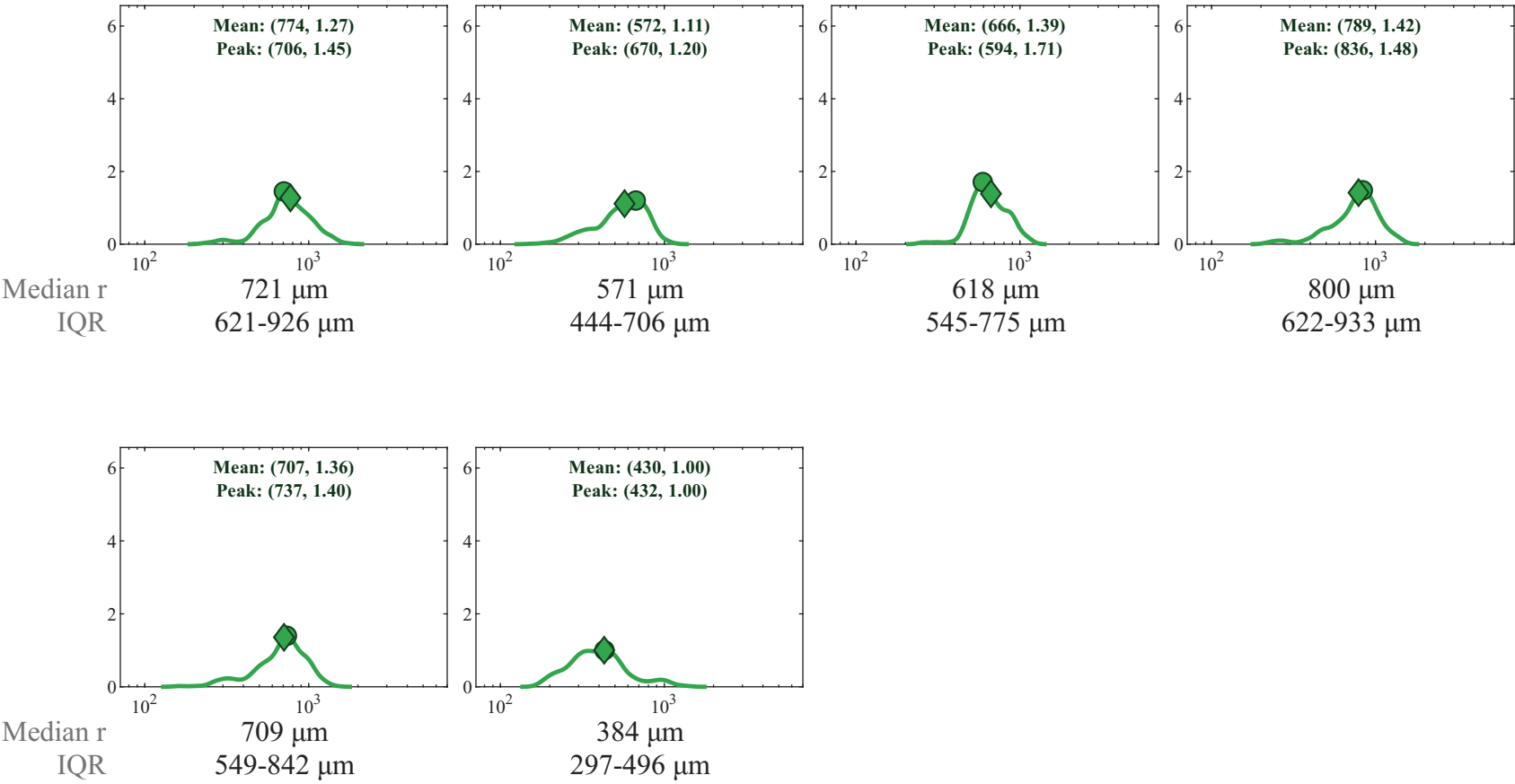


cdc13-2, Passage 1

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

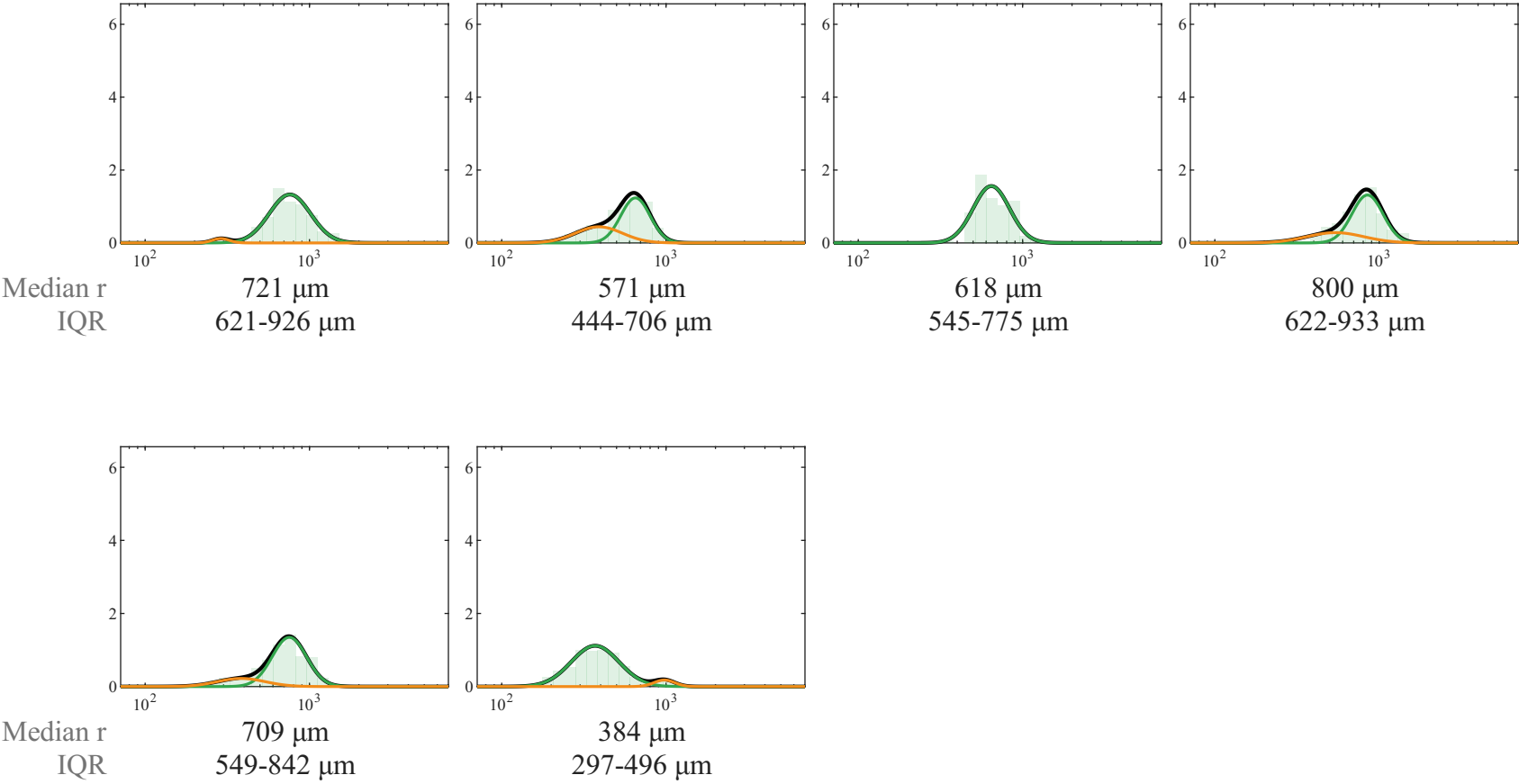


cdc13-2, Passage 1

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

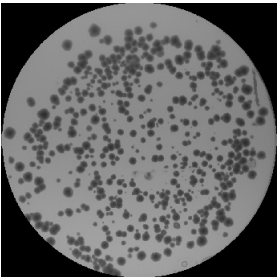
Std 6d



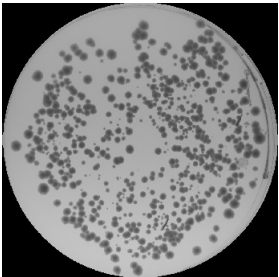
cdc13-2, Passage 2

Original Source Images

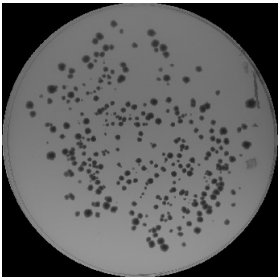
Std 6d



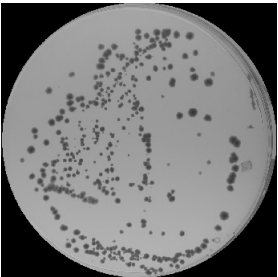
Colonies 202
Capture Ratio 35.3%
Rep A
HCE 176
Plate 90 mm



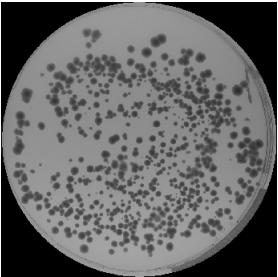
202
29.9%
A
864
90 mm



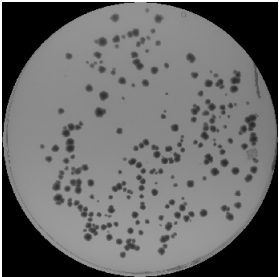
187
54.9%
A
4352
90 mm



162
36.2%
B
176
90 mm



Colonies 216
Capture Ratio 28.7%
Rep B
HCE 864
Plate 90 mm

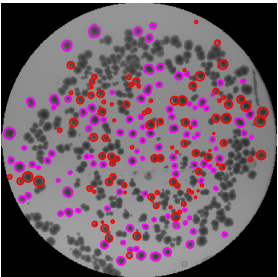


130
52.6%
B
4352
90 mm

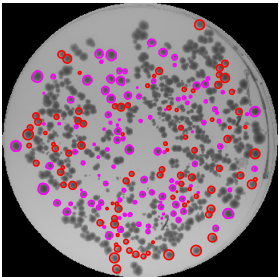
cdc13-2, Passage 2

Source Images with Colony Overlays

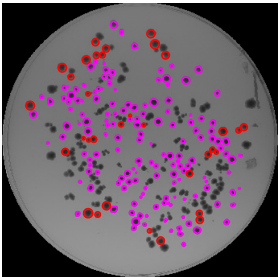
Std 6d



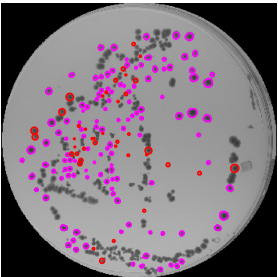
Colonies 202
Capture Ratio 35.3%
Rep A
HCE 176
Plate 90 mm



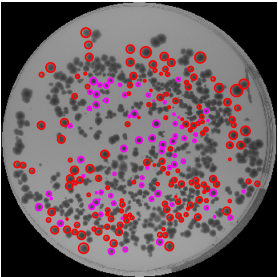
Colonies 202
Capture Ratio 29.9%
Rep A
HCE 864
Plate 90 mm



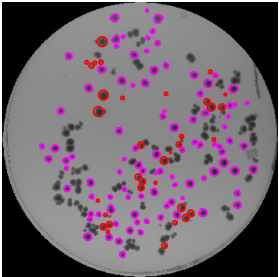
Colonies 187
Capture Ratio 54.9%
Rep A
HCE 4352
Plate 90 mm



Colonies 162
Capture Ratio 36.2%
Rep B
HCE 176
Plate 90 mm



Colonies 216
Capture Ratio 28.7%
Rep B
HCE 864
Plate 90 mm



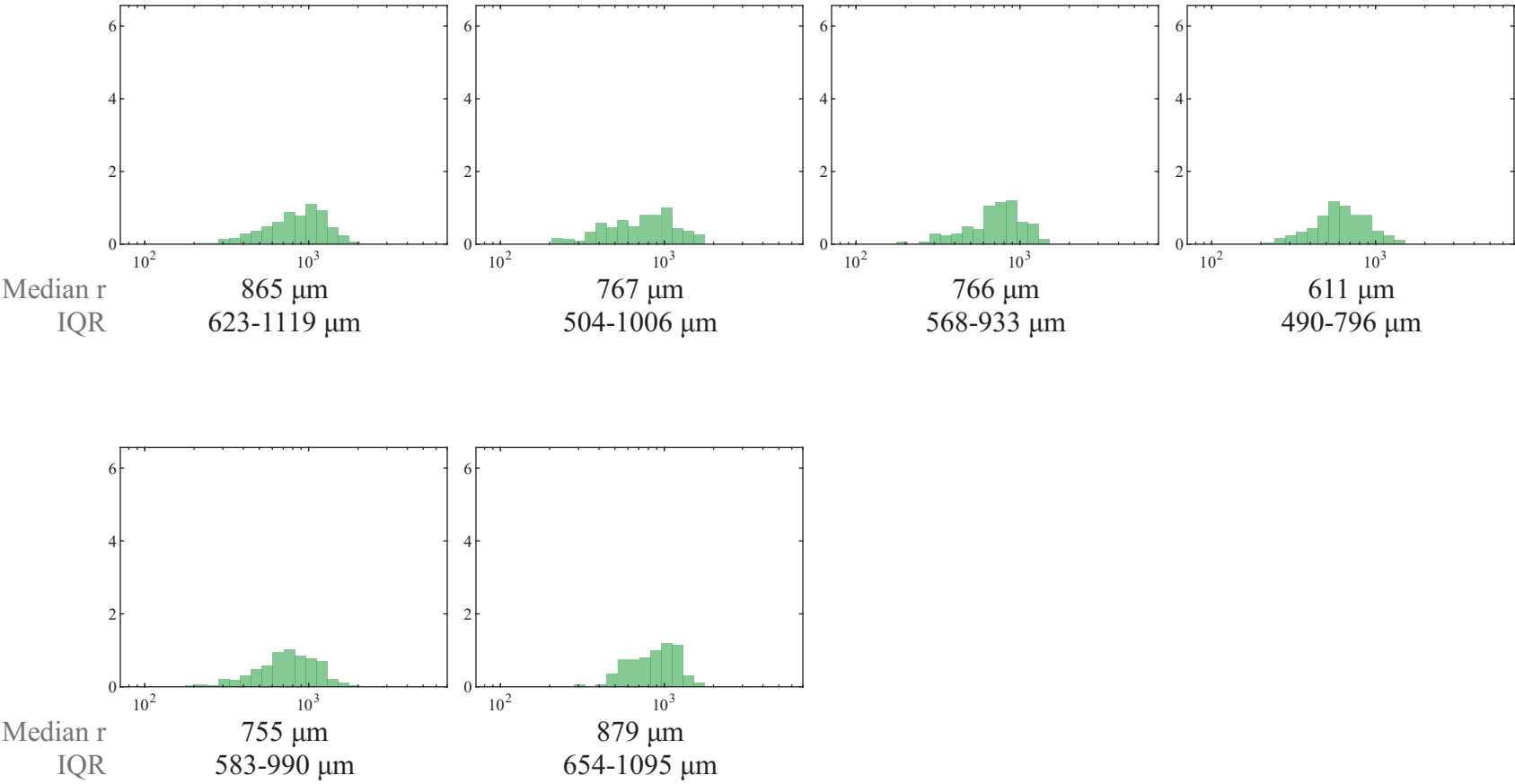
Colonies 130
Capture Ratio 52.6%
Rep B
HCE 4352
Plate 90 mm

cdc13-2, Passage 2

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

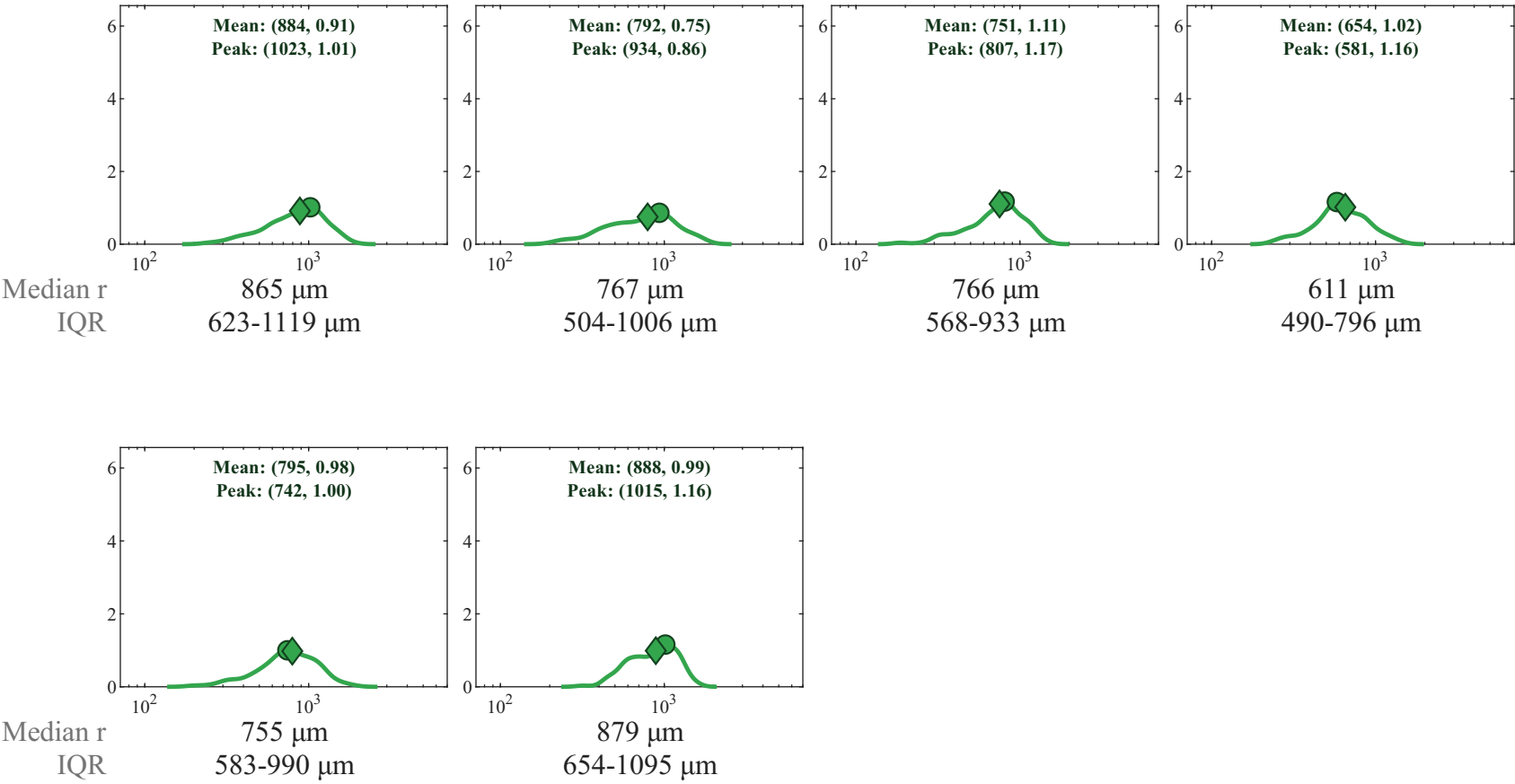


cdc13-2, Passage 2

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

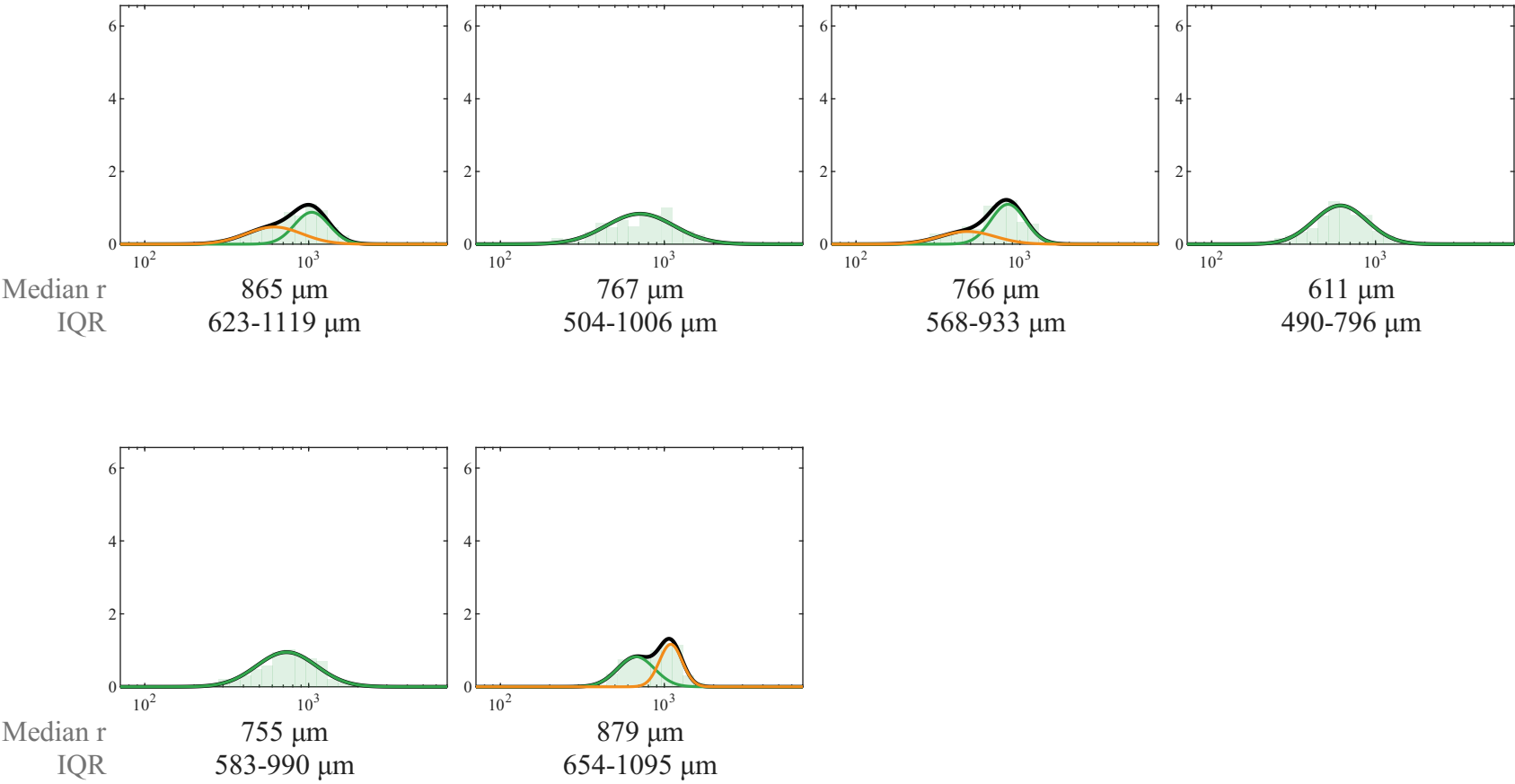


cdc13-2, Passage 2

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

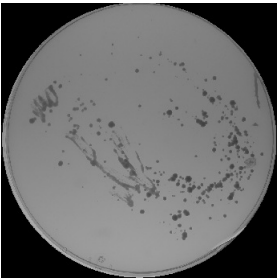
Std 6d



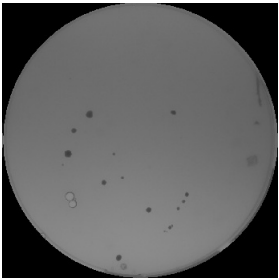
cdc13-2, Passage 3

Original Source Images

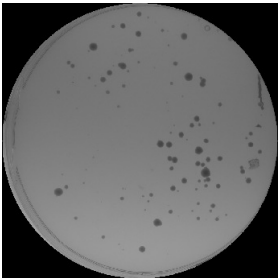
Std 6d



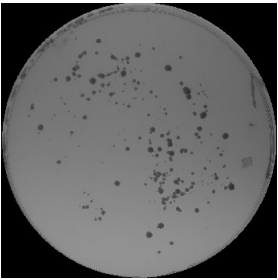
Colonies 190
Capture Ratio 30.1%
Rep A
HCE 160
Plate 90 mm



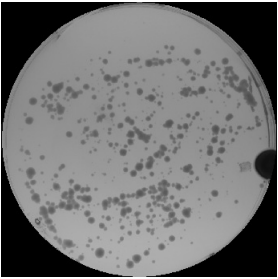
17
27.3%
A
768
90 mm



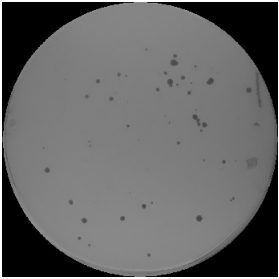
92
80.1%
A
3872
90 mm



139
38.6%
B
160
90 mm



Colonies 285
Capture Ratio 35.8%
Rep B
HCE 768
Plate 90 mm

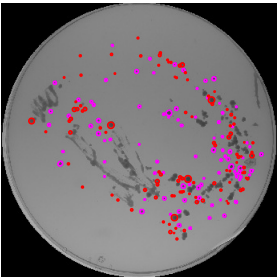


41
66.1%
B
3872
90 mm

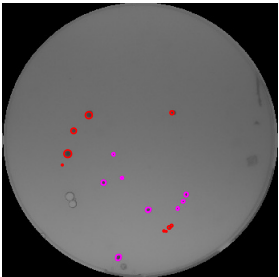
cdc13-2, Passage 3

Source Images with Colony Overlays

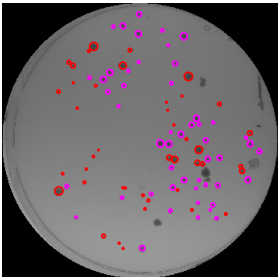
Std 6d



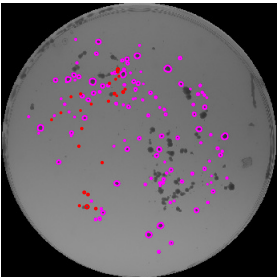
Colonies 190
Capture Ratio 30.1%
Rep A
HCE 160
Plate 90 mm



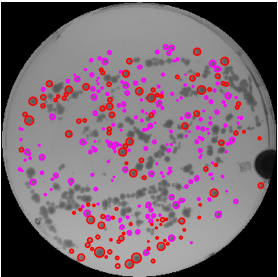
17
27.3%
A
768
90 mm



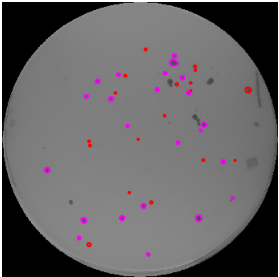
92
80.1%
A
3872
90 mm



139
38.6%
B
160
90 mm



Colonies 285
Capture Ratio 35.8%
Rep B
HCE 768
Plate 90 mm



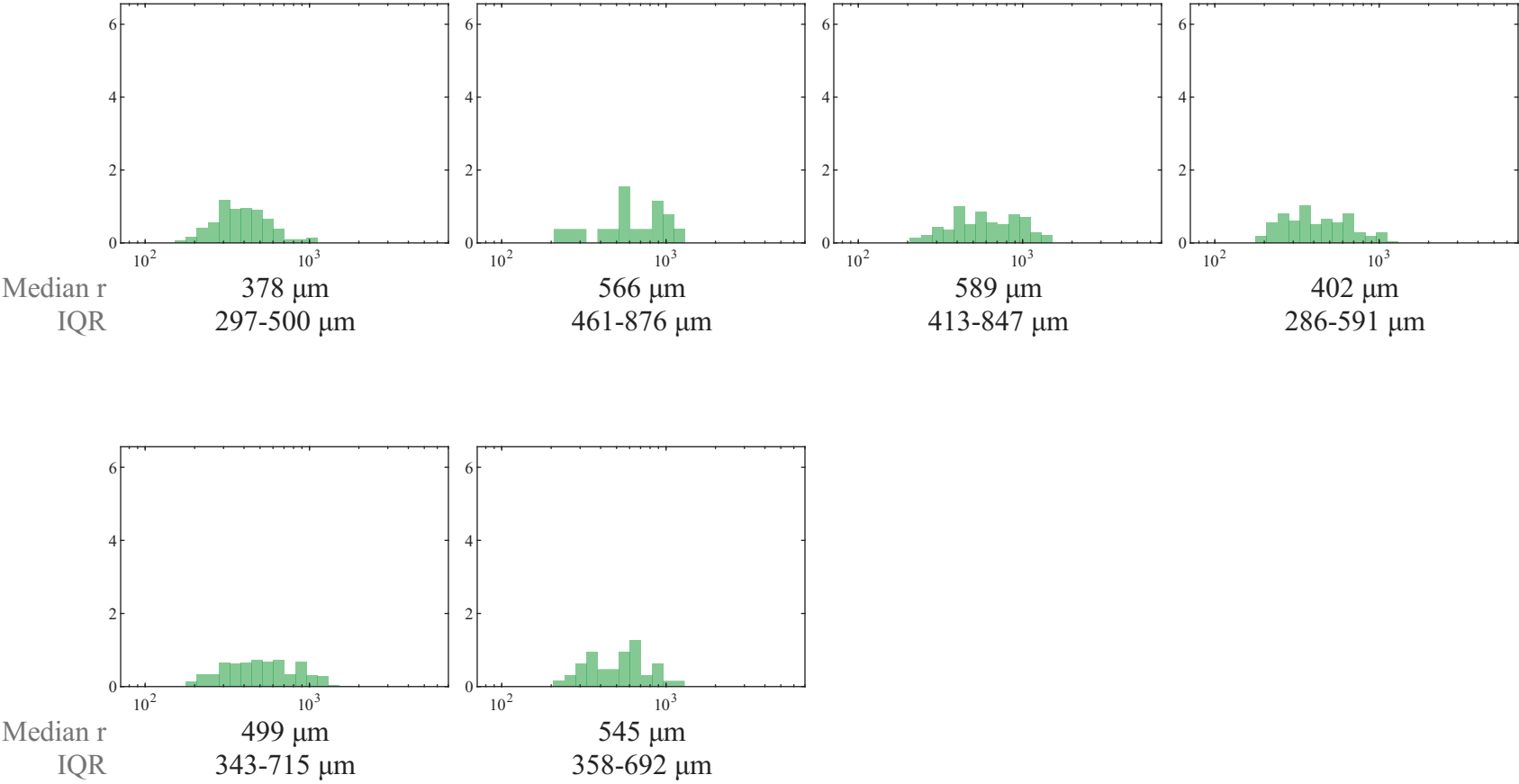
41
66.1%
B
3872
90 mm

cdc13-2, Passage 3

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

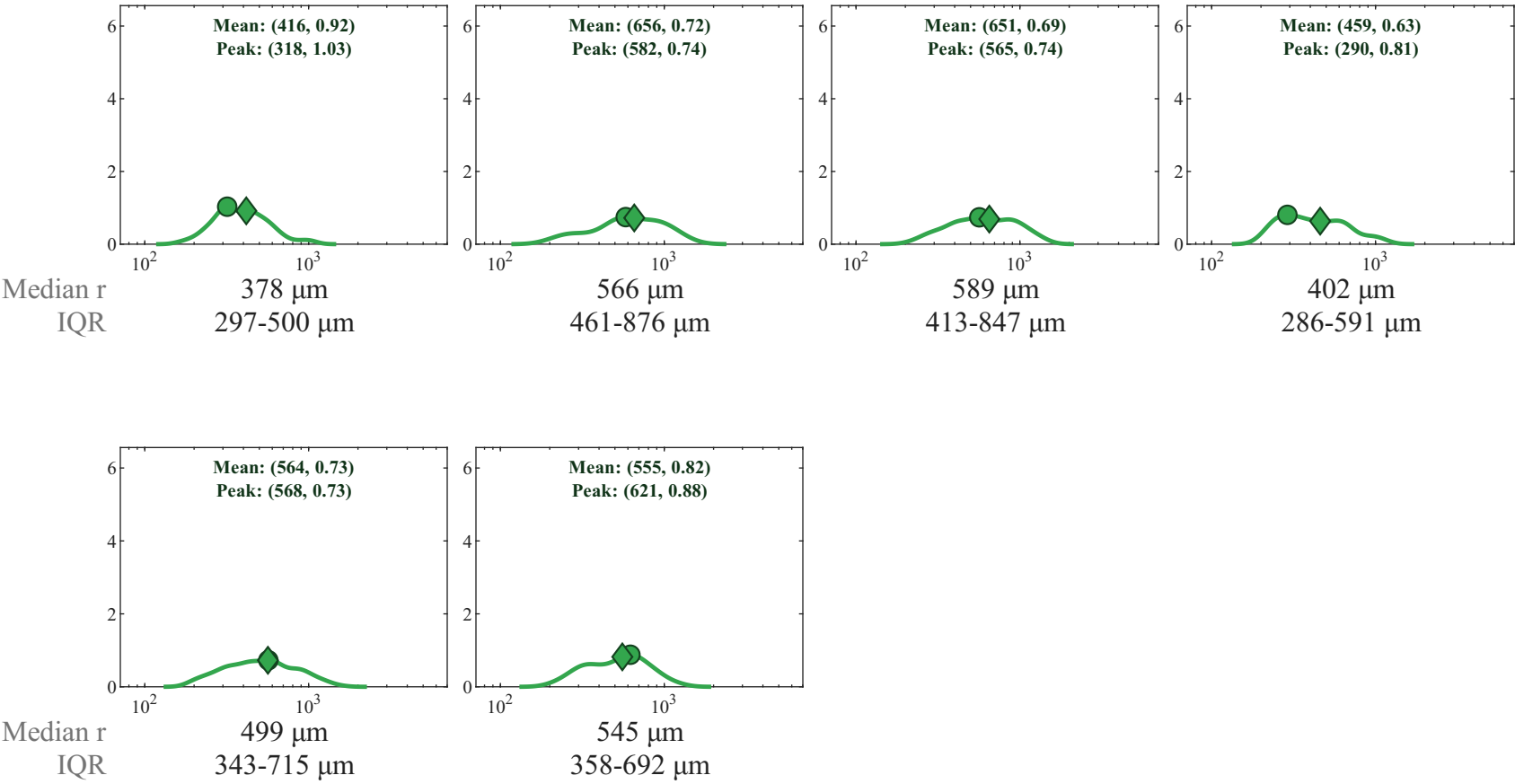


cdc13-2, Passage 3

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

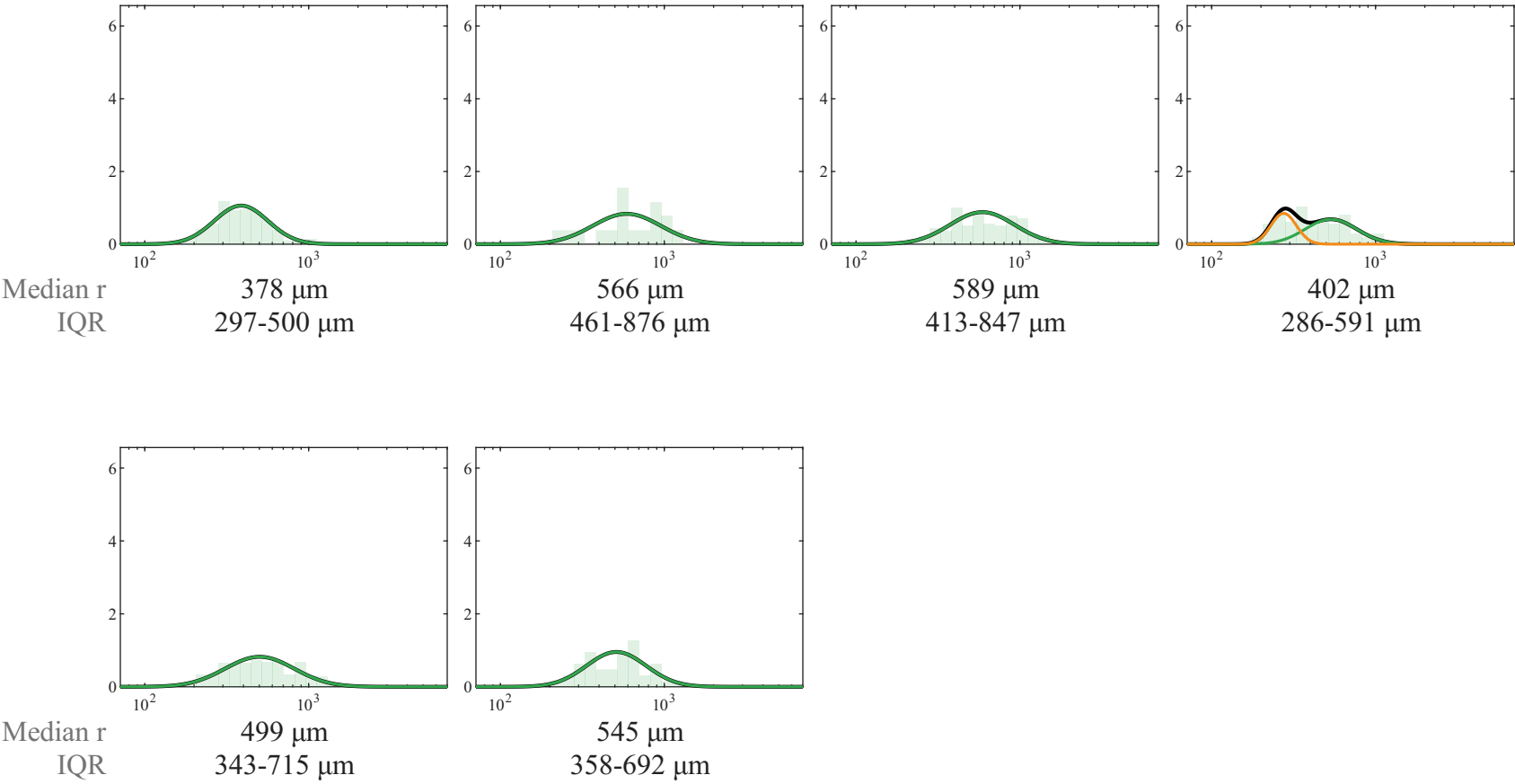


cdc13-2, Passage 3

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

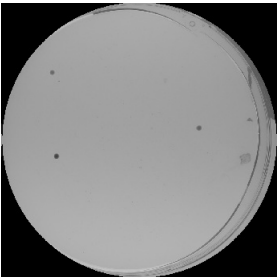
Std 6d



cdc13-2, Passage 4

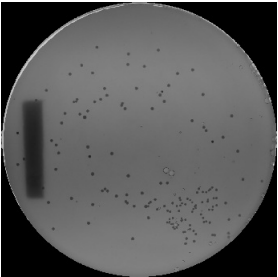
Original Source Images

Std 6d

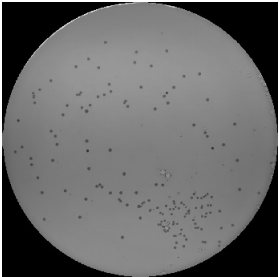


Colonies 3
Capture Ratio 4.0%
Rep B
HCE 4064
Plate 90 mm

LgSrc 6d

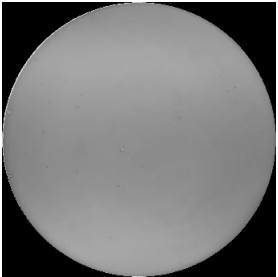


Colonies 119
Capture Ratio 25.6%
Rep A
HCE 752
Plate 140 mm



125
98.7%
B
752
140 mm

SmSrc 6d

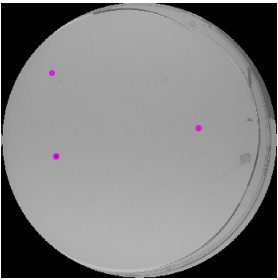


Colonies 0
Capture Ratio —
Rep —
HCE 656
Plate 140 mm

cdc13-2, Passage 4

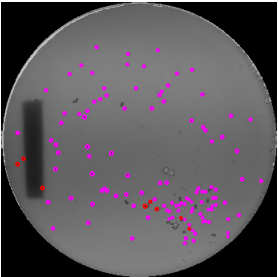
Source Images with Colony Overlays

Std 6d

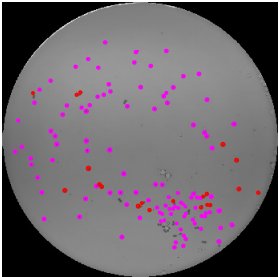


Colonies 3
Capture Ratio 4.0%
Rep B
HCE 4064
Plate 90 mm

LgSrc 6d

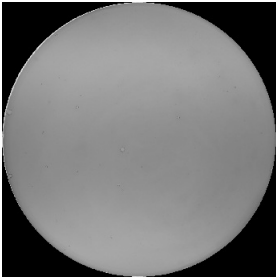


Colonies 119
Capture Ratio 25.6%
Rep A
HCE 752
Plate 140 mm



125
98.7%
B
752
140 mm

SmSrc 6d



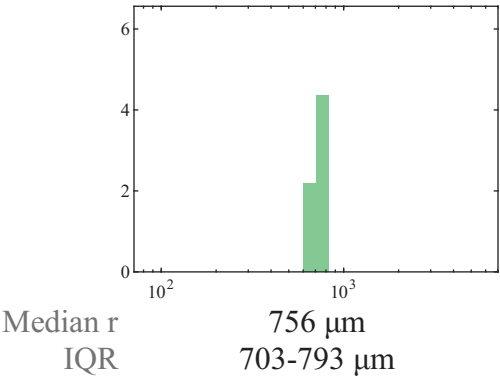
Colonies 0
Capture Ratio —
Rep —
HCE 656
Plate 140 mm

cdc13-2, Passage 4

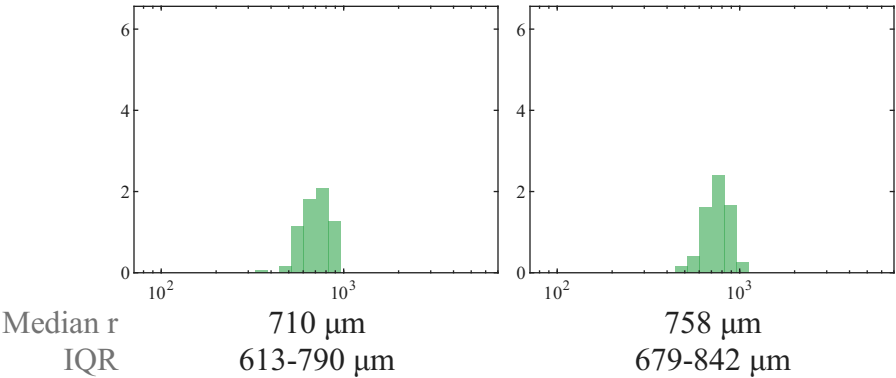
Density Histograms

X axis: Colony Radius μm , Y axis: Density

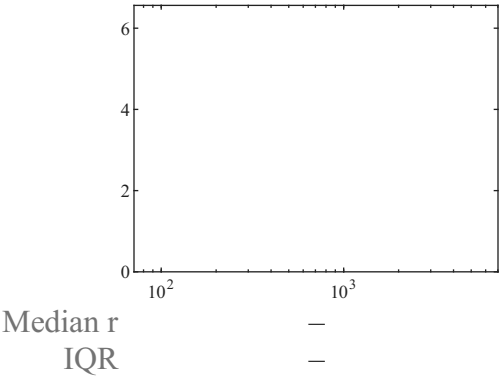
Std 6d



LgSrc 6d



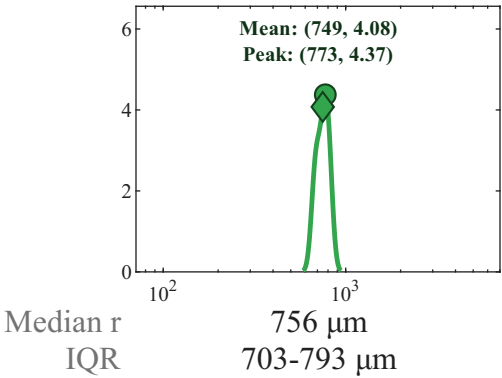
SmSrc 6d



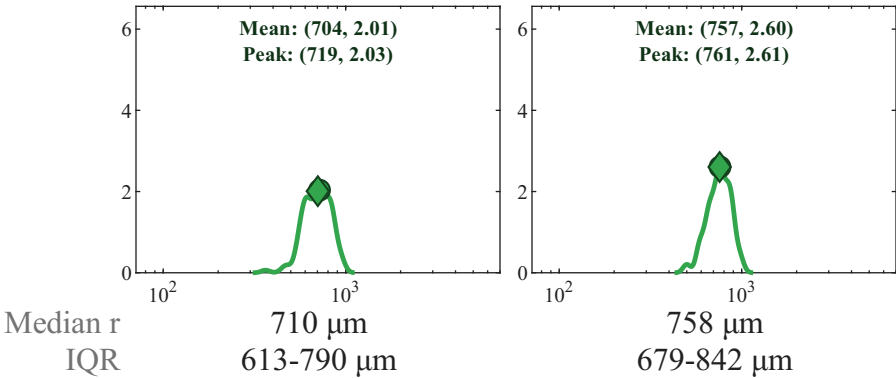
Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

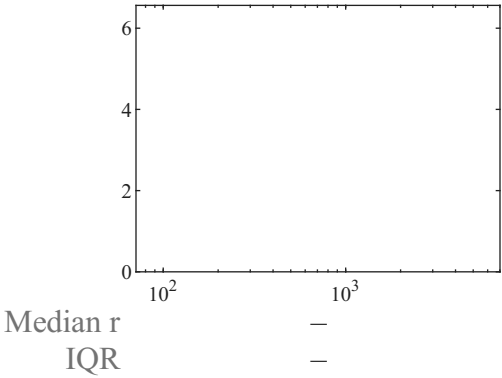
Std 6d



LgSrc 6d



SmSrc 6d

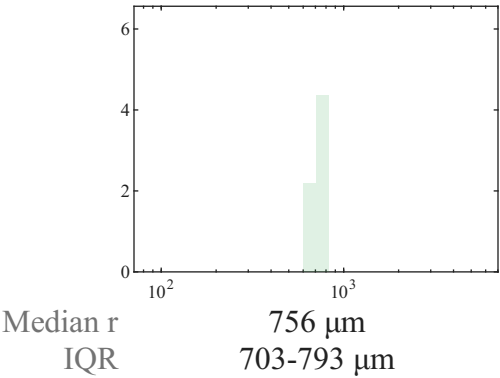


cdc13-2, Passage 4

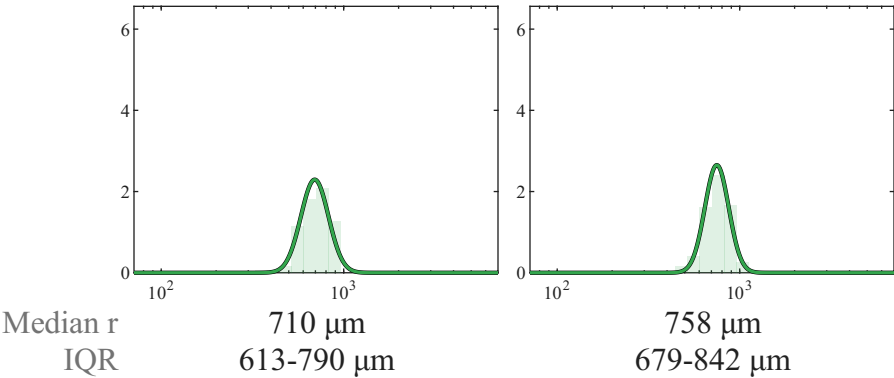
LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

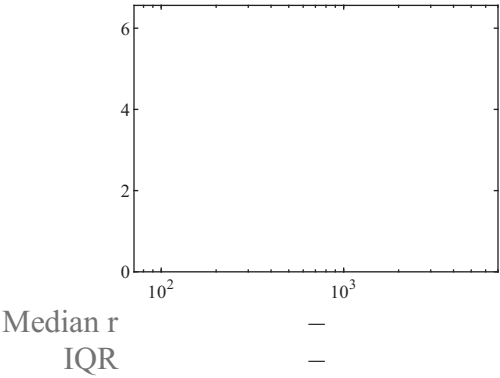
Std 6d



LgSrc 6d



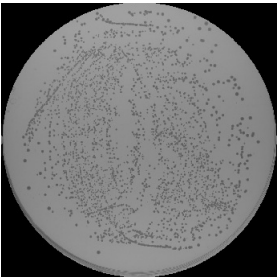
SmSrc 6d



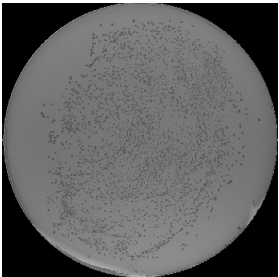
cdc13-2, Passage 5

Original Source Images

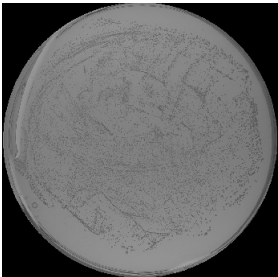
Std 6d



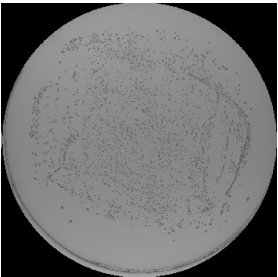
Colonies 953
Capture Ratio 43.2%
Rep A
HCE 160
Plate 90 mm



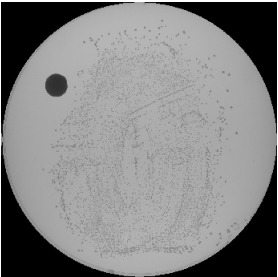
1447
38.8%
A
800
90 mm



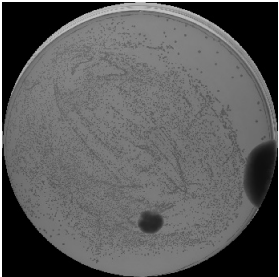
3062
20.8%
A
4032
90 mm



925
48.8%
B
160
90 mm



Colonies 1242
Capture Ratio 25.1%
Rep B
HCE 800
Plate 90 mm

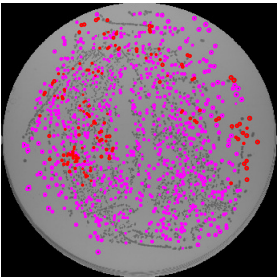


1343
10.3%
B
4032
90 mm

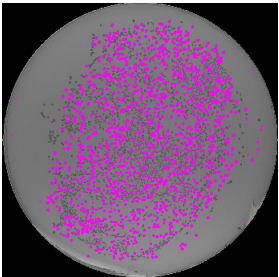
cdc13-2, Passage 5

Source Images with Colony Overlays

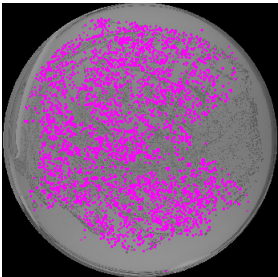
Std 6d



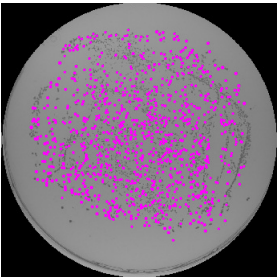
Colonies 953
Capture Ratio 43.2%
Rep A
HCE 160
Plate 90 mm



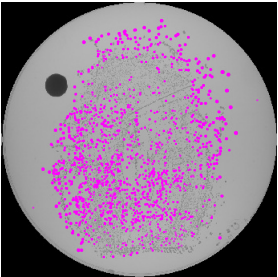
1447
38.8%
A
800
90 mm



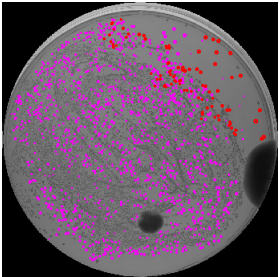
3062
20.8%
A
4032
90 mm



925
48.8%
B
160
90 mm



Colonies 1242
Capture Ratio 25.1%
Rep B
HCE 800
Plate 90 mm



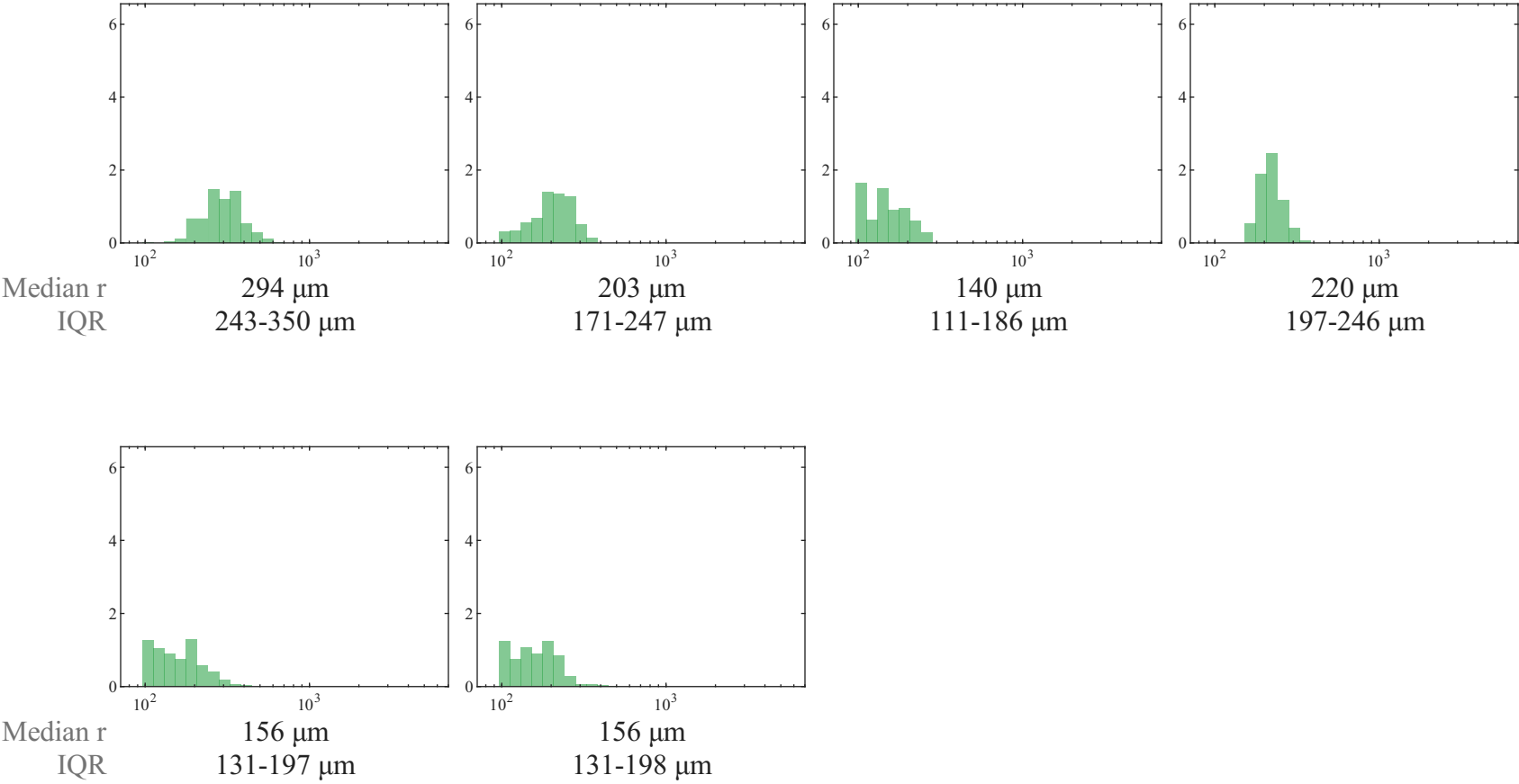
1343
10.3%
B
4032
90 mm

cdc13-2, Passage 5

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

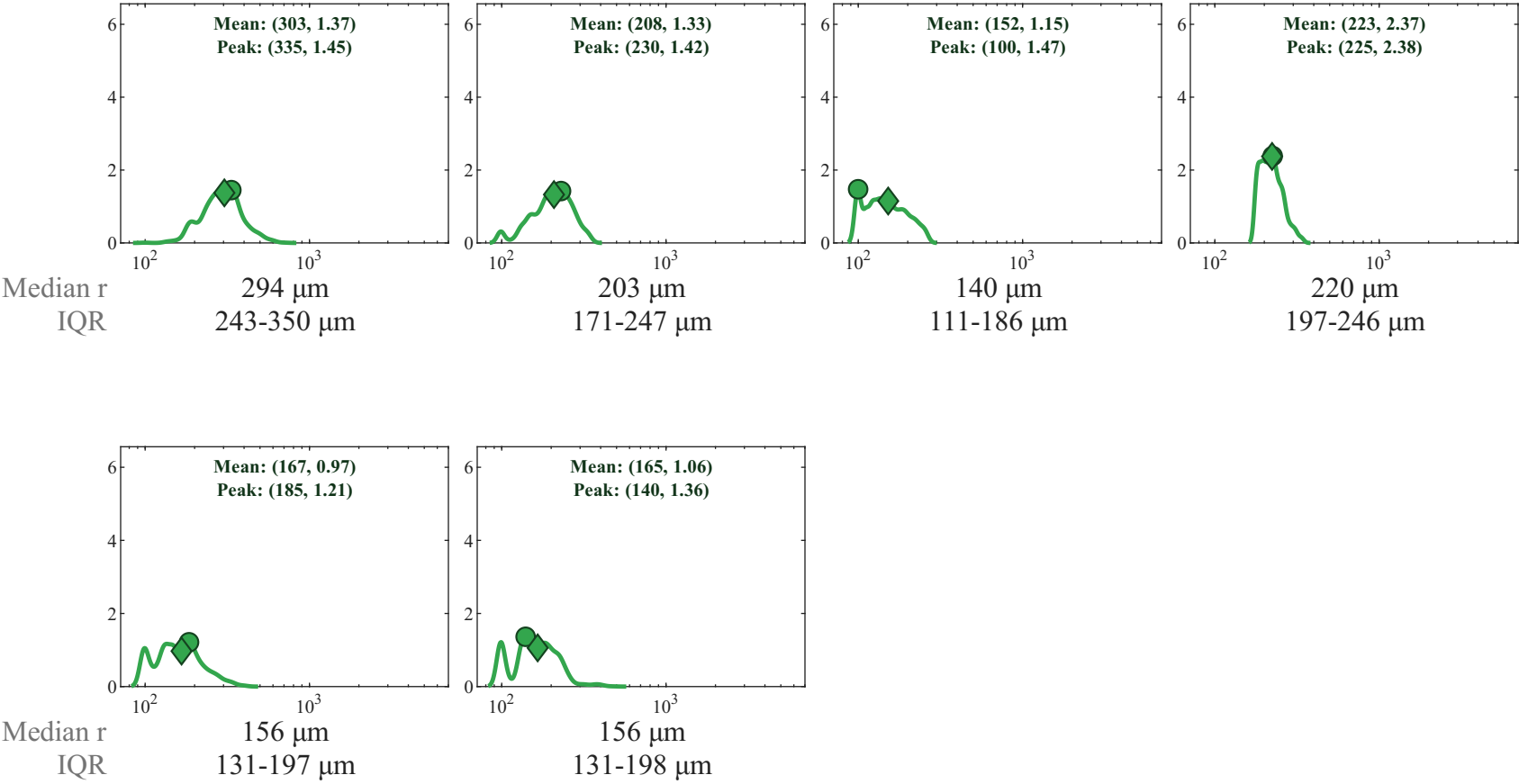


cdc13-2, Passage 5

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

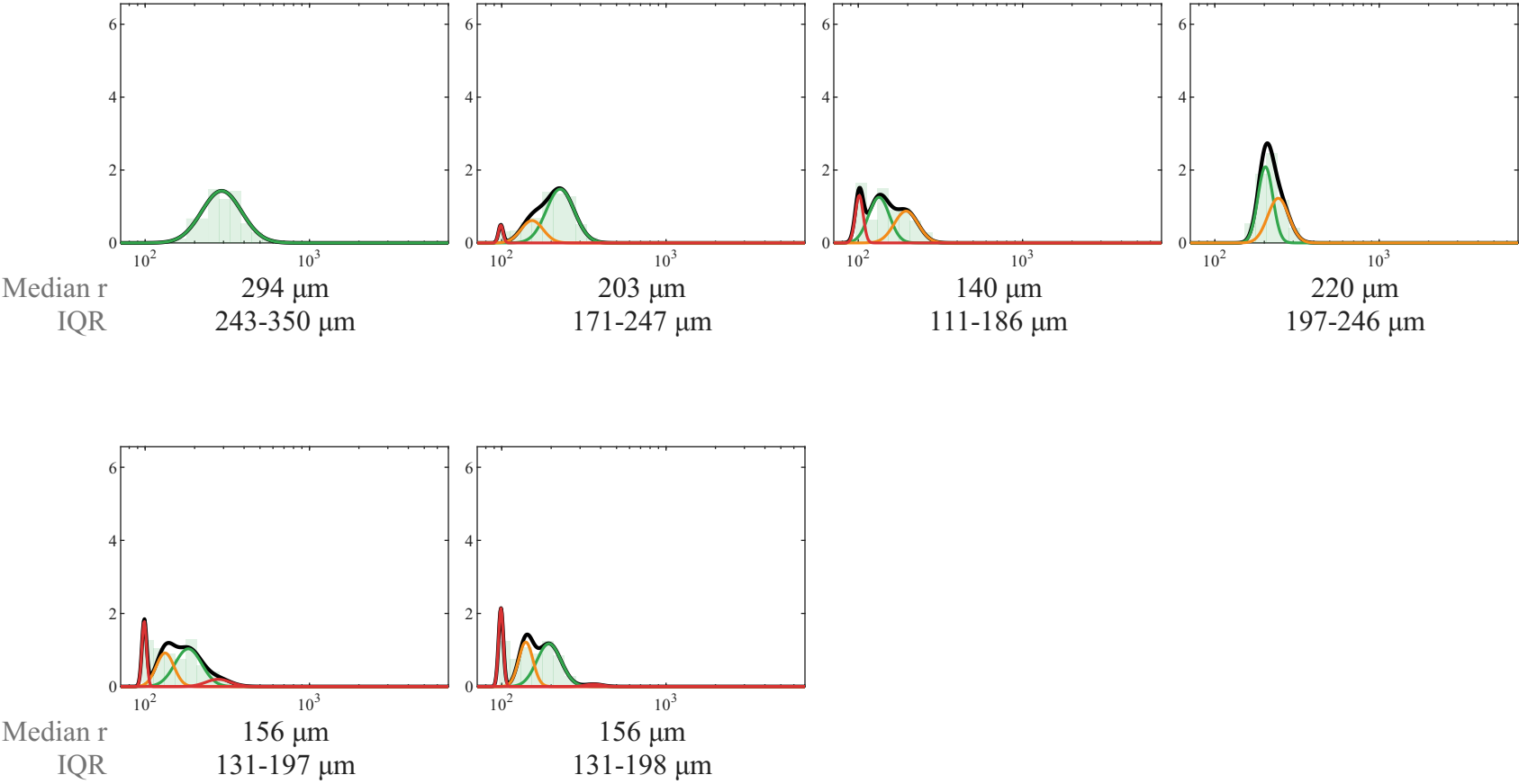


cdc13-2, Passage 5

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

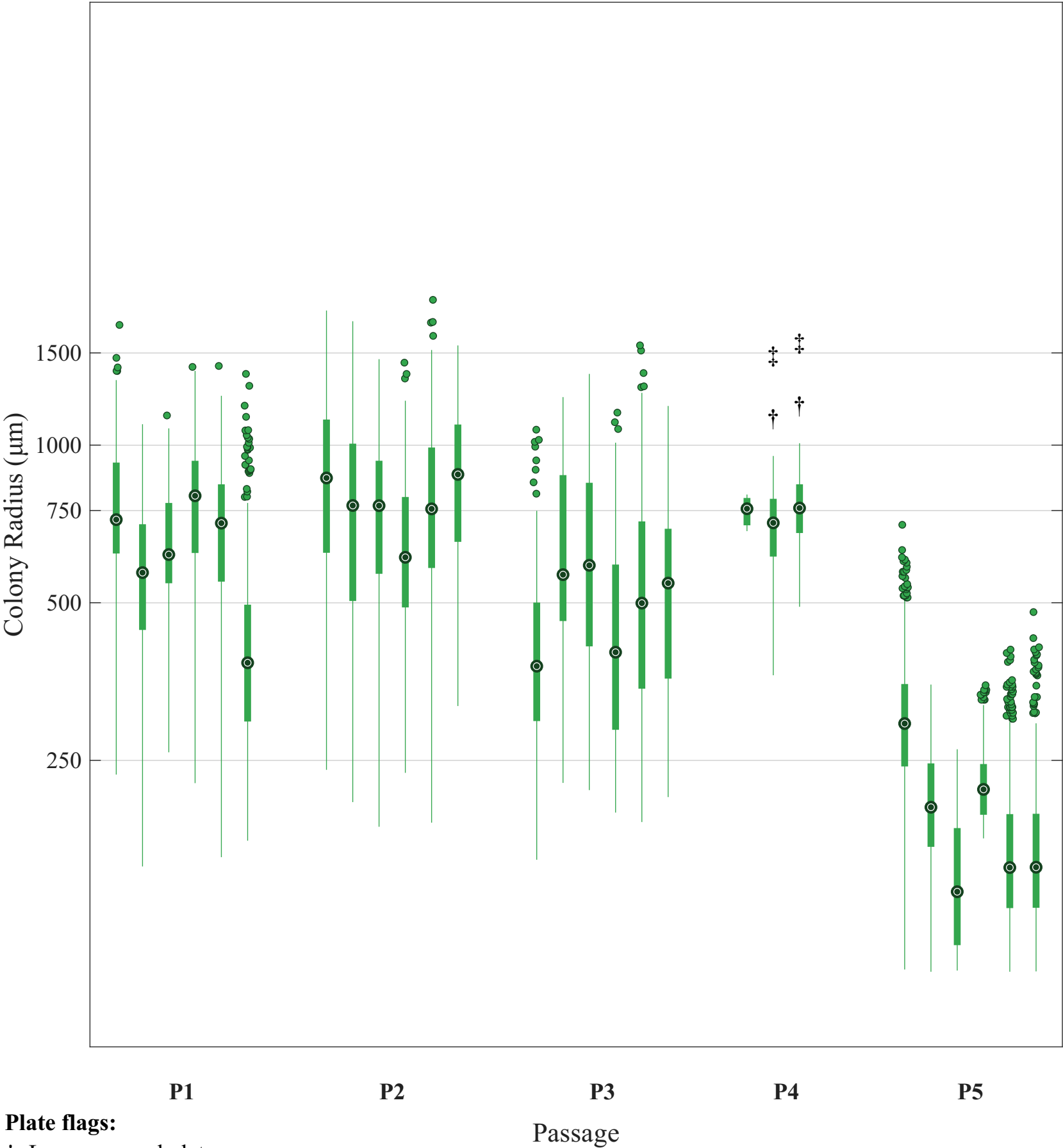
Std 6d



cdc13-2: Colony Radius by Plate, Across Passages

X axis: Passage, Y axis: Colony Radius μm

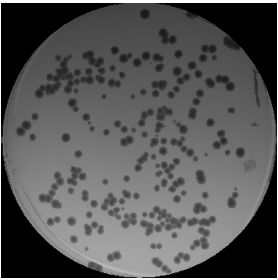
n	193	406	102	147	230	389	202	202	187	162	216	130	190	17	92	139	285	41	3	119	125	–	953	1447	706	2925	1242	1343
Median μm	721	571	618	800	709	384	865	767	766	611	755	879	378	566	589	402	499	545	756	710	758	–	294	203	140	220	156	156



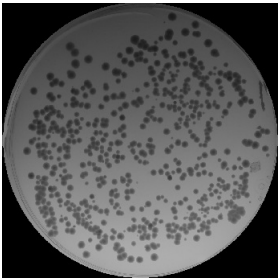
dbf4-1, cdc13-2, Passage 1

Original Source Images

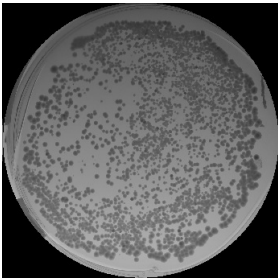
Std 6d



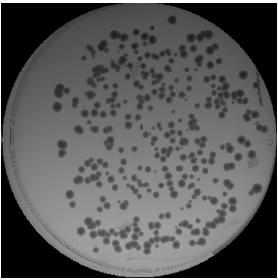
Colonies 77
Capture Ratio 30.0%
Rep A
HCE 3168
Plate 90 mm



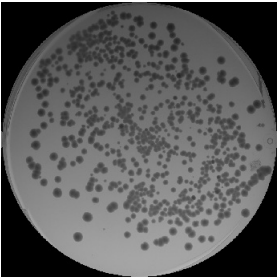
150
24.9%
A
15872
90 mm



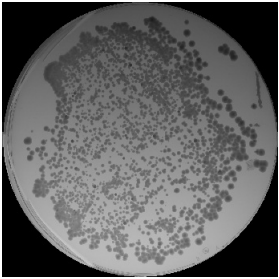
243
9.1%
A
79360
90 mm



109
30.7%
B
3168
90 mm



Colonies 159
Capture Ratio 24.3%
Rep B
HCE 15872
Plate 90 mm

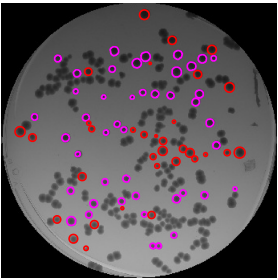


262
9.5%
B
79360
90 mm

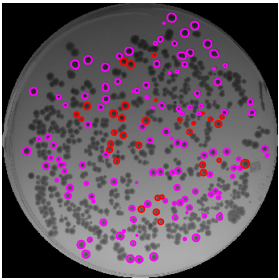
dbf4-1, cdc13-2, Passage 1

Source Images with Colony Overlays

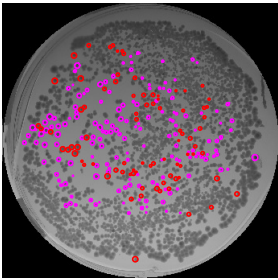
Std 6d



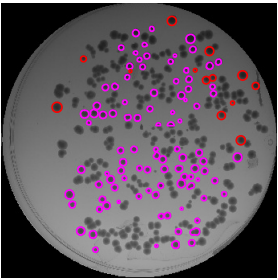
Colonies 77
Capture Ratio 30.0%
Rep A
HCE 3168
Plate 90 mm



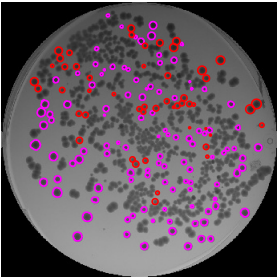
150
24.9%
A
15872
90 mm



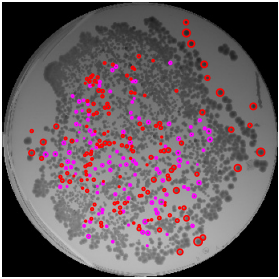
243
9.1%
A
79360
90 mm



109
30.7%
B
3168
90 mm



Colonies 159
Capture Ratio 24.3%
Rep B
HCE 15872
Plate 90 mm



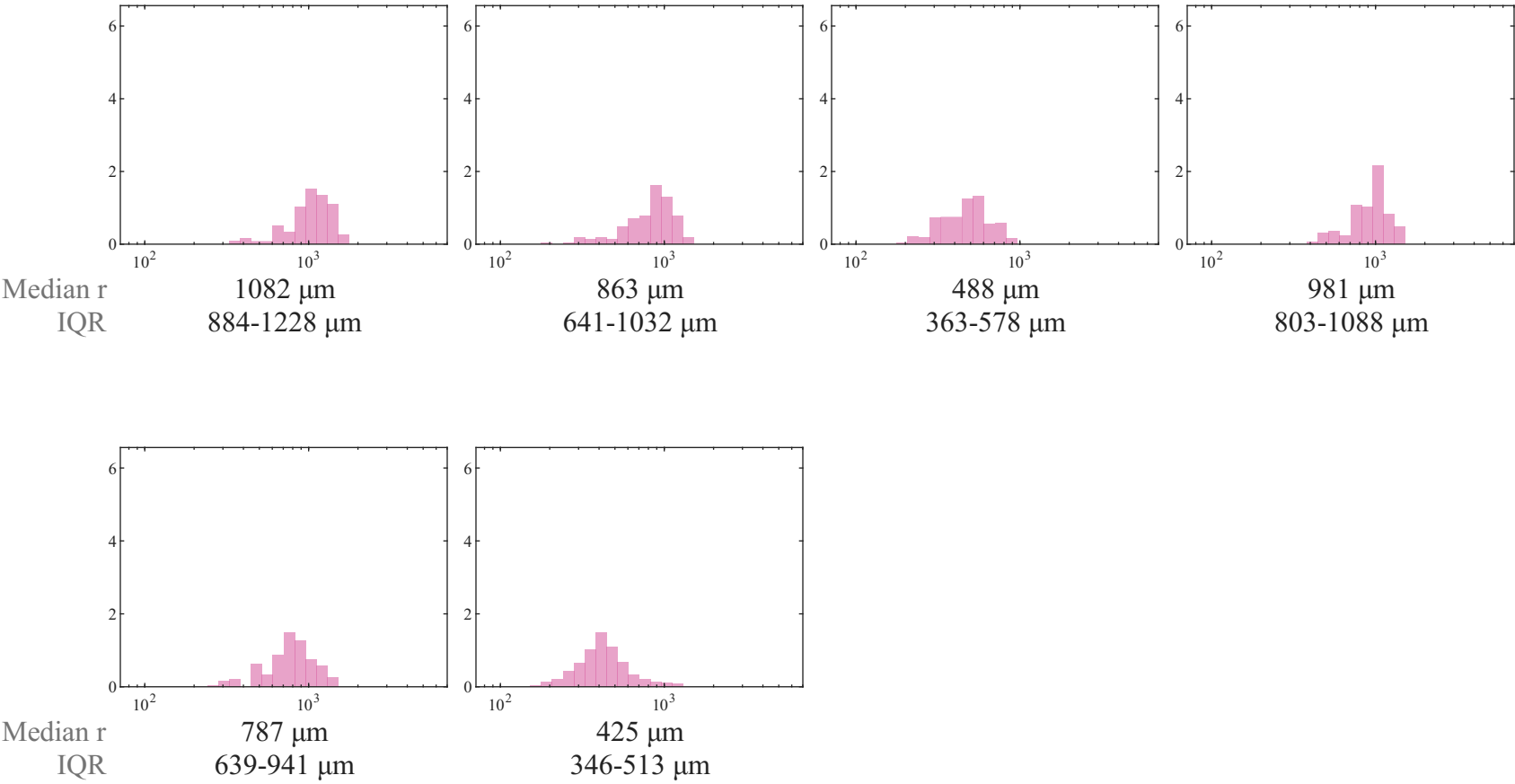
262
9.5%
B
79360
90 mm

dbf4-1, cdc13-2, Passage 1

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

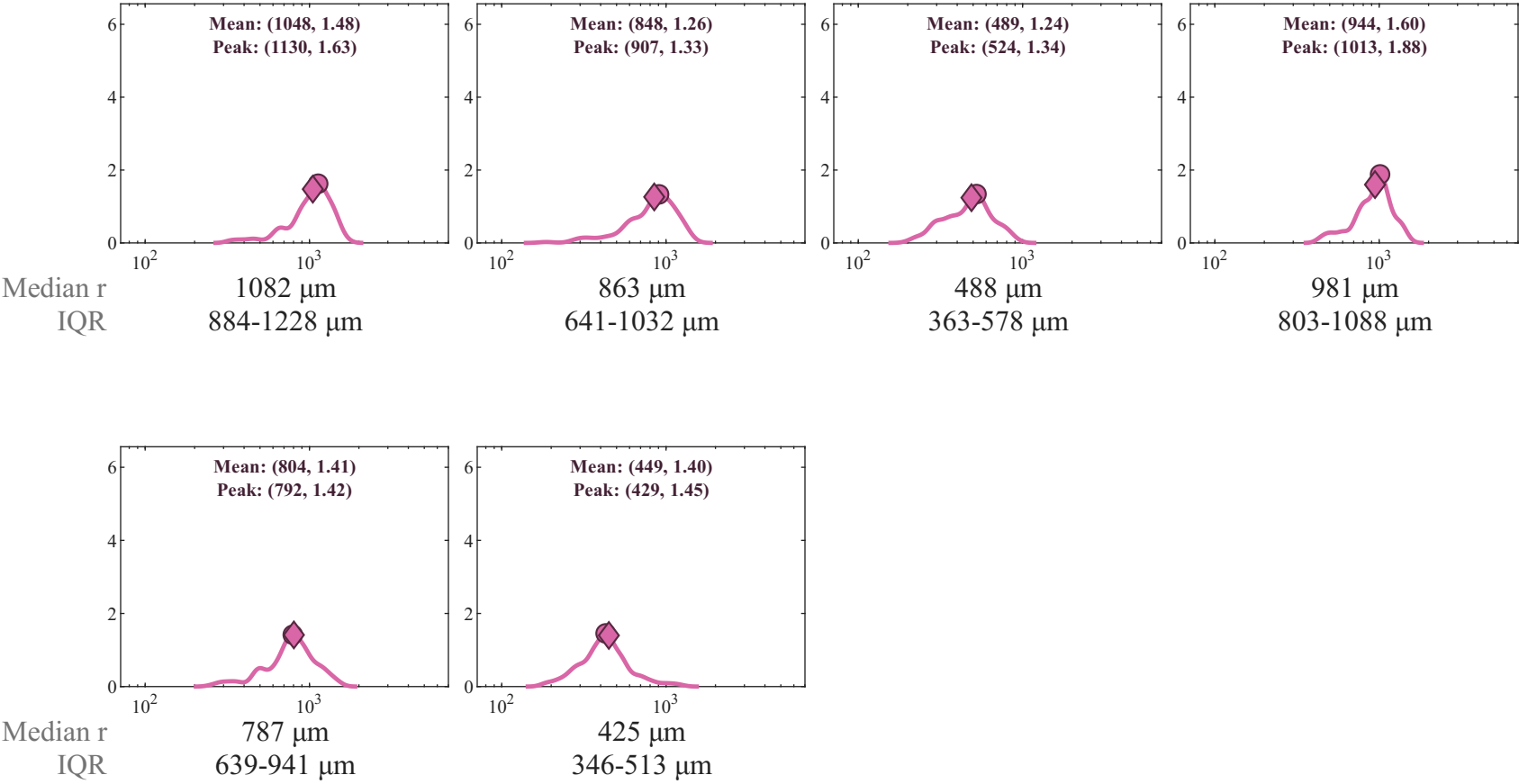


dbf4-1, cdc13-2, Passage 1

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

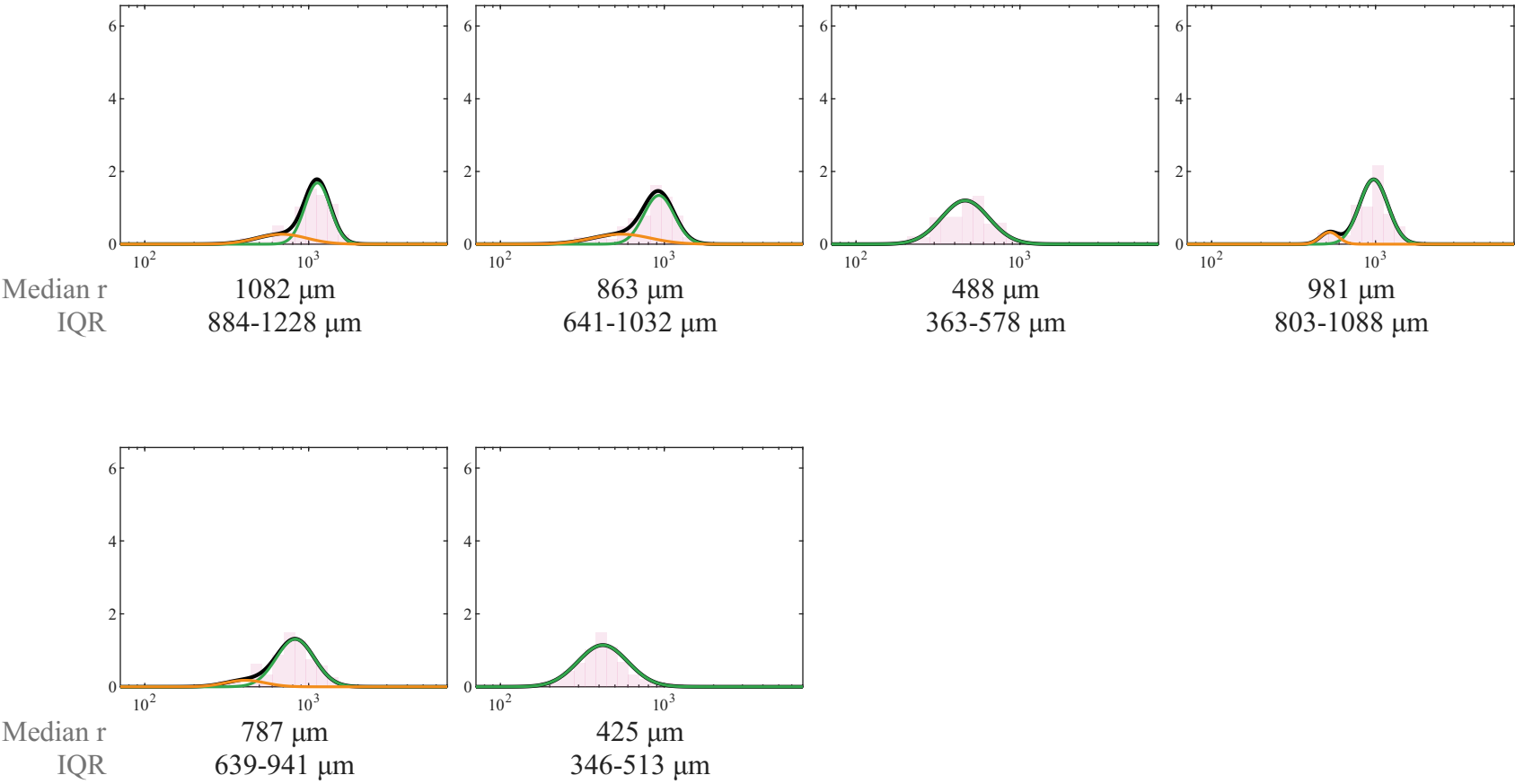


dbf4-1, cdc13-2, Passage 1

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

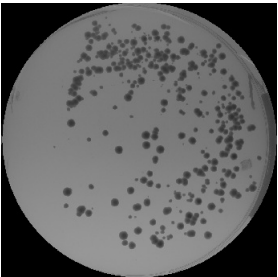
Std 6d



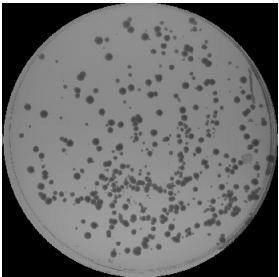
dbf4-1, cdc13-2, Passage 2

Original Source Images

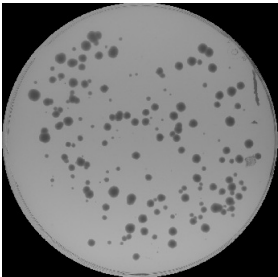
Std 6d



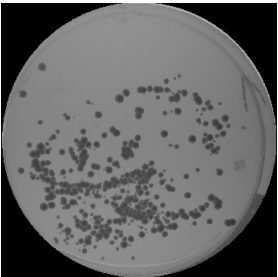
Colonies 144
Capture Ratio 36.1%
Rep A
HCE 144
Plate 90 mm



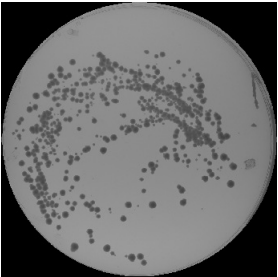
210
39.2%
A
752
90 mm



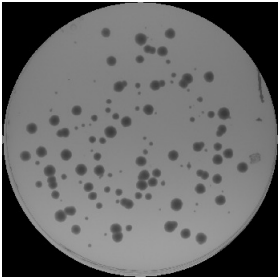
128
60.5%
A
3744
90 mm



104
21.5%
B
144
90 mm



Colonies 153
Capture Ratio 26.5%
Rep B
HCE 752
Plate 90 mm

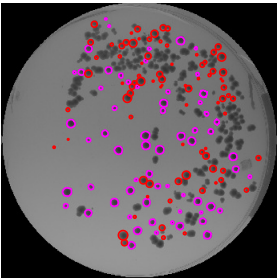


86
68.2%
B
3744
90 mm

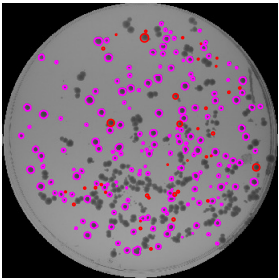
dbf4-1, cdc13-2, Passage 2

Source Images with Colony Overlays

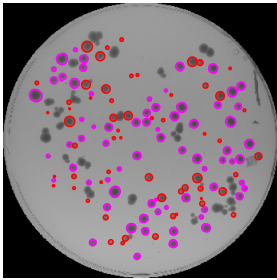
Std 6d



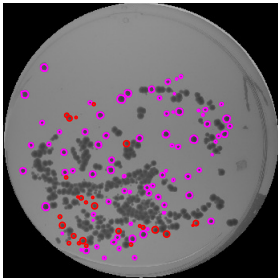
Colonies 144
Capture Ratio 36.1%
Rep A
HCE 144
Plate 90 mm



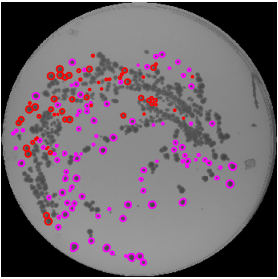
210
39.2%
A
752
90 mm



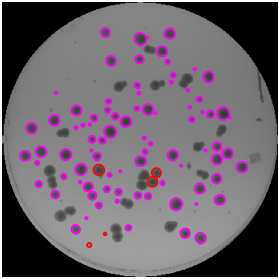
128
60.5%
A
3744
90 mm



104
21.5%
B
144
90 mm



Colonies 153
Capture Ratio 26.5%
Rep B
HCE 752
Plate 90 mm



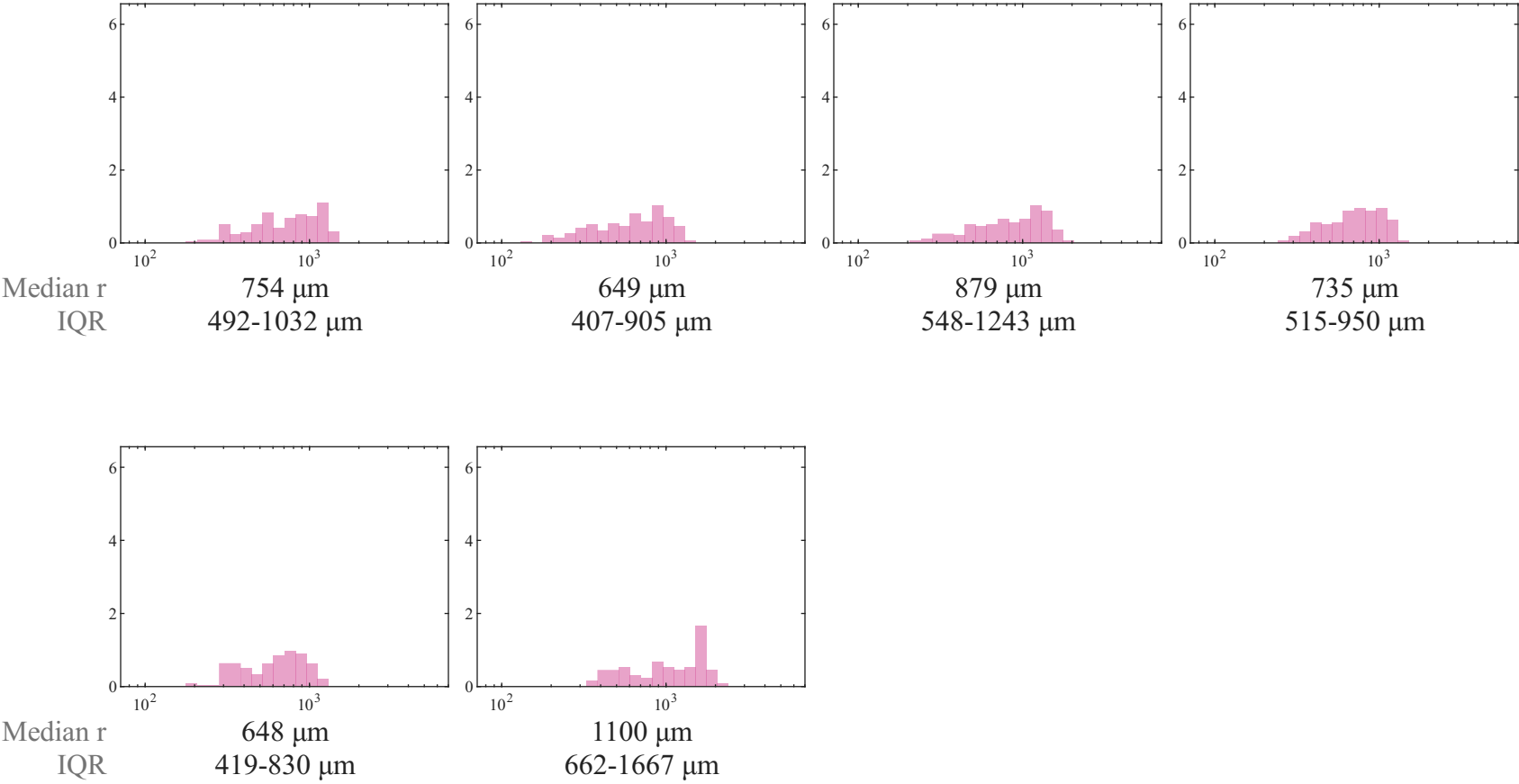
86
68.2%
B
3744
90 mm

dbf4-1, cdc13-2, Passage 2

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

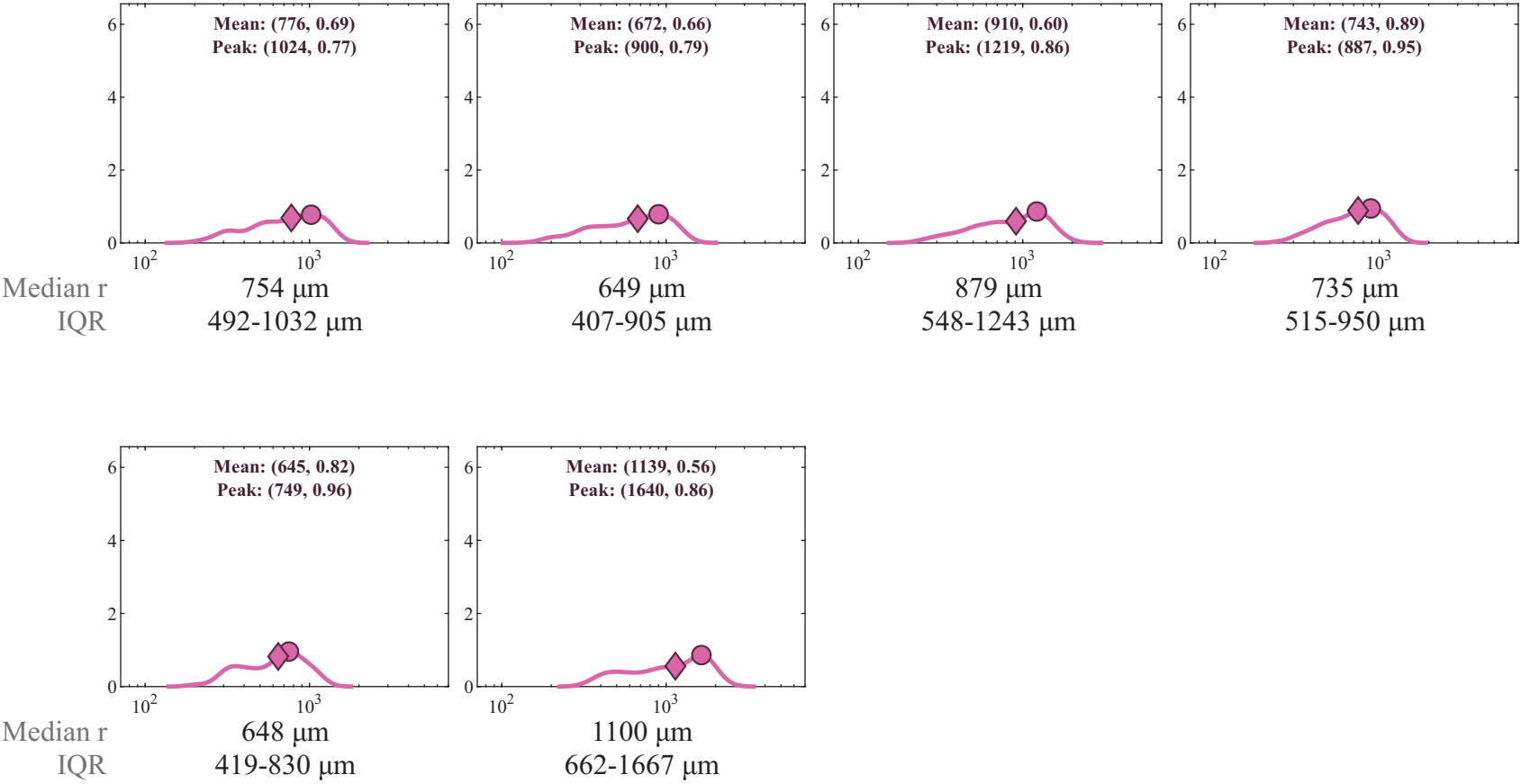


dbf4-1, cdc13-2, Passage 2

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

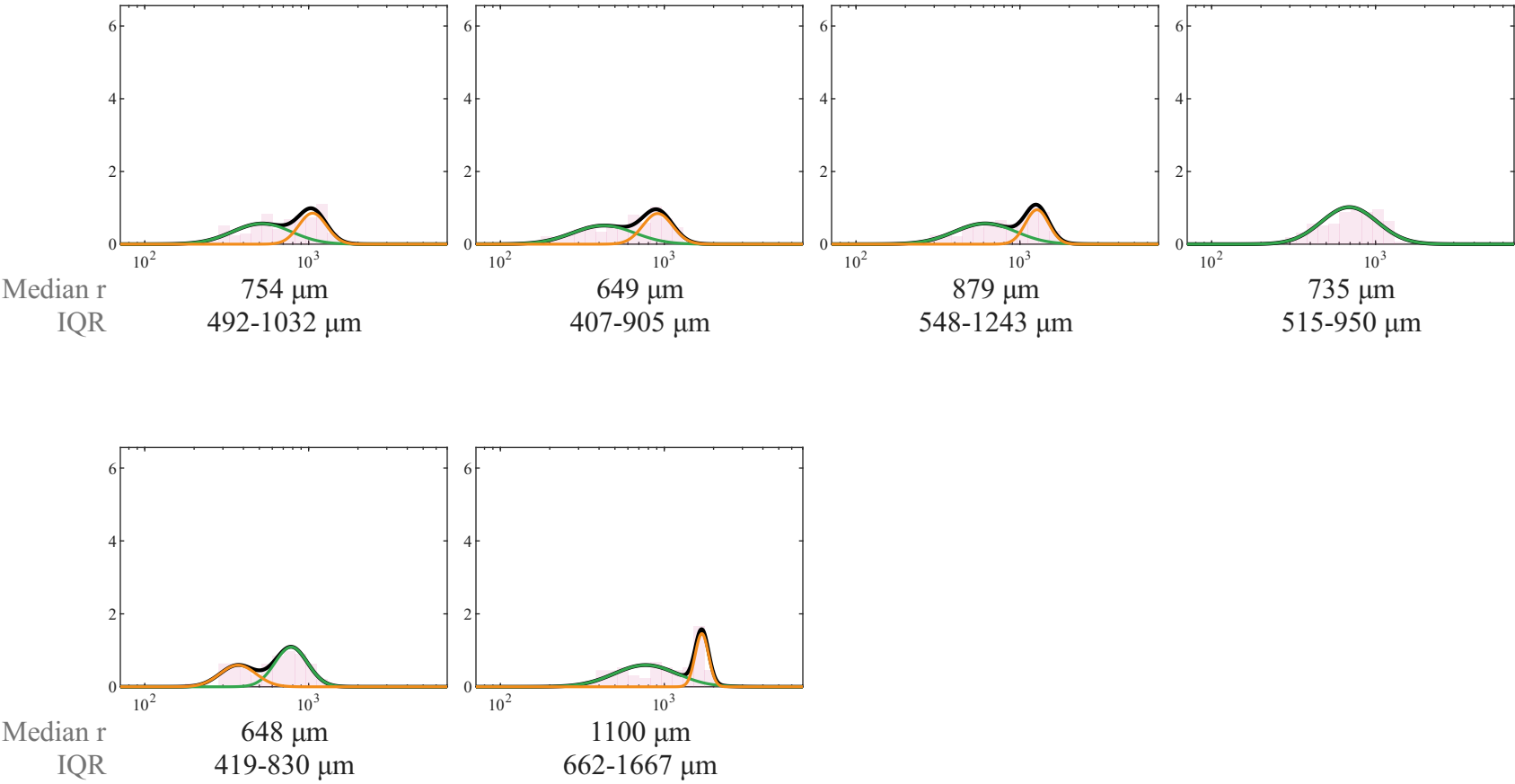


dbf4-1, cdc13-2, Passage 2

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

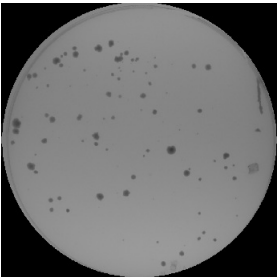
Std 6d



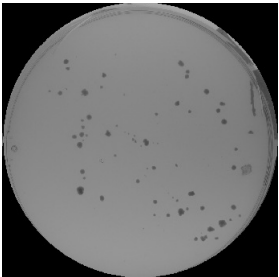
dbf4-1, cdc13-2, Passage 3

Original Source Images

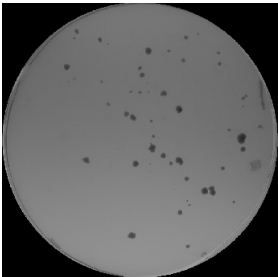
Std 6d



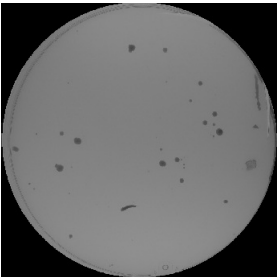
Colonies 64
Capture Ratio 74.3%
Rep A
HCE 816
Plate 90 mm



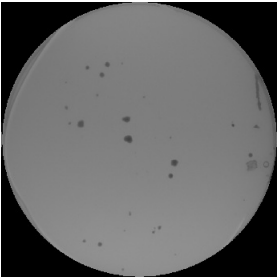
56
30.5%
A
816
90 mm



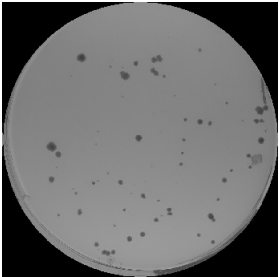
35
56.1%
A
4096
90 mm



19
37.9%
B
160
90 mm



Colonies 15
Capture Ratio 35.9%
Rep B
HCE 160
Plate 90 mm

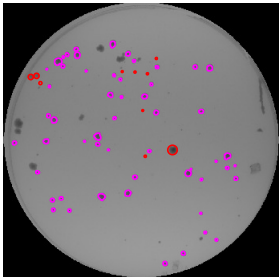


32
36.8%
B
4096
90 mm

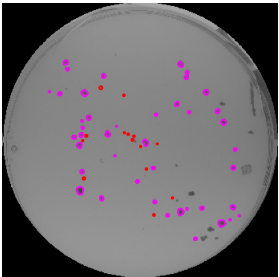
dbf4-1, cdc13-2, Passage 3

Source Images with Colony Overlays

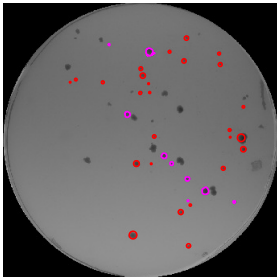
Std 6d



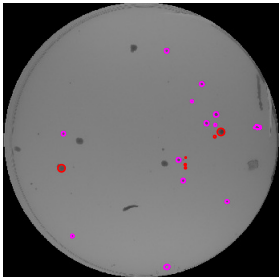
Colonies 64
Capture Ratio 74.3%
Rep A
HCE 816
Plate 90 mm



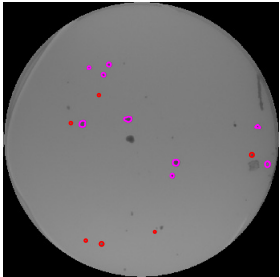
56
30.5%
A
816
90 mm



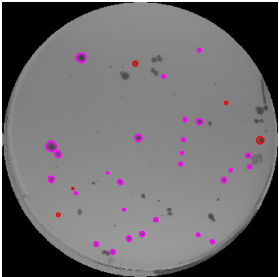
35
56.1%
A
4096
90 mm



19
37.9%
B
160
90 mm



Colonies 15
Capture Ratio 35.9%
Rep B
HCE 160
Plate 90 mm



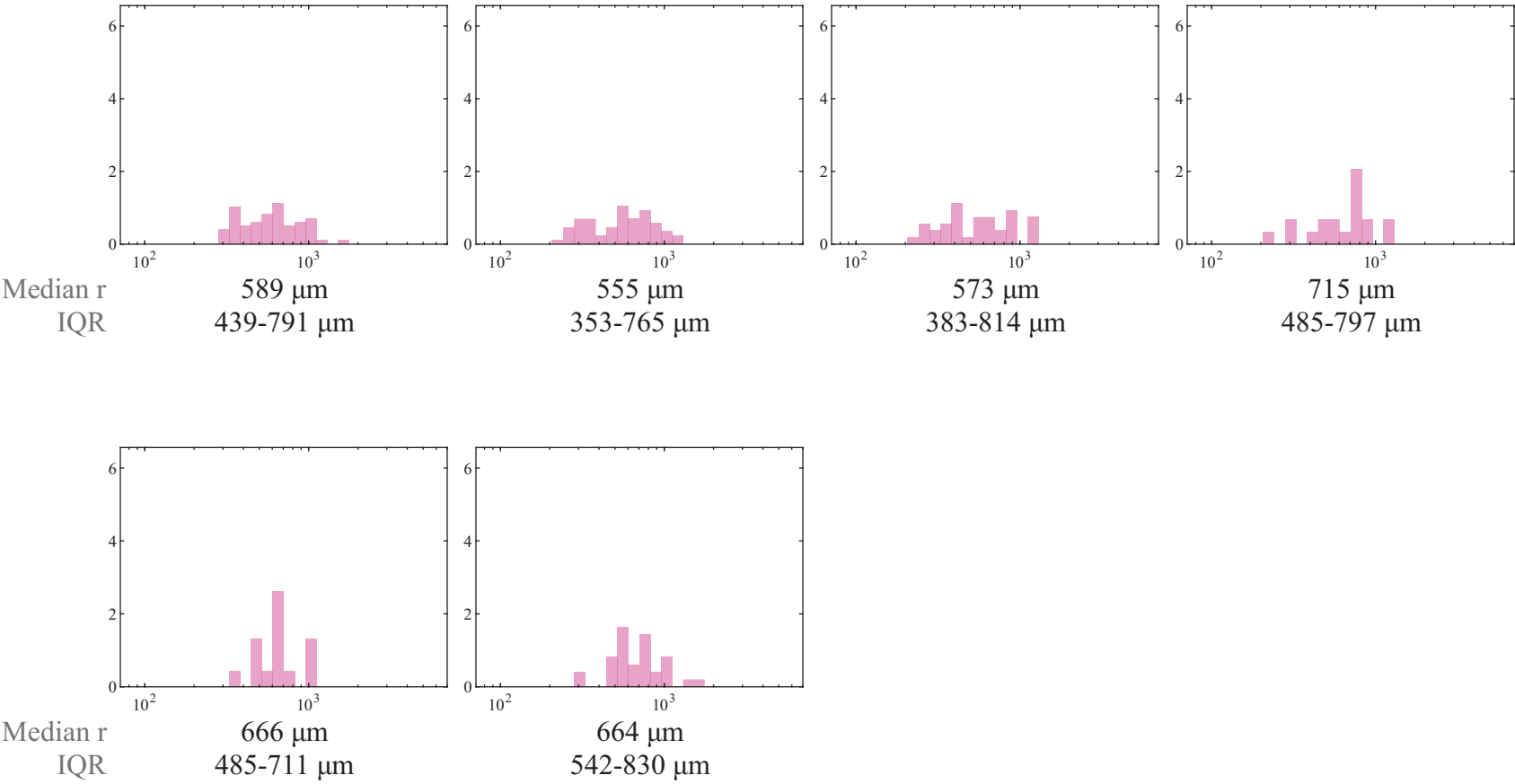
32
36.8%
B
4096
90 mm

dbf4-1, cdc13-2, Passage 3

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

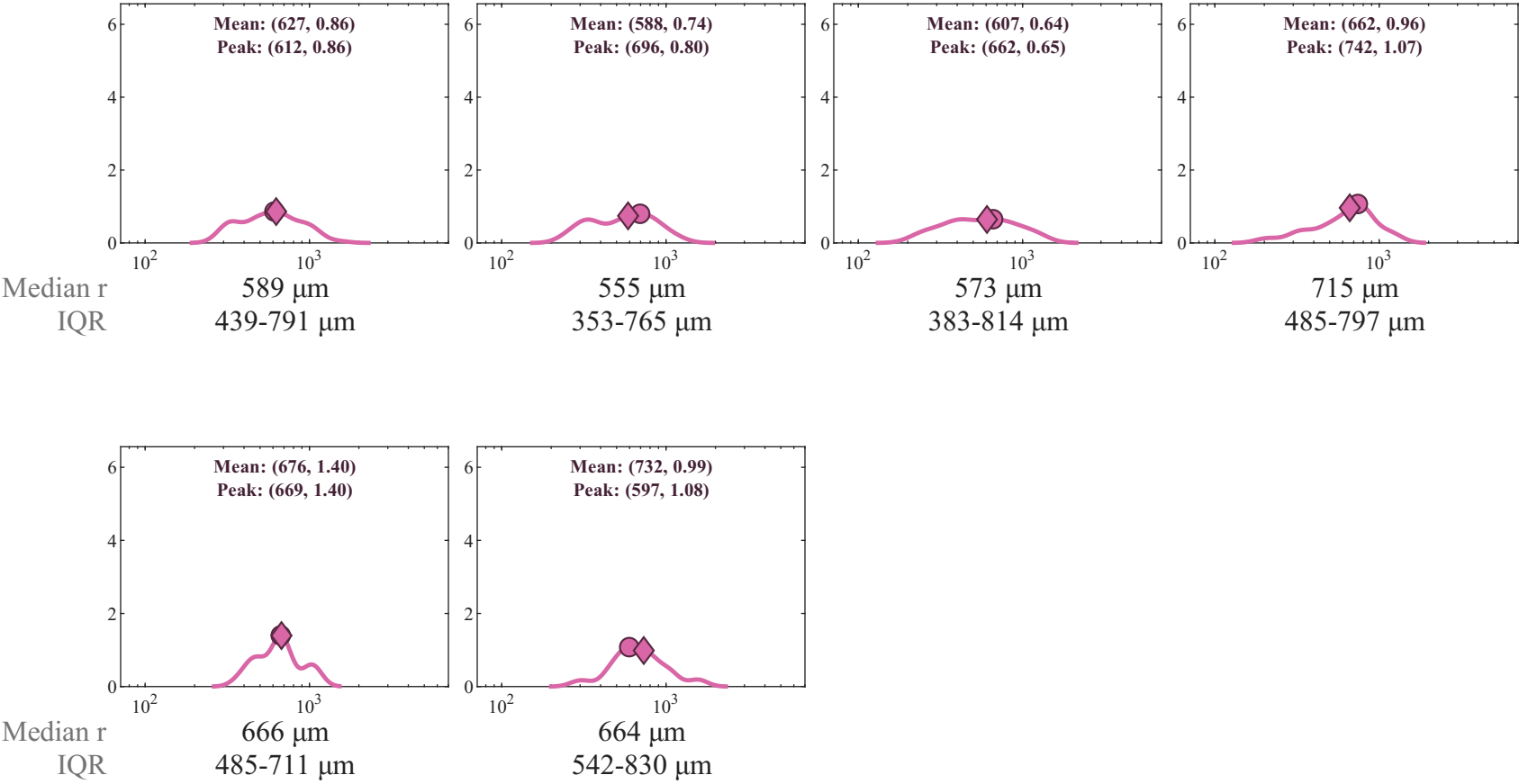


dbf4-1, cdc13-2, Passage 3

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

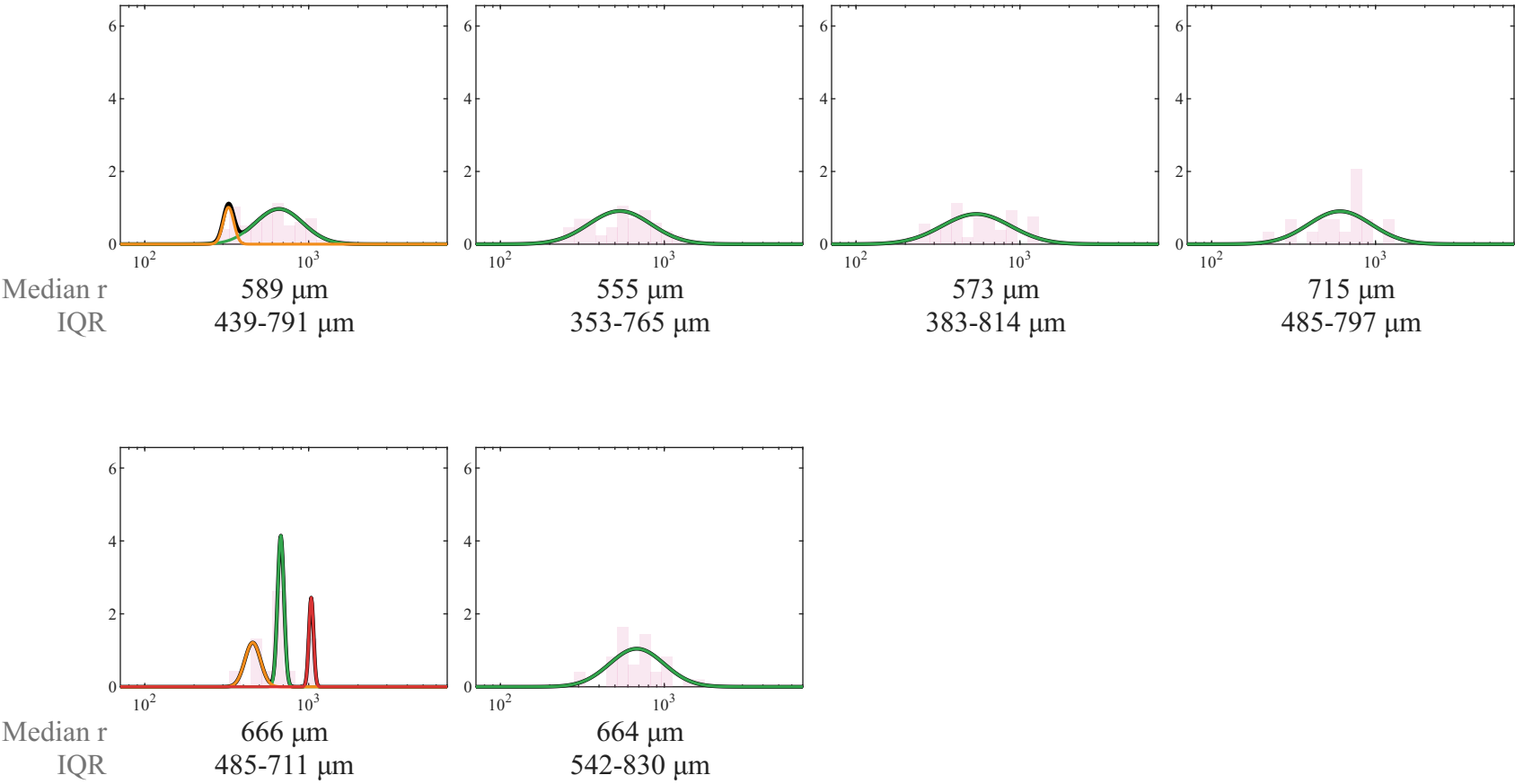


dbf4-1, cdc13-2, Passage 3

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

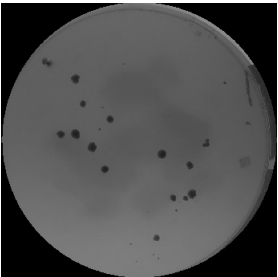
Std 6d



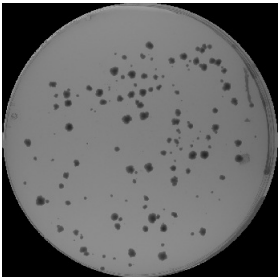
dbf4-1, cdc13-2, Passage 4

Original Source Images

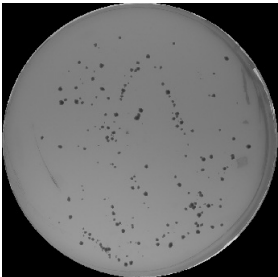
Std 6d



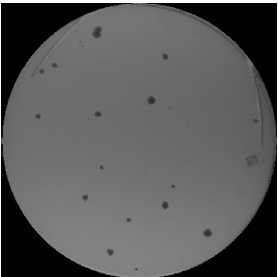
Colonies 18
Capture Ratio 27.1%
Rep A
HCE 176
Plate 90 mm



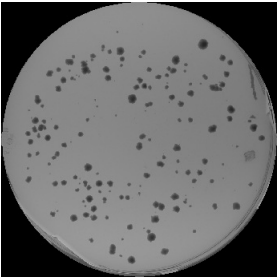
110
60.3%
A
848
90 mm



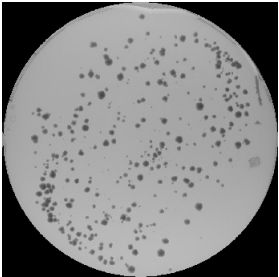
144
54.9%
A
4256
90 mm



13
45.2%
B
176
90 mm

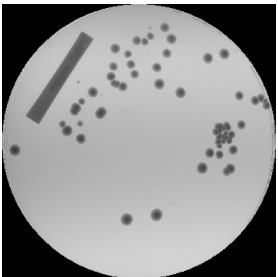


Colonies 155
Capture Ratio 68.7%
Rep B
HCE 848
Plate 90 mm



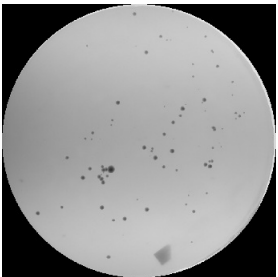
252
65.9%
B
4256
90 mm

LgSrc 6d

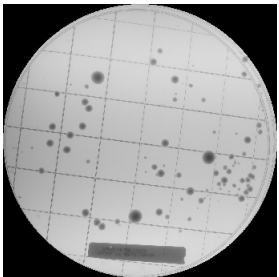


Colonies 33
Capture Ratio 46.2%
Rep —
HCE 928
Plate 140 mm

SmSrc 6d



Colonies 70
Capture Ratio 55.8%
Rep A
HCE 656
Plate 140 mm

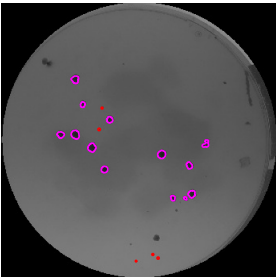


74
39.9%
B
656
140 mm

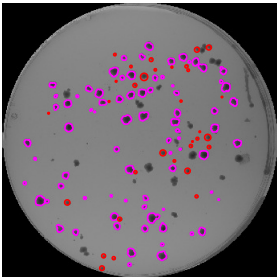
dbf4-1, cdc13-2, Passage 4

Source Images with Colony Overlays

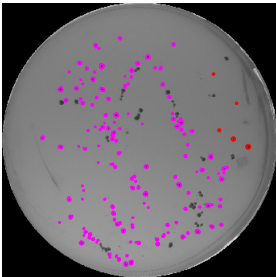
Std 6d



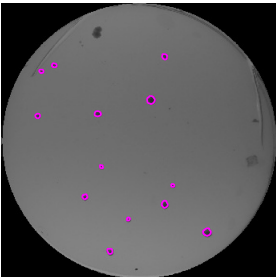
Colonies 18
Capture Ratio 27.1%
Rep A
HCE 176
Plate 90 mm



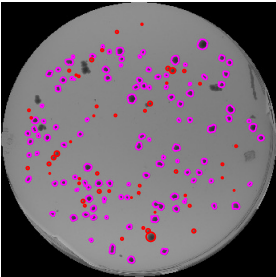
110
60.3%
A
848
90 mm



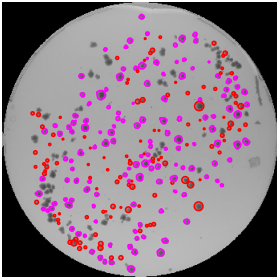
144
54.9%
A
4256
90 mm



13
45.2%
B
176
90 mm

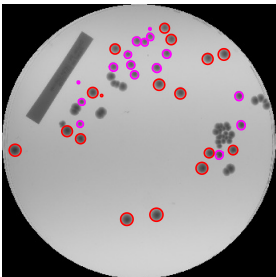


Colonies 155
Capture Ratio 68.7%
Rep B
HCE 848
Plate 90 mm



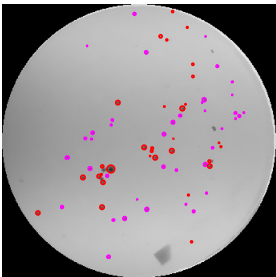
252
65.9%
B
4256
90 mm

LgSrc 6d

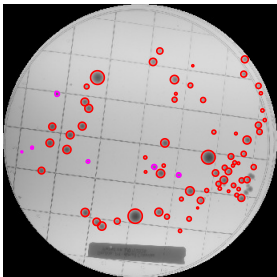


Colonies 33
Capture Ratio 46.2%
Rep —
HCE 928
Plate 140 mm

SmSrc 6d



Colonies 70
Capture Ratio 55.8%
Rep A
HCE 656
Plate 140 mm



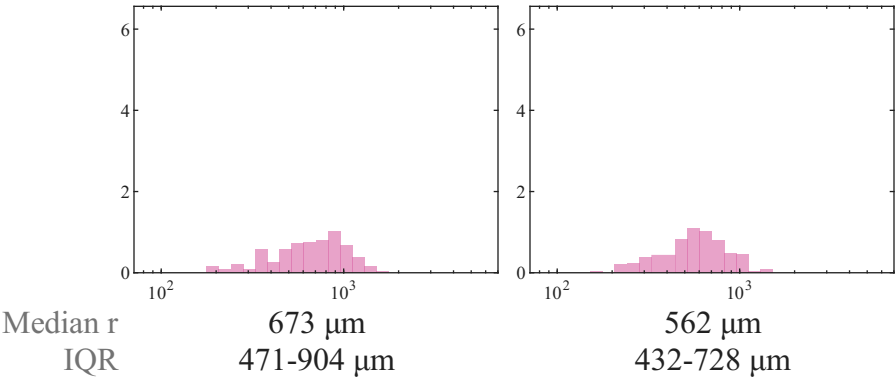
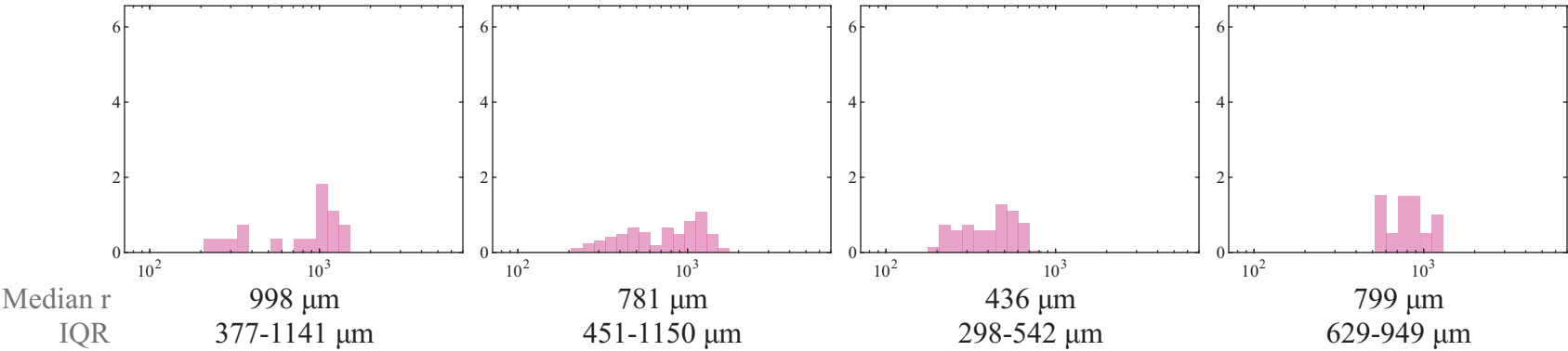
74
39.9%
B
656
140 mm

dbf4-1, cdc13-2, Passage 4

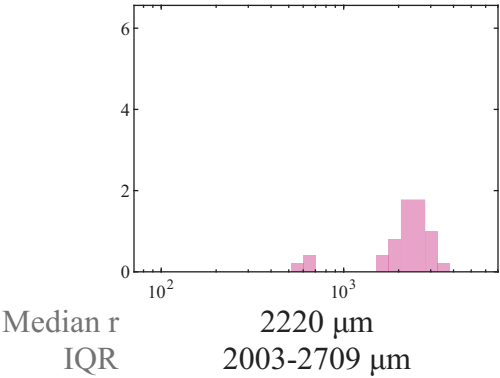
Density Histograms

X axis: Colony Radius μm , Y axis: Density

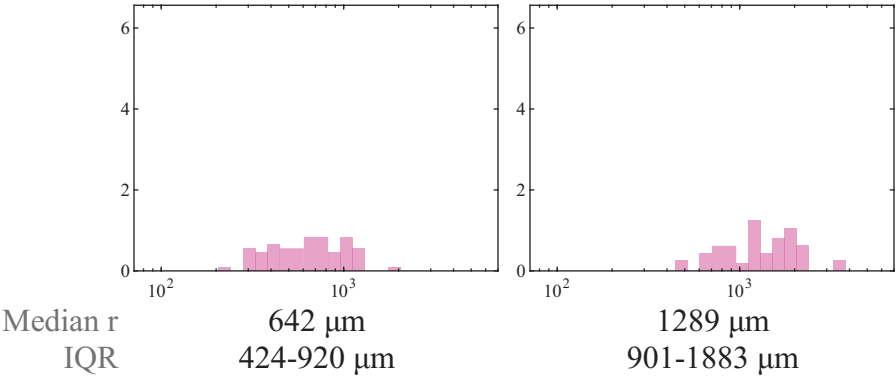
Std 6d



LgSrc 6d



SmSrc 6d

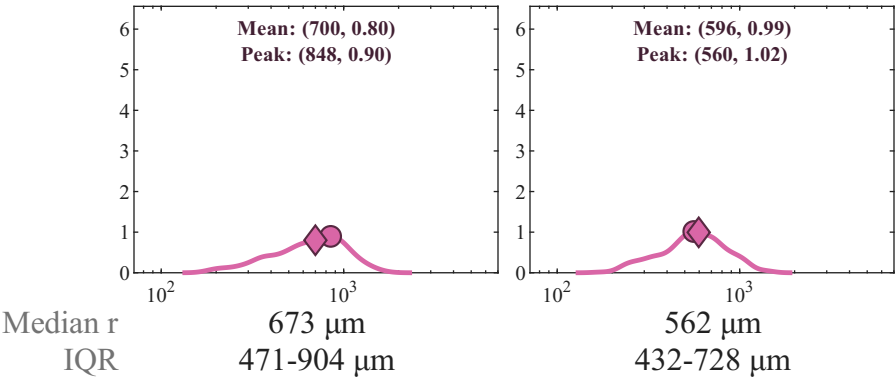
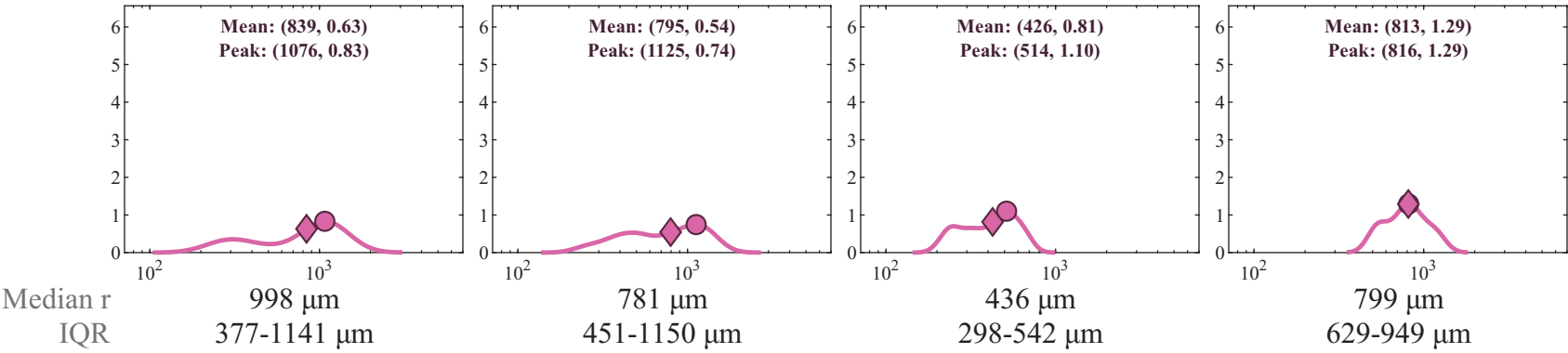


dbf4-1, cdc13-2, Passage 4

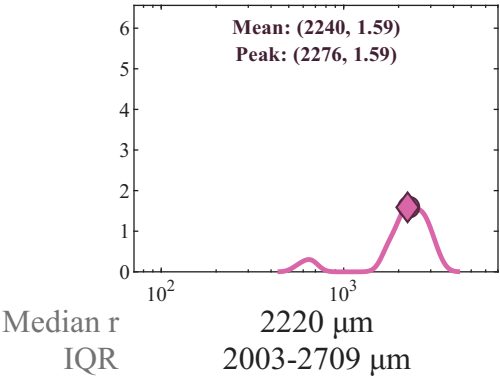
Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

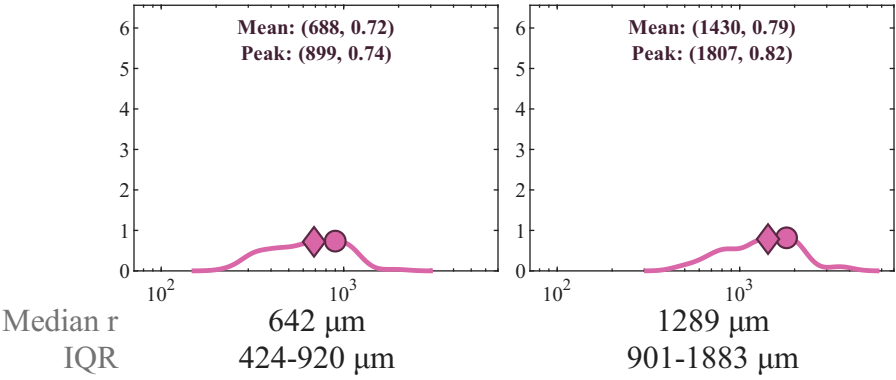
Std 6d



LgSrc 6d



SmSrc 6d

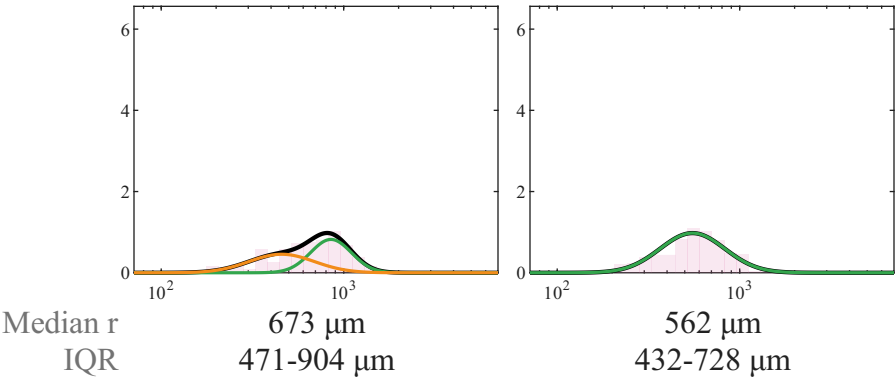
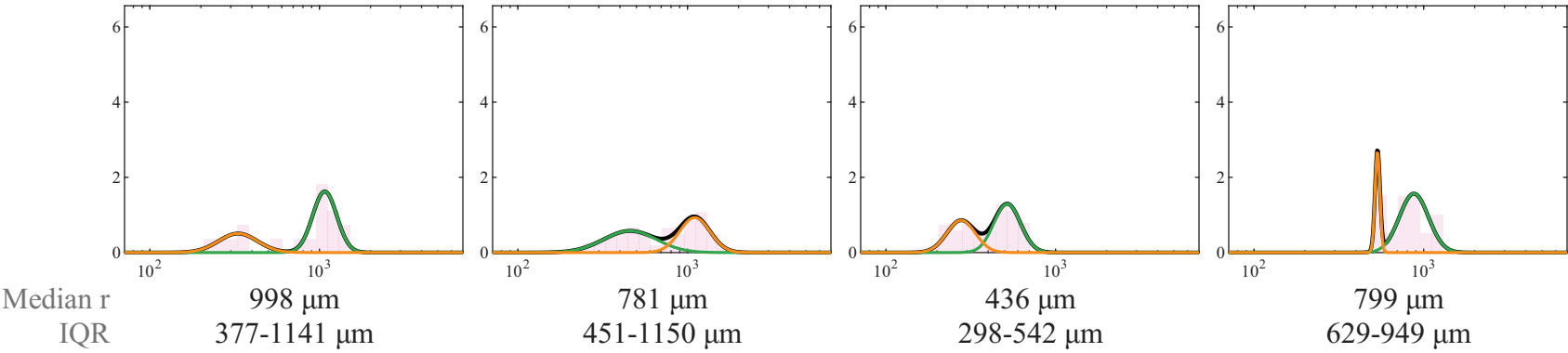


dbf4-1, cdc13-2, Passage 4

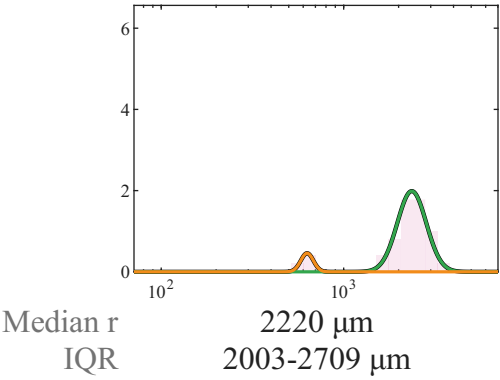
LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

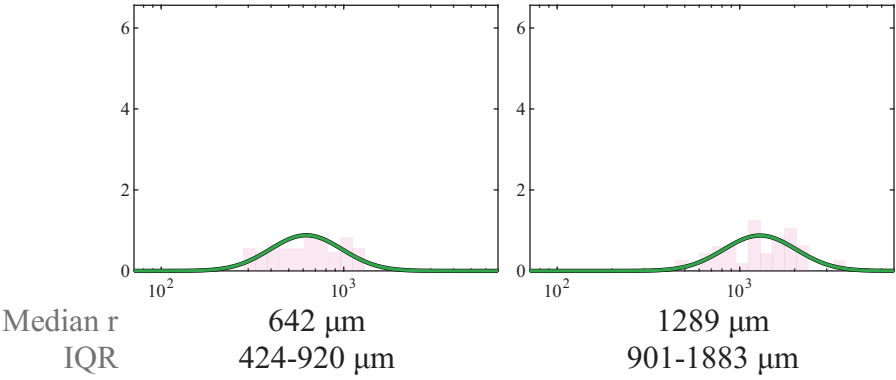
Std 6d



LgSrc 6d



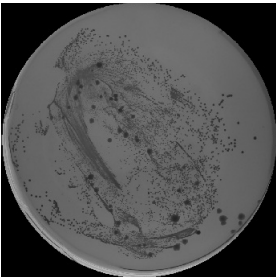
SmSrc 6d



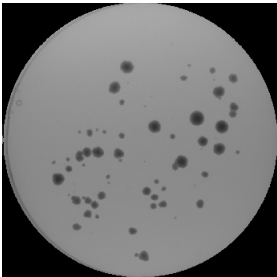
dbf4-1, cdc13-2, Passage 5

Original Source Images

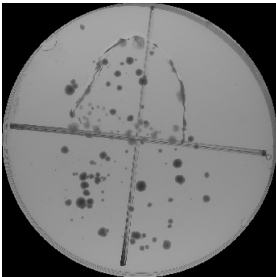
Std 6d



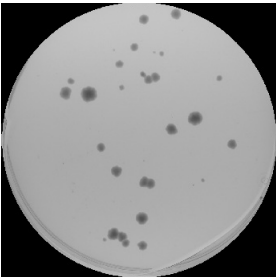
Colonies 378
Capture Ratio 8.5%
Rep A
HCE 144
Plate 90 mm



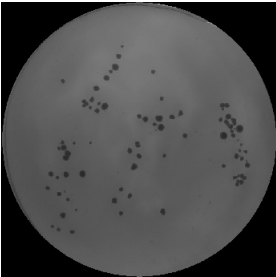
55
83.7%
A
688
90 mm



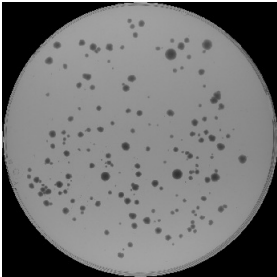
62
28.0%
A
3456
90 mm



18
68.6%
B
144
90 mm



Colonies 57
Capture Ratio 64.3%
Rep B
HCE 688
Plate 90 mm

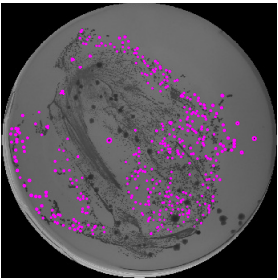


167
72.2%
B
3456
90 mm

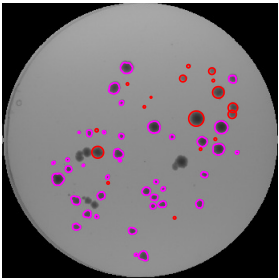
dbf4-1, cdc13-2, Passage 5

Source Images with Colony Overlays

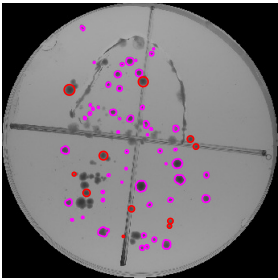
Std 6d



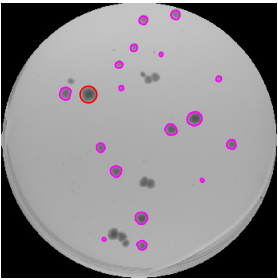
Colonies 378
Capture Ratio 8.5%
Rep A
HCE 144
Plate 90 mm



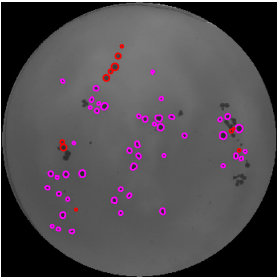
55
83.7%
A
688
90 mm



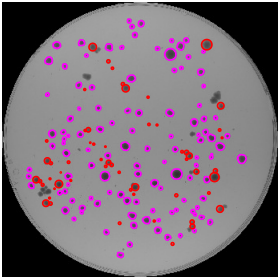
62
28.0%
A
3456
90 mm



18
68.6%
B
144
90 mm



Colonies 57
Capture Ratio 64.3%
Rep B
HCE 688
Plate 90 mm



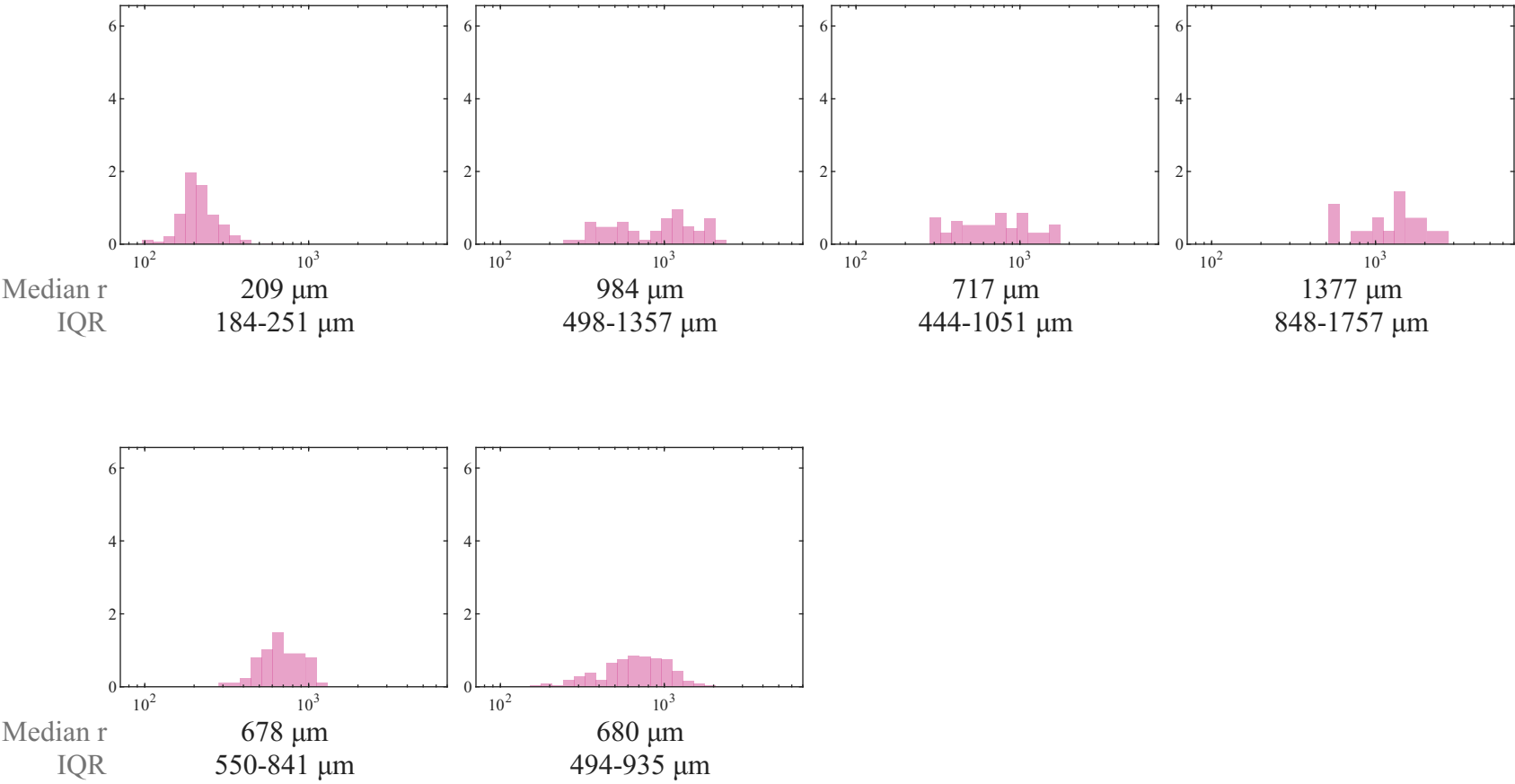
167
72.2%
B
3456
90 mm

dbf4-1, cdc13-2, Passage 5

Density Histograms

X axis: Colony Radius μm , Y axis: Density

Std 6d

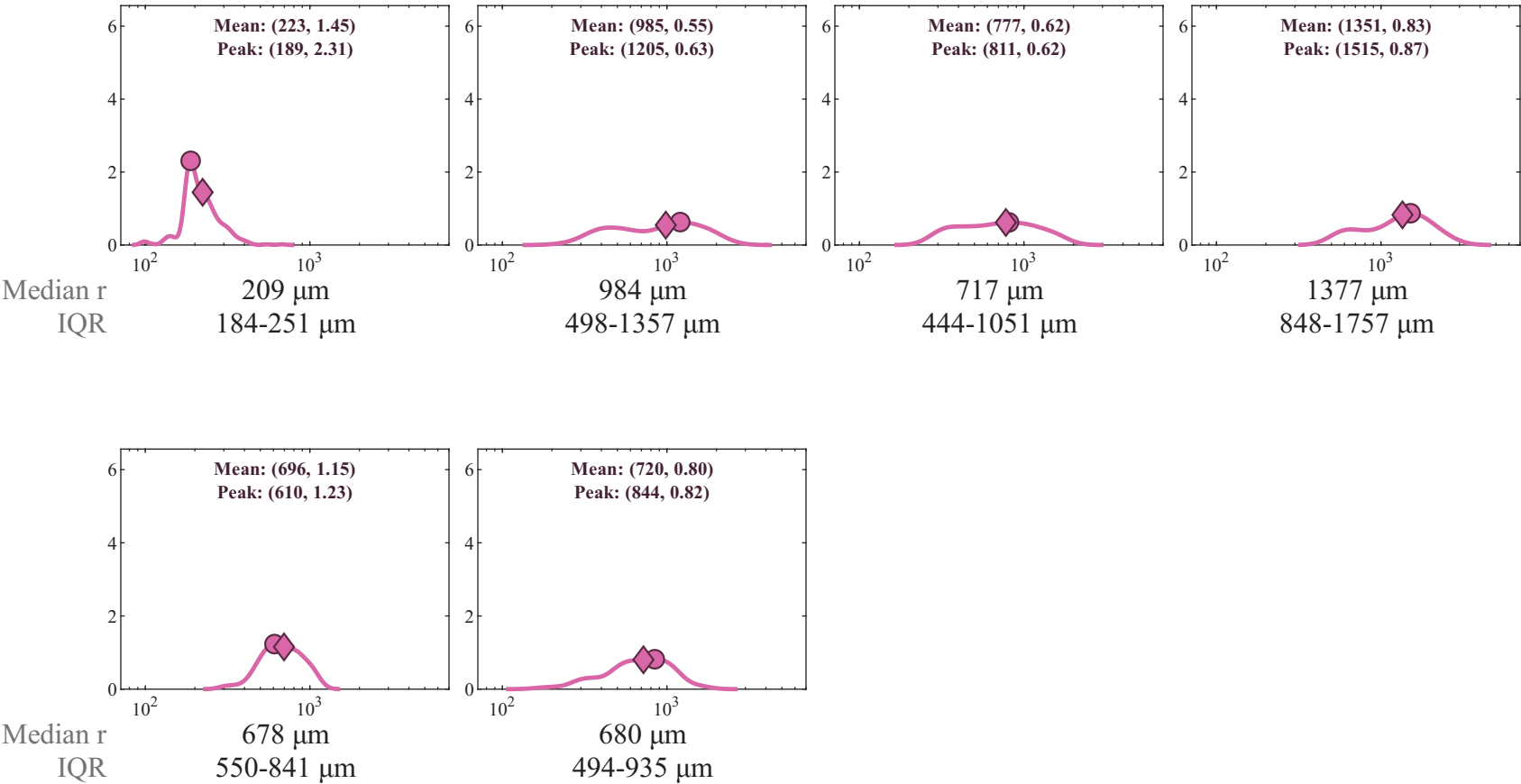


dbf4-1, cdc13-2, Passage 5

Kernel Density Estimates

X axis: Colony Radius μm , Y axis: Density

Std 6d

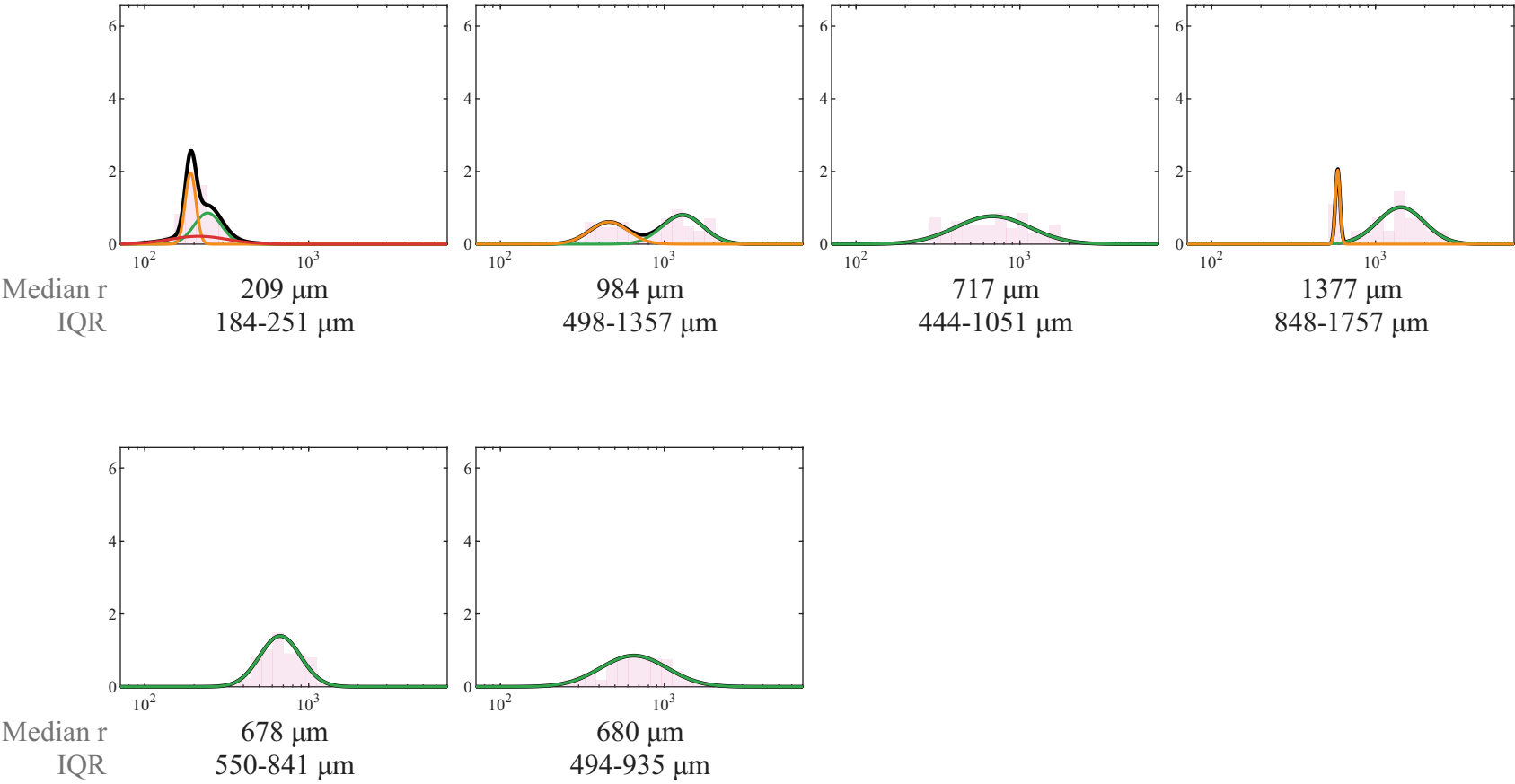


dbf4-1, cdc13-2, Passage 5

LogNormal Mixture Fits

X axis: Colony Radius μm , Y axis: Density

Std 6d



dbf4-1, cdc13-2: Colony Radius by Plate, Across Passages

X axis: Passage, Y axis: Colony Radius μm

n	77	150	243	109	159	262	144	210	128	104	153	86	64	56	35	19	15	32	18	110	144	13	155	252	33	70	74	378	55	62	18	57	167	
Median μm	1088	863	488	981	178	742	5	464	98	797	356	481	100	589	555	573	715	666	664	998	781	436	799	673	562	222	642	1289	209	984	717	1376	786	80

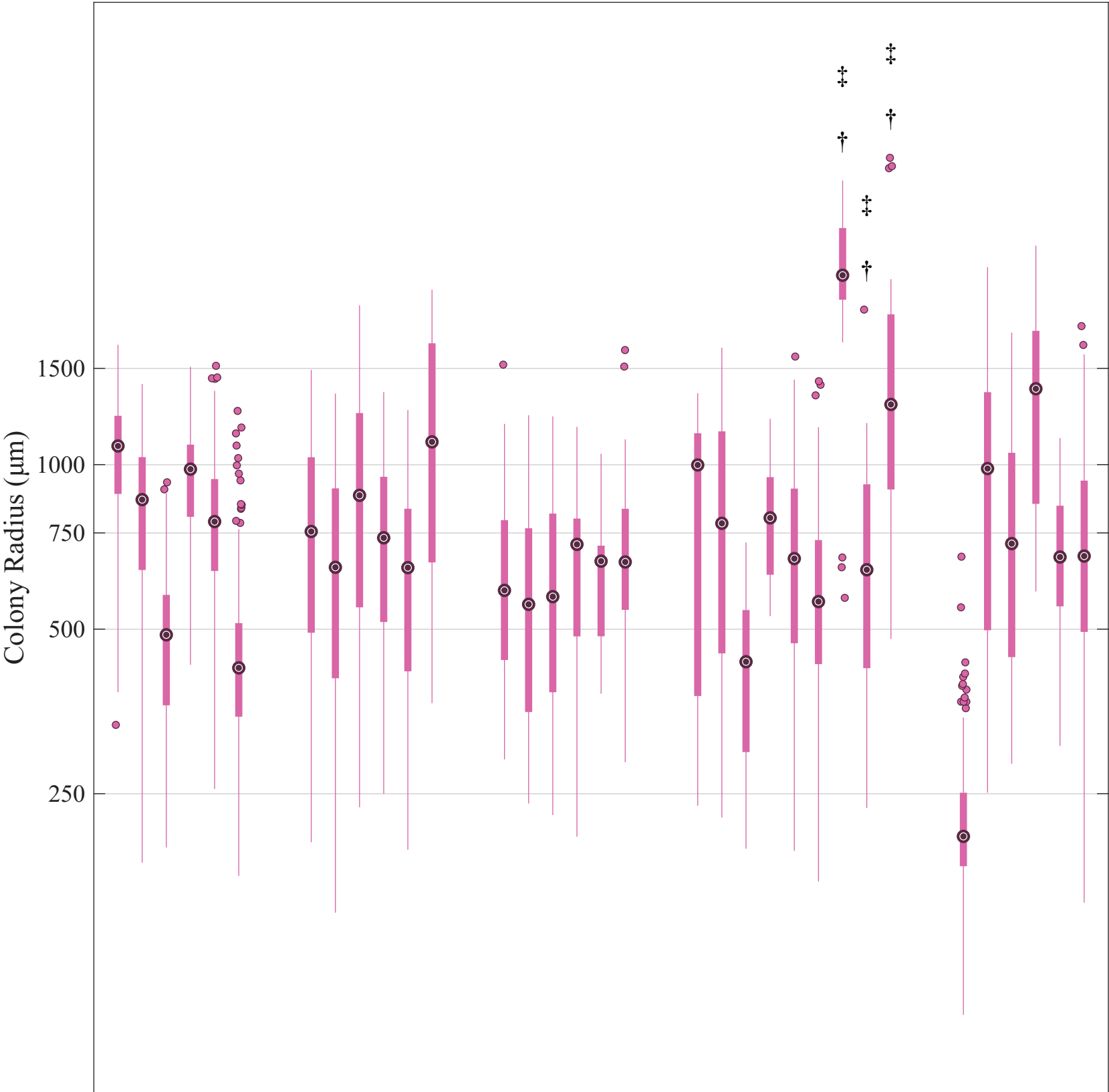


Plate flags: P1 P2 P3 P4 P5

† Large-sourced plate

‡ 140 mm plate

† Non-standard enrichment (SmallSourced)

Passage

Subpopulation Analysis Across Selected Plates

Component weight

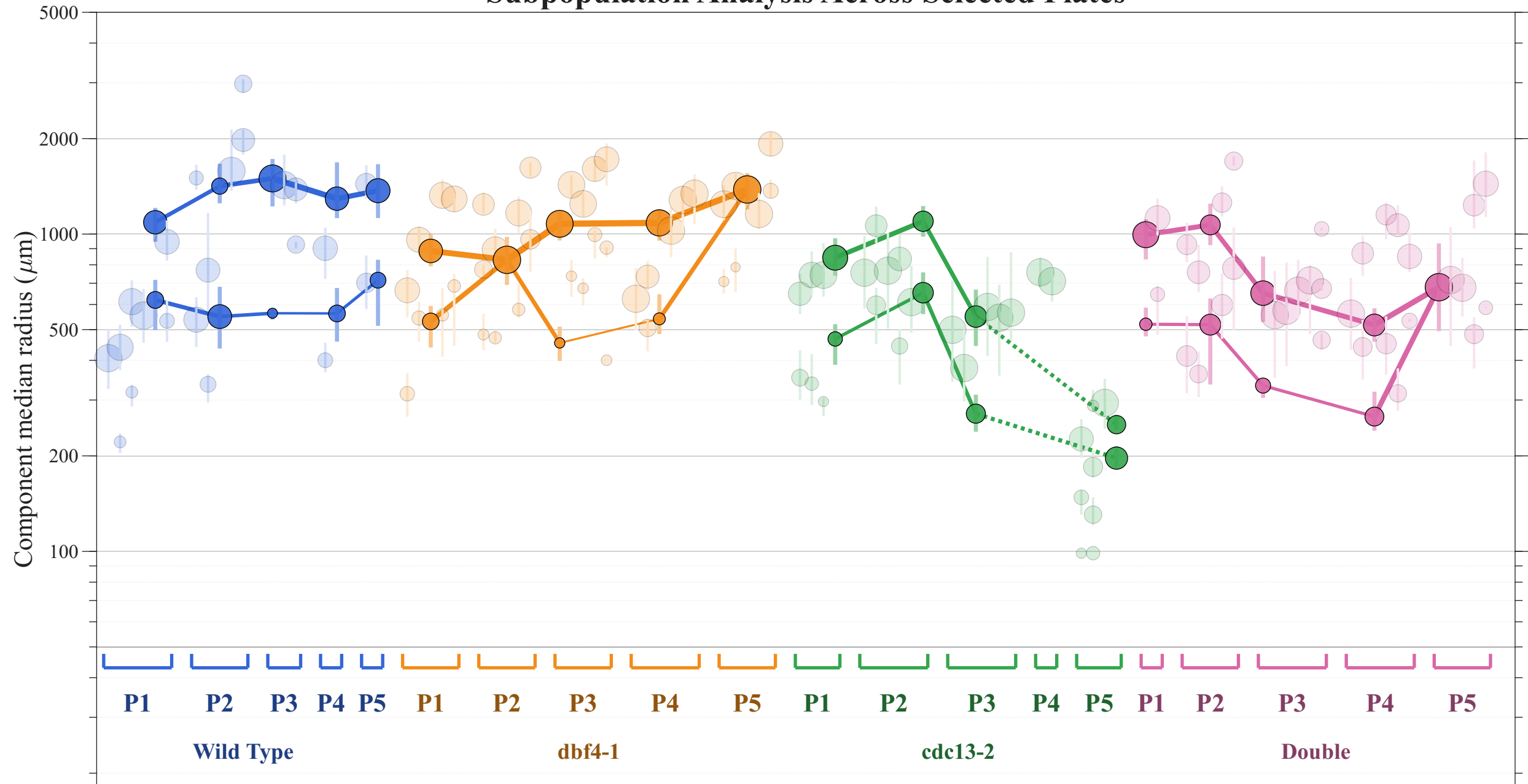
- Weight 0.1
- Weight 0.5
- Weight 1.0

Plates

- Selected plate
- Other plates
- Component IQR

Trajectory

- Consecutive passages
- Skipped passage
- Mean weight 0.1
- Mean weight 0.9



Subpopulation Analysis Across Selected Plates

Component weight

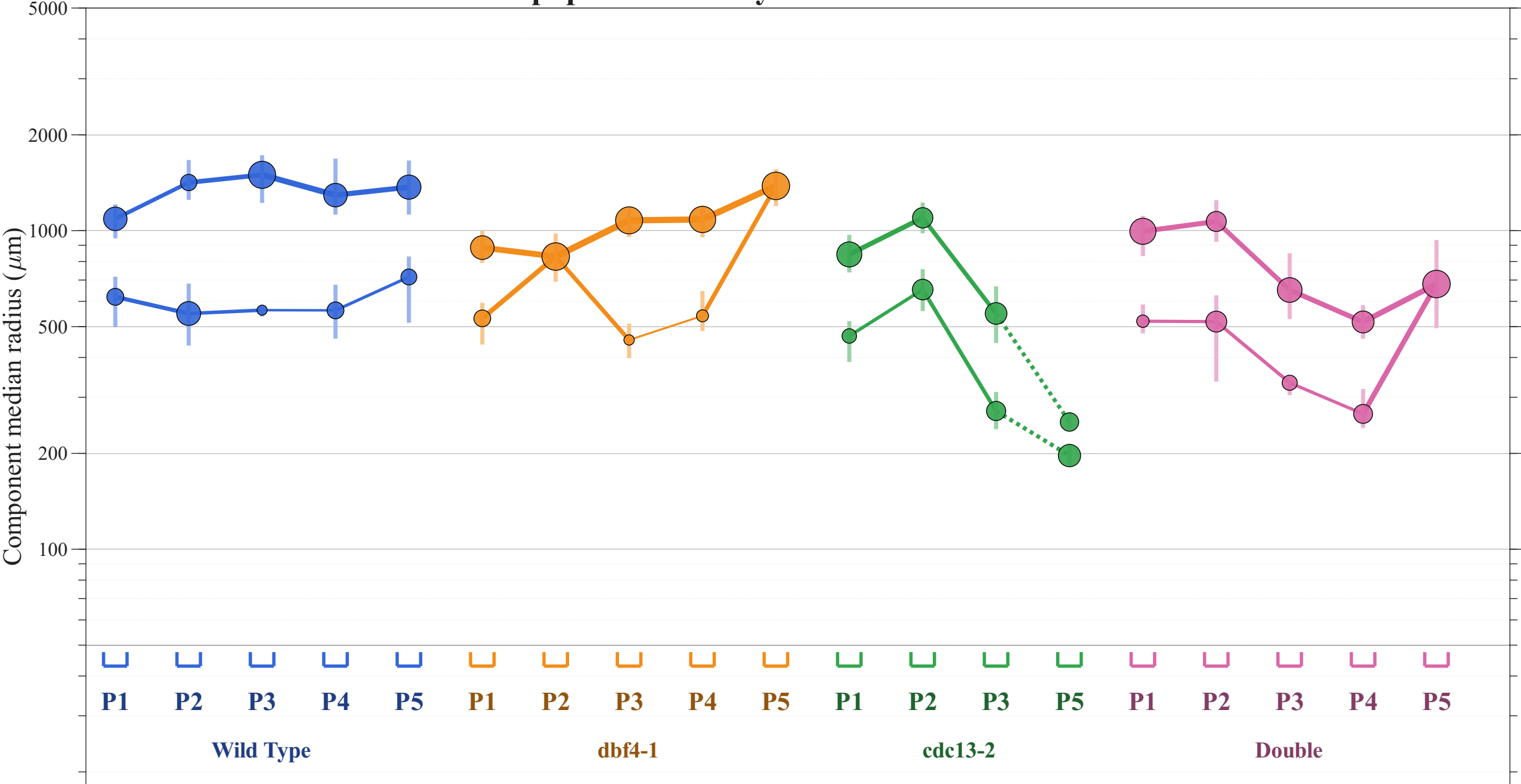
- Weight 0.1
- Weight 0.5
- Weight 1.0

Plates

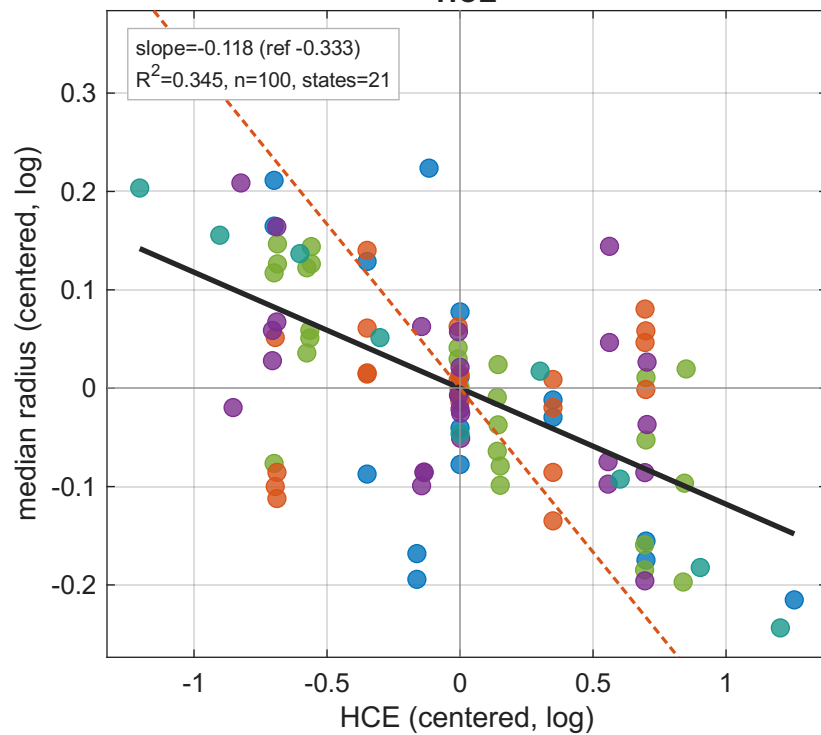
- Selected plate
- Component IQR

Trajectory

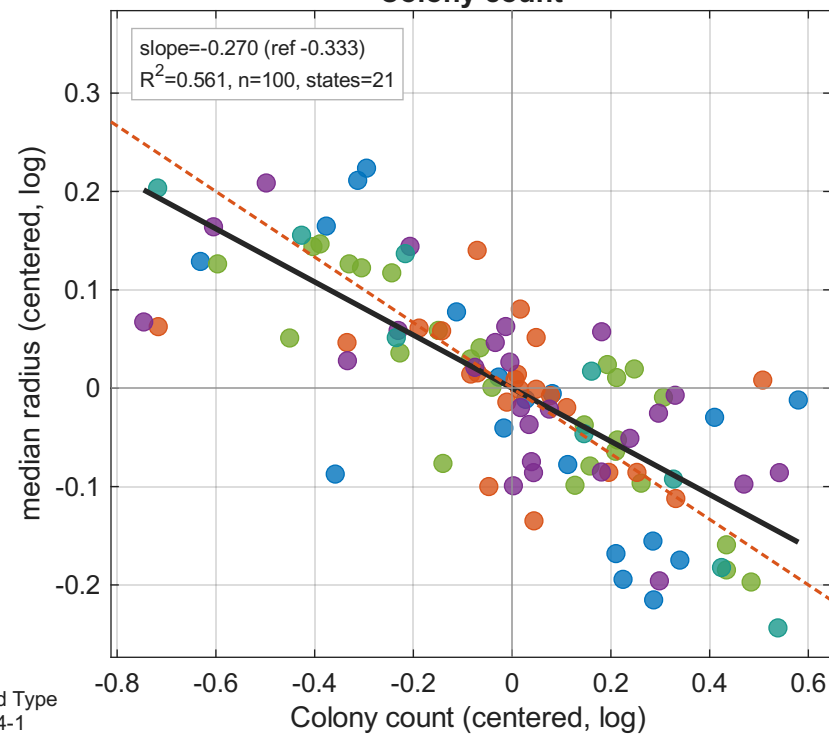
- Consecutive passages
- Skipped passage
- Mean weight 0.1
- Mean weight 0.9



HCE

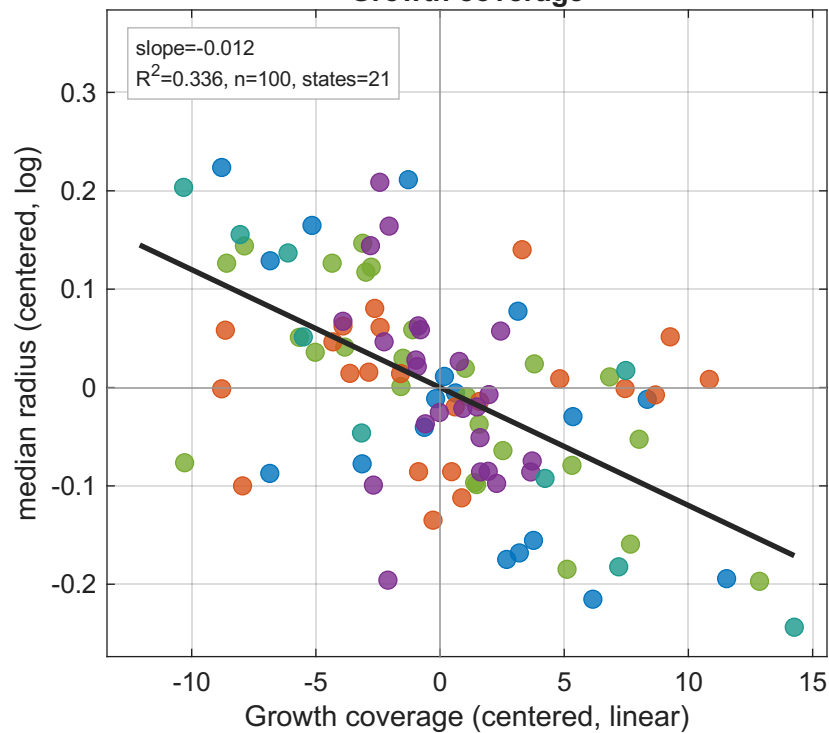


Colony count

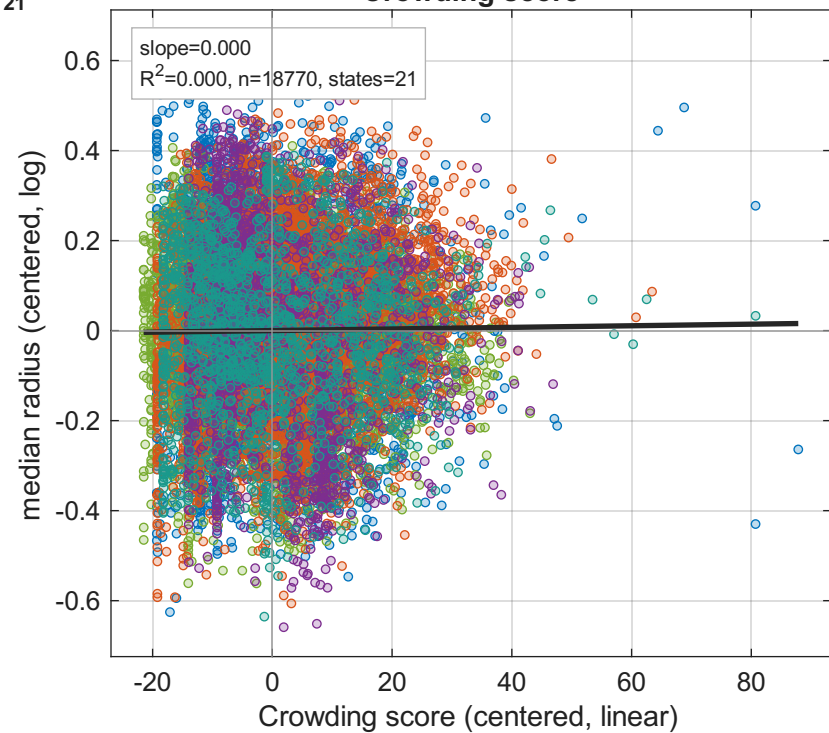


● Wild Type
● dbf4-1
● cdc13-2
● dbf4-1 cdc13-2
● WT Dilution (Part 2)
states = 21

Growth coverage



Crowding score



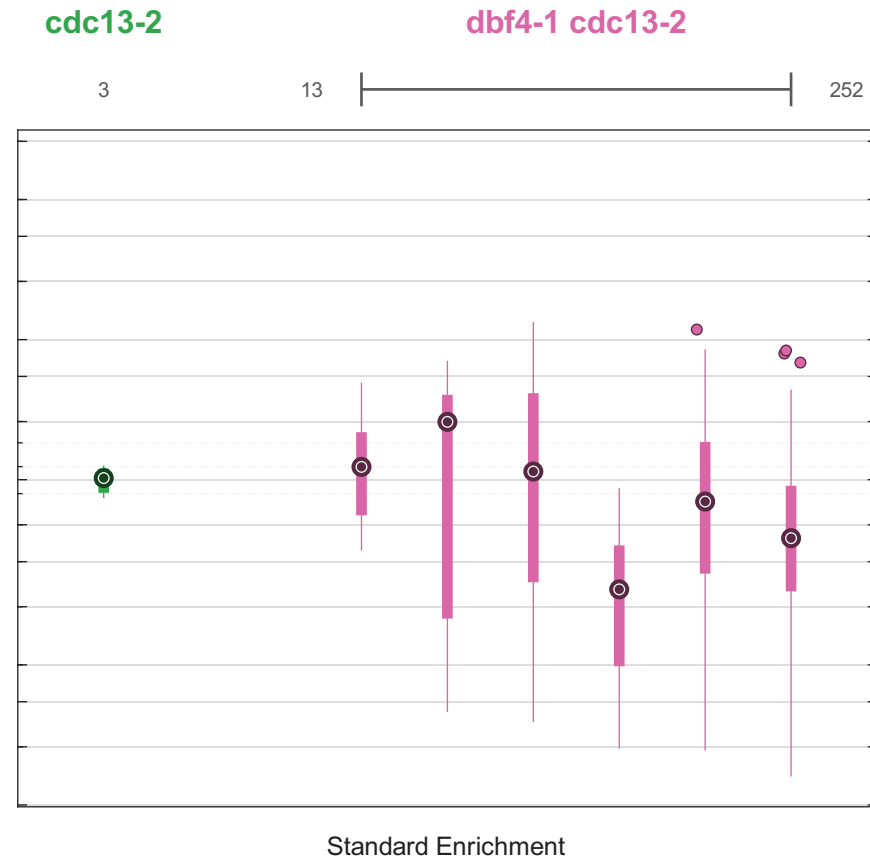
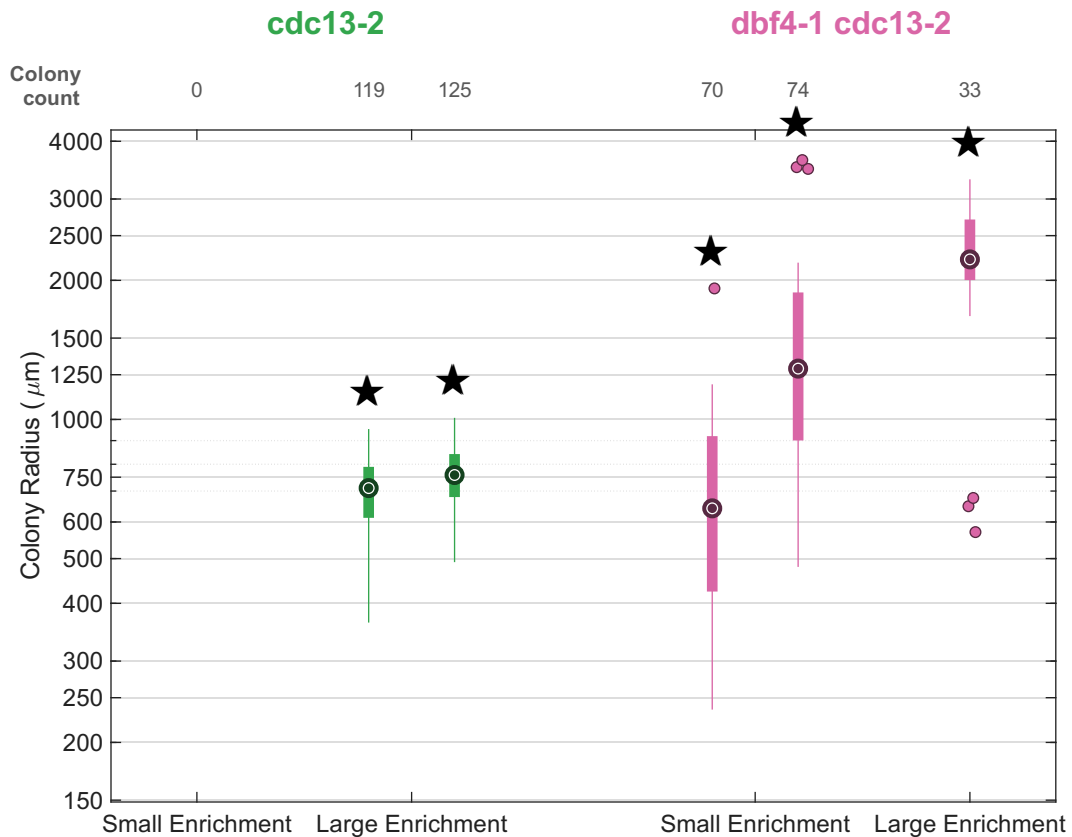


Plate flags:
★ 140 mm plate

